

# PedalScope Methods

Technical reference: how each measurement is made, and how to check it

## PedalScope

### Abstract

This document states, for each measurement PedalScope makes, exactly what the shipping code computes: the stimulus, the analysis pipeline step by step with its equations, every parameter that determines the result, and the limits of what the result can claim. Every number in it is generated from a machine-readable [methods manifest](#) emitted by the shipped code (the source is open: <https://github.com/PhaseDog/PedalScope>), every method is reimplemented independently in Python from that manifest and the published definition, and a continuous-integration [parity test](#) holds the two implementations to each other and to analytic ground truth. A method change in the app that the Python does not follow is a failing build, not a stale page.

© 2026 Richard Hoge. This document is licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). The accompanying scripts are licensed under MIT (<https://opensource.org/license/mit>). PedalScope and its mark are not part of either grant.

## Contents

|          |                                                             |           |
|----------|-------------------------------------------------------------|-----------|
| <b>1</b> | <b>How to read this document, and how to trust it</b>       | <b>3</b>  |
| 1.1      | The pipeline behind the pages . . . . .                     | 3         |
| 1.2      | What this document does not do . . . . .                    | 3         |
| <b>2</b> | <b>Notation and conventions</b>                             | <b>3</b>  |
| <b>3</b> | <b>Harmonic Distortion</b>                                  | <b>4</b>  |
| 3.1      | Purpose, and what the measurement cannot tell you . . . . . | 4         |
| 3.2      | The stimulus . . . . .                                      | 4         |
| 3.3      | The pipeline . . . . .                                      | 5         |
| 3.4      | Read-time predicates on the result . . . . .                | 8         |
| 3.5      | Parameters . . . . .                                        | 9         |
| 3.6      | The result . . . . .                                        | 10        |
| 3.7      | Worked example and parity . . . . .                         | 11        |
| 3.8      | Limits and failure modes . . . . .                          | 14        |
| <b>4</b> | <b>Transfer Curve</b>                                       | <b>14</b> |
| 4.1      | Purpose, and what the measurement cannot tell you . . . . . | 14        |
| 4.2      | The stimulus . . . . .                                      | 15        |
| 4.3      | The pipeline . . . . .                                      | 15        |
| 4.4      | Read-time predicates on the result . . . . .                | 16        |

|          |                                                             |           |
|----------|-------------------------------------------------------------|-----------|
| 4.5      | Parameters . . . . .                                        | 18        |
| 4.6      | The result . . . . .                                        | 19        |
| 4.7      | Worked example and parity . . . . .                         | 19        |
| 4.8      | Limits and failure modes . . . . .                          | 24        |
| <b>5</b> | <b>Chord IMD</b>                                            | <b>26</b> |
| 5.1      | Purpose, and what the measurement cannot tell you . . . . . | 26        |
| 5.2      | The stimulus . . . . .                                      | 26        |
| 5.3      | The pipeline . . . . .                                      | 28        |
| 5.4      | Read-time predicates on the result . . . . .                | 30        |
| 5.5      | Parameters . . . . .                                        | 34        |
| 5.6      | The result . . . . .                                        | 35        |
| 5.7      | Worked example and parity . . . . .                         | 37        |
| 5.8      | Limits and failure modes . . . . .                          | 41        |
| <b>6</b> | <b>Gain Map</b>                                             | <b>42</b> |
| 6.1      | Purpose, and what the measurement cannot tell you . . . . . | 42        |
| 6.2      | The stimulus . . . . .                                      | 43        |
| 6.3      | The pipeline . . . . .                                      | 43        |
| 6.4      | Read-time predicates on the result . . . . .                | 44        |
| 6.5      | Parameters . . . . .                                        | 46        |
| 6.6      | The result . . . . .                                        | 47        |
| 6.7      | Worked example and parity . . . . .                         | 47        |
| 6.8      | Limits and failure modes . . . . .                          | 53        |
| <b>7</b> | <b>Glossary</b>                                             | <b>55</b> |
| <b>A</b> | <b>Scripts, tests and how to reproduce</b>                  | <b>62</b> |

# 1 How to read this document, and how to trust it

## 1.1 The pipeline behind the pages

1. **The manifest.** `analysisdump manifest` (a package tool built on the app’s own measurement kernel) emits `Docs/Methods/generated/manifest.json`: one object per measurement, every parameter obtained by calling the shipped function that owns it, never by transcription. Where a parameter is a function rather than a number — a window shape, a gate — the manifest records its name, its defining constants and a one-line rule.
2. **The reimplementation.** `Docs/Methods/py/harmonic_distortion.py` is the method written again, from the manifest and the published definition [1, 3], not from the Swift; each further chapter has its own (`transfer_curve.py` for the Transfer Curve, `chord_imd.py` for Chord IMD, `gain_map.py` for the Gain Map, which imports the first). It produces the same table the shipped tool produces.
3. **The parity test.** `test_parity_hd.py` (and, per chapter, its sibling) runs the shipped code and the Python on identical synthetic devices and asserts, at every valid point, that they agree to a stated tolerance and that both agree with analytic truth to a stated tolerance. Tolerances were set from measured distributions and are quoted in §3.7, §4.7, §5.7 and §6.7.
4. **The figures** are generated by the Python, seeded and byte-stable; the UI captures are rendered headlessly from the app’s own chart views by the user-guide pipeline.
5. **The drift check.** CI regenerates the manifest, the fragments and the figures and fails on any difference from the committed copies, so what this document states is what the build computed.

## 1.2 What this document does not do

It does not describe what any measurement sounds like. A measured difference is a measured difference; whether it can be heard is a question the instrument does not answer, and no sentence here claims otherwise. It names no third-party product in a finding.

# 2 Notation and conventions

| Symbol       | Meaning                                                                                     |
|--------------|---------------------------------------------------------------------------------------------|
| $f_s$        | the sample rate, Hz                                                                         |
| $f_1, f_2$   | the sweep’s start and end frequencies, Hz                                                   |
| $L$          | the sweep’s <a href="#">rate constant</a> , s: $f(t) = f_1 e^{t/L}$                         |
| $T$          | the sweep’s actual duration, $T = L \ln(f_2/f_1)$                                           |
| $A$          | the sweep amplitude, linear, 1 = digital full scale                                         |
| $\varphi(t)$ | the sweep’s instantaneous phase, rad                                                        |
| $f(t)$       | the sweep’s <a href="#">instantaneous frequency</a> , Hz                                    |
| $k$          | a <a href="#">harmonic order</a> ; $k = 1$ is the fundamental                               |
| $K$          | the highest order extracted                                                                 |
| $f_0$        | a <a href="#">fundamental</a> frequency, Hz — the excitation whose $k$ -th harmonic is read |
| $H_k(f)$     | the <a href="#">harmonic response</a> of order $k$ on the output-frequency axis (§5)        |
| $\Delta t_k$ | the time advance of order $k$ ’s <a href="#">harmonic pulse</a> : $L \ln k$                 |
| $N$          | the deconvolution DFT length; $M$ the per-order response DFT length                         |

| Symbol               | Meaning                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------|
| $T_k$                | the <a href="#">extraction window</a> duration for order $k$                                                    |
| $h[n]$               | the deconvolved <a href="#">impulse response</a> , circular over $N$ samples                                    |
| dB re input          | $20 \log_{10}  H_k $ — <a href="#">input-normalized</a> : a unit-gain linear device reads $H_1 = 1$ , i.e. 0 dB |
| <a href="#">dBFS</a> | dB relative to digital full scale                                                                               |
| <a href="#">dBc</a>  | dB relative to a carrier (here, the louder played tone or the fundamental)                                      |

Angles are radians. Logarithms are natural unless written  $\log_{10}$ . Time-domain indices are integers  $n = 0 \dots N - 1$ ; “circular” means modulo the buffer length. A raised cosine is  $\frac{1}{2}(1 - \cos \theta)$  rising or  $\frac{1}{2}(1 + \cos \theta)$  falling.

## 3 Harmonic Distortion

### 3.1 Purpose, and what the measurement cannot tell you

Harmonic Distortion characterises how a device converts a single input frequency into output at integer multiples of it: for every fundamental  $f_0$  in the swept band and every order  $k \leq K$ , the amplitude the device emits at  $k f_0$ , normalized by the input amplitude. One [synchronized swept sine](#) captures the whole band at one drive level; the result is  $K$  frequency responses  $H_1(f) \dots H_K(f)$  and the derived [total harmonic distortion](#) curve.

What it cannot tell you, stated once: (a) anything about behaviour at other drive levels — one sweep is one operating point; the Gain Map and Compression measurements own the level axis; (b) anything about intermodulation between two simultaneous notes — Chord IMD owns that; (c) whether a read in an order is content the circuit shaped or noise that landed in that order’s analysis window — the [harmonic shaping](#) predicate reads that separately (§3.4); (d) anything below the rig’s own [measurement floor](#) (§3.4); and (e) anything about what the distortion sounds like.

### 3.2 The stimulus

The stimulus is a synchronized exponential sine sweep [2, 3]:

$$x[n] = A \sin \varphi(t_n), \quad \varphi(t) = 2\pi f_1 L \left( e^{t/L} - 1 \right), \quad t_n = n/f_s, \quad n = 0 \dots \lfloor T f_s + \frac{1}{2} \rfloor - 1. \quad (1)$$

Its instantaneous frequency is  $f(t) = f_1 e^{t/L}$ , so the sweep reaches  $f_2$  at  $T = L \ln(f_2/f_1)$ . The rate constant is not free: the [synchronization condition](#) requires  $f_1 L$  to be an integer, so for a requested duration  $\hat{T}$

$$L = \frac{\max(1, \text{round}(f_1 \hat{T} / \ln(f_2/f_1)))}{f_1}, \quad (2)$$

which is what makes the  $k$ -th harmonic generated by a memoryless nonlinearity an exact time-advanced copy of the sweep, advanced by

$$\Delta t_k = L \ln k, \quad (3)$$

and therefore separable in the deconvolved response with meaningful phase. The first  $\lfloor t_{\text{in}} f_s \rfloor$  samples are scaled by a rising raised cosine and the last  $\lfloor t_{\text{out}} f_s \rfloor$  by a falling one. The fades bound the [excited band](#): below  $f(t_{\text{in}})$  and above  $f(T - t_{\text{out}})$  the device was driven under a ramp, and reads there are properties of the measurement, not the device.

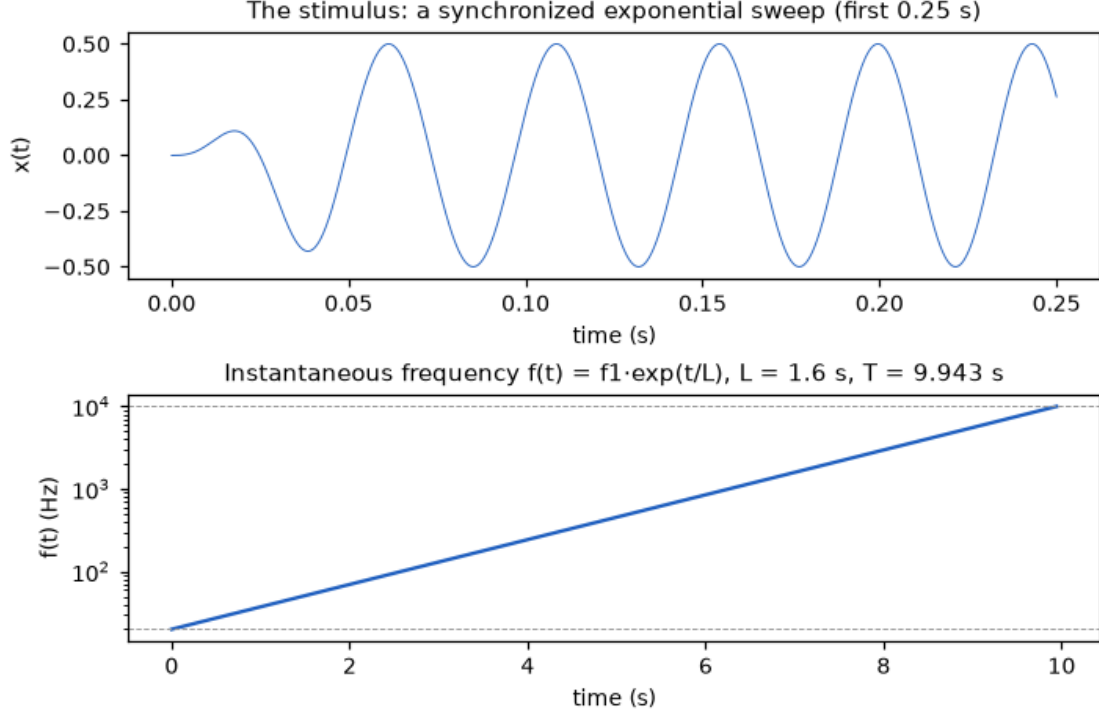


Figure 1: The stimulus, from the reimplemention: the waveform’s first quarter second and the instantaneous frequency over the full sweep.

### 3.3 The pipeline

Given the captured response  $y[n]$ , time-aligned with the sweep’s start (the app measures and removes the loop latency per capture before analysis; the oracle runs at zero latency):

1. **deconvolution with the analytic inverse.** Zero-pad  $y$  to  $N$ , the smallest power of two not less than the capture length plus  $\lceil 0.1f_s \rceil$ , and take its DFT  $Y[m]$ . The **inverse filter** is the stationary-phase closed form [3]

$$\tilde{X}^{-1}(f) = 2\sqrt{f/L} \exp\left(j\left(\frac{\pi}{4} - 2\pi fL(1 - \ln(f/f_1))\right)\right), \quad (4)$$

evaluated at  $f = mf_s/N$  for  $0 < m < N/2$  and mirrored to the negative bins, tapered to zero below  $f_1$  (a rising raised cosine from  $f_1/2$  to  $f_1$ ) with the DC and Nyquist bins zero. Using the closed form rather than inverting the sweep’s own spectrum extrapolates the phase law above  $f_2$ , which is what lets harmonics of high fundamentals — whose output frequencies exceed  $f_2$  — deconvolve into clean pulses all the way to Nyquist. The impulse response is

$$h[n] = \frac{1}{Nf_s} \text{IDFT}\{Y[m] \tilde{X}^{-1}(mf_s/N)\}[n], \quad (5)$$

the  $1/N$  being the inverse DFT’s scaling and the  $1/f_s$  converting a sampled signal’s DFT to the continuous-time spectrum the analytic inverse expects. A unit-gain linear device yields a pulse of height  $A$  at  $n = 0$ .

2. **Harmonic separation.** Order  $k$ ’s pulse arrives  $\Delta t_k$  earlier than the linear one, i.e. at circular index  $N - \text{round}(\Delta t_k f_s)$  (Figure 2).
3. **Windowing.** Each order is cut out with a raised-cosine window of half-widths  $\ell_k$  (earlier

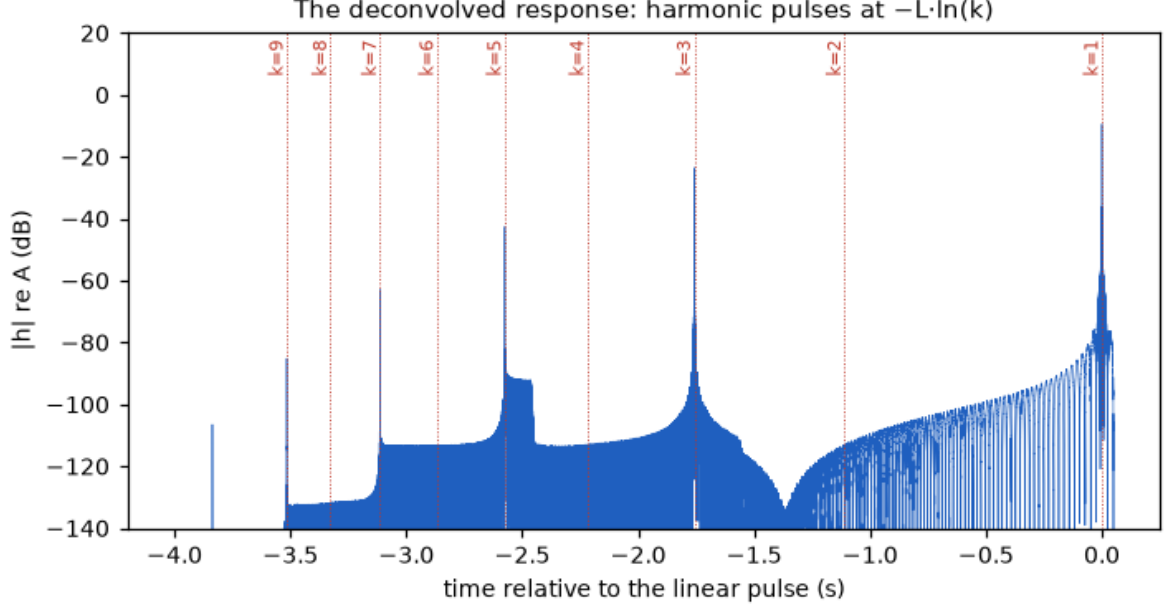


Figure 2: The deconvolved response of the worked example (a symmetric soft clipper), on a negative-delay axis. Odd orders carry content; the even orders' pulses are absent, as they must be for a symmetric device.

in time, toward order  $k + 1$ ) and  $r_k$  (toward order  $k - 1$ ), in samples:

$$c = \min\left(\frac{M}{2} - 1, t_{\max} f_s\right), \quad \ell_k = \left\lfloor \min\left(c, \frac{(\Delta t_{k+1} - \Delta t_k) f_s}{2}\right) \right\rfloor, \quad (6)$$

$$r_k = \begin{cases} \lfloor c \rfloor & k = 1 \\ \left\lfloor \min\left(c, \frac{(\Delta t_k - \Delta t_{k-1}) f_s}{2}\right) \right\rfloor & k > 1, \end{cases}$$

with  $t_{\max}$  the half-width cap. The window is  $w(o) = \frac{1}{2}(1 + \cos(\pi|o|/\ell_k))$  for  $-\ell_k \leq o < 0$  and  $\frac{1}{2}(1 + \cos(\pi o/r_k))$  for  $0 \leq o < r_k$ : unity at the pulse, zero at each edge. An order is emitted only while  $\ell_k \geq 16$  and  $r_k \geq 16$ ; the analyzer stops at the first order that fails, so a short sweep simply carries fewer orders. The window duration  $T_k = (\ell_k + r_k)/f_s$  is what sets order  $k$ 's spectral resolution and the correlation width the shaping predicate must clear.

4. **Spectra.** The windowed segment is placed with the pulse at circular index 0 of an  $M$ -point buffer (so no linear-phase correction is needed) and transformed:

$$H_k[b] = \frac{1}{A} \text{DFT}_M \{ h[(p_k + o) \bmod N] w(o) \}[b], \quad b = 0 \dots M/2, \quad (7)$$

with  $p_k$  the pulse position. The  $1/A$  is the [input-normalized](#) convention:  $H_1 = 1$  for a unit-gain linear device.

5. **Reading.**  $H_k$  lives on the *output* frequency axis. The  $k$ -th harmonic of a fundamental  $f_0$  is read at  $f = k f_0$ ,  $|H_k|$  interpolated linearly between the two bins adjacent to  $f/(f_s/M)$ , and reported as  $20 \log_{10} |H_k(k f_0)|$ .
6. **Validity.** A read is trustworthy only inside the [validity bound](#):  $f_1 \leq f_0 \leq f_2$  and

$$f_0 \leq \frac{f_s}{2k + 1}, \quad (8)$$

beyond which the capture folds order  $k + 1$  onto the read at  $k f_0$  (the [alias image](#) bound), and on a loopback-compensated record additionally  $f_0 \leq f_{\text{cal}}/k$  with  $f_{\text{cal}}$  the calibration

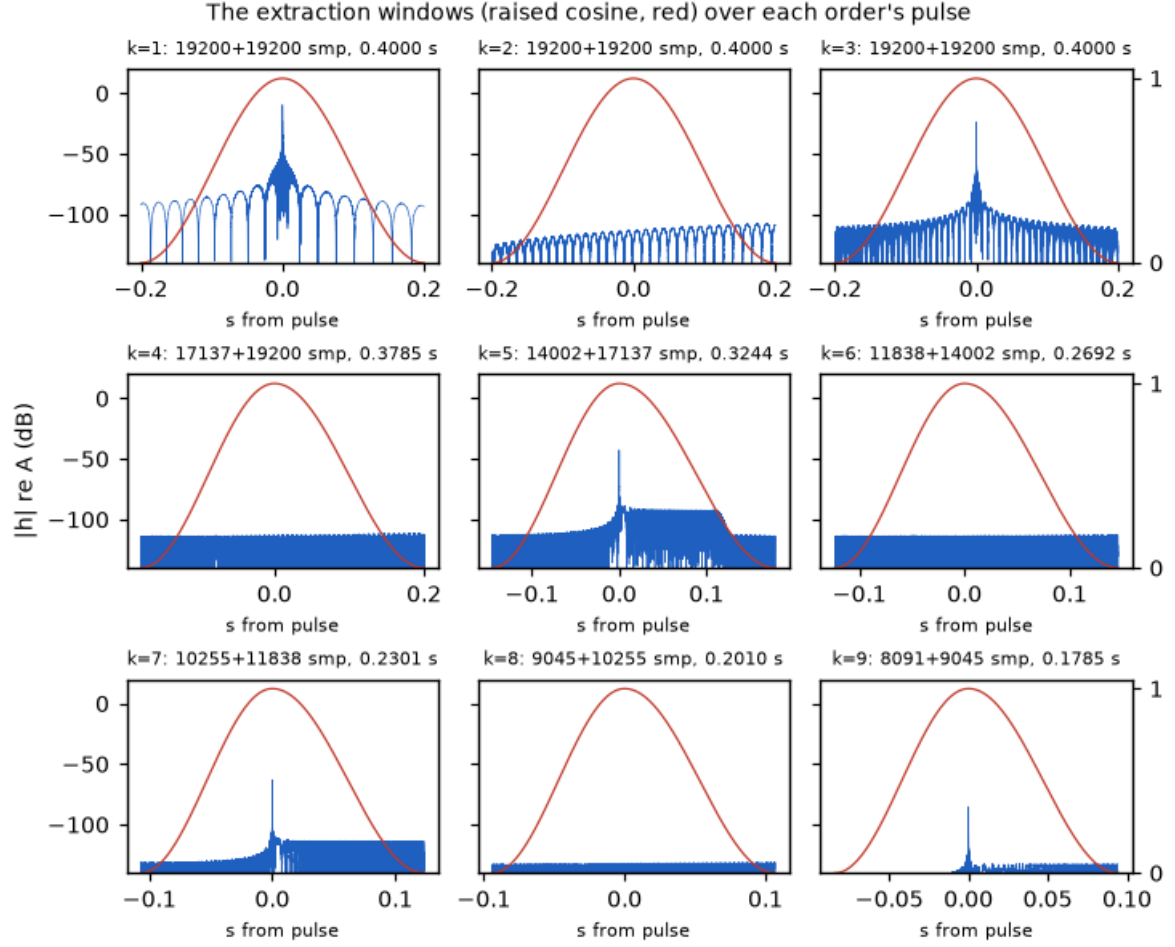


Figure 3: The nine extraction windows over their pulses. Orders 1–3 take the full cap; from order 4 the gap to the neighbouring pulse binds.

sweep’s excited-band top. Invalid reads are excluded from every fit, feature, summary and comparison.

7. **THD.** With the audible band  $B$ ,

$$\text{THD}(f_0) = \frac{\sqrt{\sum_{k \geq 2, kf_0 \in B, \text{ valid}} |H_k(kf_0)|^2}}{|H_1(f_0)|}, \quad (9)$$

THD and not THD+N; a point where the validity bound removed an in-band order is flagged truncated. The curve is sampled on a log-spaced grid of fundamentals from  $1.5f_1$  (the first fraction of the sweep is fade-in territory) to  $f_2$ .

### 3.4 Read-time predicates on the result

Three further judgements are computed from the stored spectra when a record is displayed or compared; none is persisted, so fixing a predicate fixes every stored record.

**The floor.** The rig’s own floor comes from the embedded calibration’s pooled THD-vs- frequency stamp (one loopback sweep), read at  $f_0$  by linear interpolation on the  $(\log f, \text{dB})$  plane, then adjusted: per track by  $-10 \log_{10} N_{\text{tracks}}$  (the pooled figure spread over the harmonic tracks), for drive by the calibration sweep level minus the record’s drive, and for [sweep averaging](#) by  $-10 \log_{10}(N/N_{\text{cal}})$  — applied only where the residual the stamp measures is noise (inside the validity bound, above the fade-in edge). A plugin record’s operative floor enters as a second, never-averaged bound and the applicable floor is the larger. A read at or below its floor is a bound, not a figure.

**Fundamental presence.** Every ratio to  $H_1$  (THD above all) is arithmetic unless the fundamental is a measurement: a fundamental sitting at or beyond the analyzer’s separation bound below the loudest valid, in-band order is [fundamental presence-absent](#) and nothing is divided by it.

**Shaping.** Anything inside order  $k$ ’s window is reported as  $H_k$ , the device’s own hiss included. Whether a read is content the circuit deterministically shaped is decided by the [coherence](#) of the complex response along frequency: with a lag of  $\lceil \gamma / (T_k \cdot f_s / M) \rceil$  bins (clear of the window’s own spectral correlation) and  $n$  samples,

$$z_i = H(b_i + \text{lag}) \overline{H(b_i)}, \quad \Gamma = \frac{|\sum_i z_i|}{\sum_i |z_i|}, \quad (10)$$

which for pure noise is  $1/\sqrt{n}$ .  $\Gamma$  maps to a signal-to-noise ratio through a pinned Monte-Carlo table, that ratio to the read’s own  $1\text{-}\sigma$  dB [read scatter](#) (the Rician mean-of-dB uncertainty [4]), and a read is shaped when its ratio clears the bar. The predicate’s constants and both tables are in the parameter table below.

**Averaging.**  $N$  back-to-back sweeps in one capture are analyzed separately, each repeat [derotated](#) against the first by one fitted delay for all orders (the delay  $\delta$  maximizing  $\text{Re} \sum z e^{j\omega\delta}$  over cross-spectral terms sampled per order), and the complex spectra are averaged bin by bin. Magnitudes are never averaged:  $N$  reads of a biased quantity average to the same biased number more precisely.



### 3.5 Parameters

Every value below is read from the manifest, which read it from the shipped code. The per-order table follows.

| Parameter                                    | Value                                   |
|----------------------------------------------|-----------------------------------------|
| <b>Stimulus</b>                              |                                         |
| start frequency $f_1$                        | 20 Hz                                   |
| end frequency $f_2$                          | 10 000 Hz                               |
| requested duration $\hat{T}$                 | 10 s                                    |
| sample rate $f_s$                            | 96 000 Hz                               |
| amplitude $A$ (re full scale)                | 0.5                                     |
| rate constant $L$                            | 1.6 s                                   |
| synchronization cycles $f_1 L$               | 32                                      |
| actual duration $T = L \ln(f_2/f_1)$         | 9.943372957 s                           |
| sample count                                 | 954 564                                 |
| fade-in / fade-out                           | 0.05 s / 0.01 s                         |
| excited band (fade edges)                    | 20.635 Hz – 9937.7 Hz                   |
| <b>Deconvolution</b>                         |                                         |
| inverse-filter taper                         | 10 Hz – 20 Hz                           |
| DFT length $N$ for one synth capture         | 1 048 576                               |
| <b>Extraction</b>                            |                                         |
| harmonic count $K$                           | 9                                       |
| response DFT length $M$                      | 65 536                                  |
| window half-width cap                        | 0.2 s                                   |
| minimum half-width                           | 16 samples (a literal in the analyzer)  |
| bin width $f_s/M$                            | 1.4648438 Hz                            |
| bins per response                            | 32 769                                  |
| <b>Grids</b>                                 |                                         |
| distance grid (§11)                          | 96 log-spaced points, 30–10 000 Hz      |
| THD curve grid                               | 127 of 128 requested, 30–9552.89 Hz     |
| oracle grid ( <b>synth</b> )                 | 48 points, 30–10 000 Hz                 |
| distance fixed floor clamp                   | -80 dB                                  |
| distance minimum resolved share              | 0.2                                     |
| audible band                                 | 20 Hz – 20 000 Hz                       |
| <b>Floor</b>                                 |                                         |
| calibration analyzed sweeps $N_{\text{cal}}$ | 1                                       |
| averaging term at 1 / 2 / 4 / 8 sweeps       | 0 dB / -3.01 dB / -6.021 dB / -9.031 dB |
| stamp read (median points)                   | 1                                       |
| at-floor epsilon                             | 0.1 dB                                  |
| <b>Averaging</b>                             |                                         |
| default / offered sweep counts               | 4 / 1, 2, 4, 8                          |
| pre-roll / gap / tail                        | 0.5 s / 0.5 s / 1 s                     |
| delay fit: samples per order                 | 48                                      |
| delay fit: scan limit / coarse / fine        | $\pm 8$ / 0.02 / 0.0005 samples         |
| <b>Fundamental presence</b>                  |                                         |
| analyzer separation bound                    | 76 dB                                   |
| verdict grid points                          | 128                                     |
| <b>Shaping</b>                               |                                         |
| window samples                               | 32                                      |
| pure-noise coherence $1/\sqrt{n}$            | 0.17678                                 |
| decorrelation factor                         | 2.4                                     |

| Parameter                          | Value               |
|------------------------------------|---------------------|
| shaped SNR bar                     | 6 dB                |
| unshaped scatter                   | 5.57 dB             |
| <b>Summaries</b>                   |                     |
| lexicon version                    | 1.2.2               |
| <b>Oracle (analysisdump synth)</b> |                     |
| post-sweep pad                     | 48 000 samples      |
| phantom Nyquist gate               | $0.98 \times f_s/2$ |
| ground-truth quadrature points     | 262 144             |
| noise generator increment          | 0x9E3779B97F4A7C15  |

| $k$ | $\Delta t_k = L \ln k$ (s) | window $T_k$ (s) | window (samples) | max $f_0$ valid (Hz) |
|-----|----------------------------|------------------|------------------|----------------------|
| 1   | 0                          | 0.4              | 38 400           | 10 000               |
| 2   | 1.10904                    | 0.4              | 38 400           | 10 000               |
| 3   | 1.75778                    | 0.4              | 38 400           | 10 000               |
| 4   | 2.21807                    | 0.37851          | 36 337           | 10 000               |
| 5   | 2.5751                     | 0.324365         | 31 139           | 8727.27              |
| 6   | 2.86682                    | 0.269167         | 25 840           | 7384.62              |
| 7   | 3.11346                    | 0.230135         | 22 093           | 6 400                |
| 8   | 3.32711                    | 0.201042         | 19 300           | 5647.06              |
| 9   | 3.51556                    | 0.1785           | 17 136           | 5052.63              |

### 3.6 The result

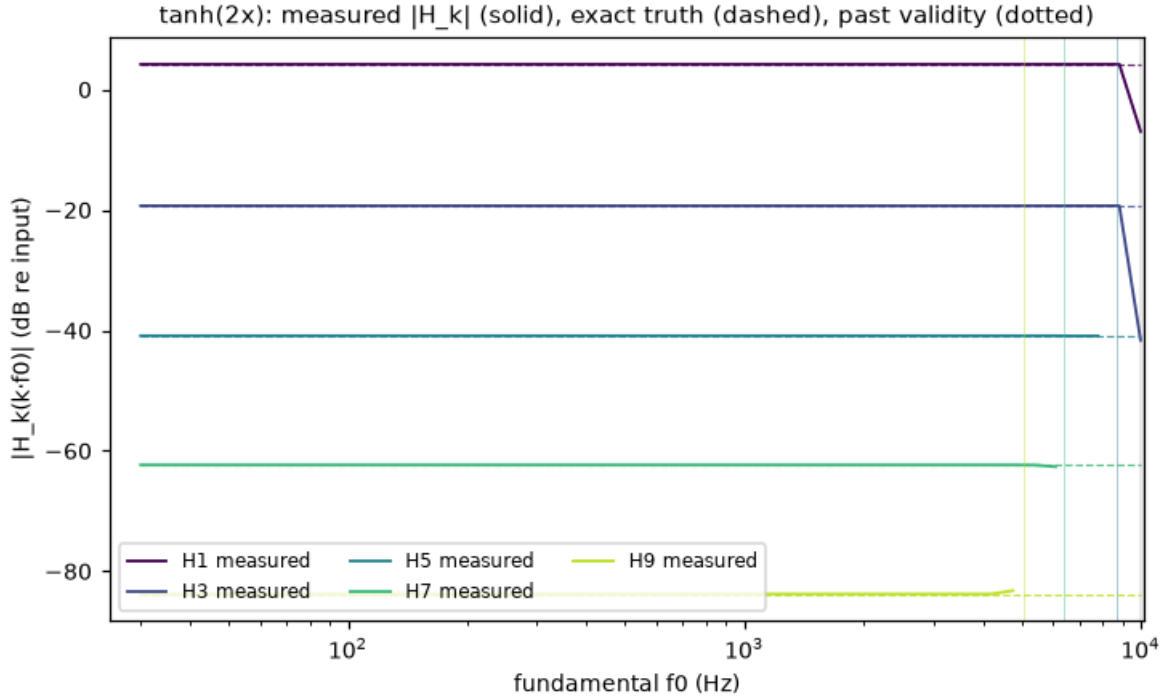


Figure 4: The worked example,  $y = \tanh(2x)$  at  $A = 0.5$ : measured  $|H_k|$  for the odd orders against the exact Fourier amplitude of the nonlinearity at the sweep amplitude. Dotted segments lie past each order's validity bound; the shaded strip past the fade-out edge is the excited band's end, where the last grid point reads the ramp, not the device.

The app’s own rendering of the same measurement kind, from the user guide’s figure pipeline (a simulated device, rendered headlessly from the chart views, never screenshotted):

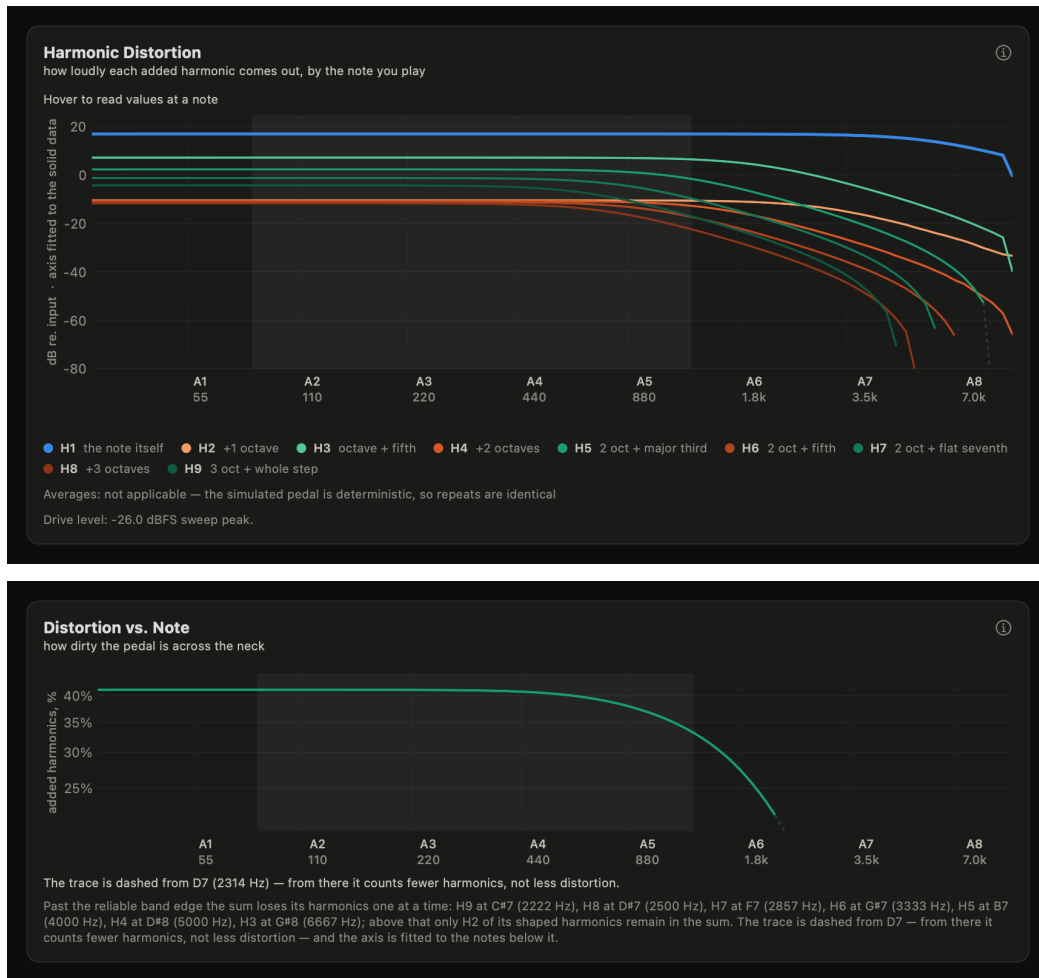


Figure 5: The Harmonic Distortion chart and the Distortion-vs-Note chart as the app draws them, from the guide’s simulated fixture.

### 3.7 Worked example and parity

The [parity test](#) runs four synthetic devices through both implementations at one seed with no noise: a [phantom capture](#) of three orders (exact by construction),  $\tanh(2x)$ , a hard clip at  $\pm 0.1$  under an amplitude of 0.5, and  $x + 0.2x^2 + 0.3x^3$ . For every valid point inside the excited band it asserts:

- $|\text{Python} - \text{Swift}| \leq 1e - 08$  dB on every order that carries content (measured maximum 2.1e-11 dB on this run; the two implementations differ only by DFT and transcendental rounding);
- both against analytic truth:  $\leq 0.01$  dB over the whole band for a phantom, and over the lower half of each order’s validity band for the per-sample nonlinearities (tanh 0.01 dB, poly 0.01 dB, hardclip 0.3 dB);
- analytically empty orders read at least 20 dB below the case’s loudest order on both sides, and the two sides agree on that residue.

The per-sample tolerances differ because the synthetic device, applied sample by sample at  $f_s$ , aliases the content it generates above Nyquist back into the band — exactly as a simulated device in the app does, and unlike a hardware device behind a converter’s anti-alias filter. The aliasing is worst in the top half-octave of each order’s band: over the full band,  $\tanh(2x)$  departs from truth by up to 0.554 dB and the hard clip (a near-square wave whose spectrum falls only as  $1/k^2$ ) by up to 3.1 dB. That this is the synthetic device and not the method is shown by each case’s *phantom twin*: the same first nine Fourier amplitudes as an alias-free phantom, which the analyzer reads to within 0.01 dB over the whole band.

| case          | $k$ | expected<br>(dB) | S–P<br>(dB) | S–T band<br>(dB) | P–T band<br>(dB) | S–T full<br>(dB) |
|---------------|-----|------------------|-------------|------------------|------------------|------------------|
| phantom       | 1   | 0                | 3.9e-15     | 0.0052           | 0.0052           | 0.0052           |
| phantom       | 2   | -10.458          | 5.3e-15     | 0.000851         | 0.000851         | 0.000851         |
| phantom       | 3   | -20              | 1.1e-14     | 0.000121         | 0.000121         | 0.000121         |
| tanh          | 1   | 4.2083           | 4.4e-15     | 0.00492          | 0.00492          | 0.00492          |
| tanh          | 3   | -19.292          | 1.1e-14     | 0.000172         | 0.000172         | 0.000218         |
| tanh          | 5   | -40.883          | 1.4e-13     | 4.17e-05         | 4.17e-05         | 0.0446           |
| tanh          | 7   | -62.328          | 2.1e-12     | 4.29e-05         | 4.29e-05         | 0.299            |
| tanh          | 9   | -83.757          | 2.1e-11     | 8.3e-05          | 8.3e-05          | 0.554            |
| tanh twin     | 1   | 4.2083           | 3.6e-15     | 0.00515          | 0.00515          | 0.00515          |
| tanh twin     | 3   | -19.292          | 1.4e-14     | 0.000169         | 0.000169         | 0.000169         |
| tanh twin     | 5   | -40.883          | 1.4e-13     | 7.21e-05         | 7.21e-05         | 7.21e-05         |
| tanh twin     | 7   | -62.328          | 1.6e-12     | 3.02e-05         | 3.02e-05         | 3.02e-05         |
| tanh twin     | 9   | -83.757          | 1e-11       | 7.73e-05         | 7.73e-05         | 7.73e-05         |
| hardclip      | 1   | -11.94           | 5.3e-15     | 0.0127           | 0.0127           | 0.0319           |
| hardclip      | 3   | -21.955          | 7.1e-15     | 0.0896           | 0.0896           | 0.548            |
| hardclip      | 5   | -27.372          | 1.4e-14     | 0.122            | 0.122            | 1.93             |
| hardclip      | 7   | -31.859          | 1.4e-14     | 0.202            | 0.202            | 2.01             |
| hardclip      | 9   | -36.348          | 1.4e-14     | 0.279            | 0.279            | 3.1              |
| hardclip twin | 1   | -11.94           | 7.1e-15     | 0.00515          | 0.00515          | 0.00515          |
| hardclip twin | 3   | -21.955          | 1.4e-14     | 0.000179         | 0.000179         | 0.000179         |
| hardclip twin | 5   | -27.372          | 1.1e-14     | 5.82e-05         | 5.82e-05         | 5.82e-05         |
| hardclip twin | 7   | -31.859          | 7.1e-15     | 2.04e-05         | 2.04e-05         | 2.04e-05         |
| hardclip twin | 9   | -36.348          | 2.1e-14     | 6.05e-05         | 6.05e-05         | 6.05e-05         |
| poly          | 1   | 0.47533          | 3.7e-15     | 0.00523          | 0.00523          | 0.00523          |
| poly          | 2   | -26.021          | 2.5e-14     | 0.000588         | 0.000588         | 0.000588         |
| poly          | 3   | -34.54           | 4.3e-14     | 0.000136         | 0.000136         | 0.000136         |
| poly twin     | 1   | 0.47533          | 3.7e-15     | 0.00516          | 0.00516          | 0.00516          |
| poly twin     | 2   | -26.021          | 1.8e-14     | 0.00106          | 0.00106          | 0.00106          |
| poly twin     | 3   | -34.54           | 5e-14       | 9.32e-05         | 9.32e-05         | 9.32e-05         |

For  $\tanh(2x)$  the exact amplitudes are  $H_1 = 4.2083$  dB,  $H_3 = -19.292$  dB and  $H_9 = -83.757$  dB re input. A fifth case adds seeded white noise at  $-80$  dBFS and four averaged repeats (the noise stream is reproduced exactly on both sides — same generator, same seeding). On the points the shipped predicate reads as shaped on both sides, the error against truth must sit within 3 times the read’s own stated scatter; the last column is that ratio:

| $k$ | expected (dB) | shaped points | S–P (dB) | S–T shaped, band (dB) | error / allowance |
|-----|---------------|---------------|----------|-----------------------|-------------------|
| 1   | 4.2083        | 42            | 4.4e-15  | 0.00491               | 0.022             |
| 3   | -19.292       | 42            | 1.1e-14  | 0.00246               | 0.036             |
| 5   | -40.883       | 41            | 7.8e-14  | 0.0283                | 0.41              |
| 7   | -62.328       | 38            | 8.7e-13  | 0.254                 | 0.87              |
| 9   | -83.757       | 36            | 1.2e-11  | 3.9                   | 0.8               |

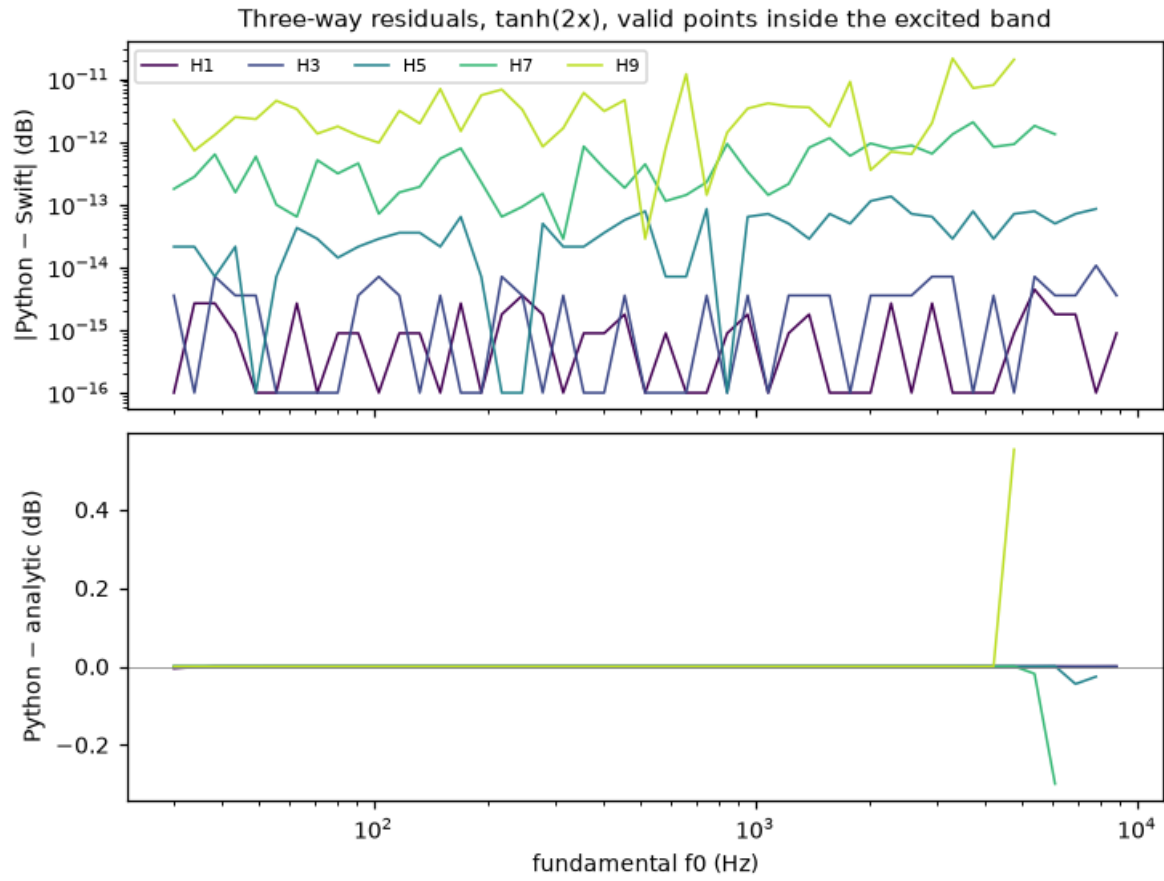


Figure 6: Three-way residuals for the worked example, every valid point inside the excited band. Top:  $|\text{Python} - \text{Swift}|$ , at the level of double-precision rounding. Bottom: Python minus analytic truth, showing the per-sample aliasing rising toward each order's band top.

### 3.8 Limits and failure modes

- **The fade edges are measurement, not device.** The last grid point sits on  $f_2$ , inside the fade-out, and reads the ramp (the phantom's  $H_1$  there is 11.7 dB low); on a compensated record the calibrated-band bound removes it, and on a raw or simulated one the [excited band](#) is the honest limit. The first fraction of the sweep is treated the same way by every grid's  $1.5f_1$  start.
- **Leakage between orders is bounded, and the bound is measured.** An order's read is floored by what the loudest order leaks into its window: worst at the band's low edge, and the value the presence predicate uses is the measured one, posted a hair below what the analyzer demonstrated.
- **A read is not automatically content.** The shaping predicate exists because a symmetric device's even orders are a read of its noise; every quantity computed from an unshaped read describes the noise.
- **Aliasing in a simulated device.** A device applied per sample at  $f_s$  folds its content above Nyquist into the band; the parity numbers above quantify it for two clippers. Hardware through a converter does not do this, and the alias-image bound (8) is about the *capture's* folding of the next order, a different mechanism.
- **Short sweeps carry fewer orders.** The 16-sample minimum half-width is a literal in the analyzer, pinned by the tool's guard rather than by the constants census; it binds only for sweeps far shorter than the standard plan.
- **The THD curve's top point.** The shipped curve requests 128 log-spaced points but its top point can land an ulp above  $f_2$  and be dropped by the band guard; the manifest reports the delivered count.

## 4 Transfer Curve

### 4.1 Purpose, and what the measurement cannot tell you

The [transfer curve](#) draws a device's clipping shape directly: while a steady low-frequency sine plays, the instantaneous output is plotted against the instantaneous input over one [averaged cycle](#). A memoryless device retraces a single curve, the [static curve](#), and the figure's bends are its [clipping](#); anything that makes the output depend on where the signal has just been — a shifting bias point, stored charge — opens the figure into a loop. The result is the averaged cycle itself (two vectors of  $n$  phase bins), the static curve, and three dimensionless loop areas: the total, the part a linear filter's phase rotation explains, and the part it does not.

What it cannot tell you, stated once: (a) anything about other drive levels — one curve is one operating point, and a clipper whose knee sits above the drive draws a straight line; (b) whether an open loop is memory or filter phase — ordinary tone filters rotate phase and open the loop without any memory, which is why the [phase removal](#) exists and why the raw figure's middle band of loop area is hedged; (c) the device's *level* asymmetry once a series [coupling capacitor](#) has removed the output's mean, which every pedal's output stage does (§4.8); (d) anything about a rig's own residual alignment, which draws a loop the measurement cannot distinguish from the device's (§4.8); and (e) anything about what the distortion sounds like.

## 4.2 The stimulus

The stimulus is a sine at the probe note  $f_0$  (the standard note is E2, at the value the parameter table gives verbatim; other notes are exact twelve-tone equal temperament with A4 = 440 Hz), at a linear peak amplitude  $A$ :

$$x[n] = A \sin(2\pi f_0 n / f_s), \quad n = 0 \dots \lfloor T f_s \rfloor - 1, \quad (11)$$

scaled over its first and last  $\lfloor t_{\text{fade}} f_s \rfloor$  samples by the raised cosine  $\frac{1}{2}(1 - \cos(\pi i / N_{\text{fade}}))$ , and preceded and followed by silence (the pre-roll and the tail). The app plays this stimulus, measures the loop latency per capture, and drops the pre-roll before analysis; the oracle renders it through a synthetic device at zero latency. The standard amplitude is the fingerprint sweep's drive ( $-26$  dBFS), so the curve and the other measurement kinds describe one operating point; the worked example below uses the loudest drive the hardware sheet offers, because at the standard drive the reference clippers draw a straight line, which is the point of the sheet's level control.

## 4.3 The pipeline

Given the tone  $x[n]$  and the captured output  $y[n]$ , aligned by the integer loop latency  $\ell$ :

1. **Phase binning.** With  $P = f_s / f_0$  samples per cycle, every sample  $i \geq \lfloor s P \rfloor$  ( $s$  the skipped opening cycles, while the circuit settles) with  $i + \ell$  inside the capture is assigned to the bin

$$b(i) = \min\left(n - 1, \left\lfloor n \cdot ((i/P) \bmod 1) \right\rfloor\right), \quad (12)$$

and the averaged cycle is the per-bin mean of  $x[i]$  and of  $y[i + \ell]$  respectively:  $\bar{x}_b$ ,  $\bar{y}_b$ ,  $b = 0 \dots n - 1$ . Noise averages; shape does not. Every sample to the end of the tone is binned, the fade-out included — only the start is skipped (§4.8).

2. **The box.** Every area is normalized by the rectangle that bounds the figure ([box normalization](#)):

$$B = 4 \max_b |\bar{x}_b| \max_b |\bar{y}_b|. \quad (13)$$

3. **The total loop area.** The closed polyline  $(\bar{x}_b, \bar{y}_b)$  is split at every proper self-crossing and the absolute shoelace areas of the resulting lobes are summed — the [unsigned lobe area](#) — so that opposite-winding lobes add instead of cancelling (a signed integral sees only the fundamental's quadrature and lets a figure-eight report zero). A lobe smaller than a fixed fraction of its own polyline's box is dropped as bin noise. The reported figure is  $\text{total} = \text{lobes}(\bar{x}, \bar{y}) / B$ .

4. **The fundamental ellipse.** Both cycles are projected onto  $\{1, \cos \theta_b, \sin \theta_b\}$ ,  $\theta_b = 2\pi b / n$ :

$$a_x = \frac{2}{n} \sum_b \bar{x}_b \cos \theta_b, \quad b_x = \frac{2}{n} \sum_b \bar{x}_b \sin \theta_b, \quad a_y, b_y \text{ likewise}, \quad \bar{y} = \frac{1}{n} \sum_b \bar{y}_b, \quad (14)$$

and the fundamental-only trajectory is the least-squares [fundamental ellipse](#), whose area is analytic:

$$\text{linear} = \frac{\pi |a_x b_y - a_y b_x|}{B}. \quad (15)$$

For an input  $A \sin \theta$  and an output whose fundamental is  $R \sin(\theta + \varphi_1)$  this is  $\pi A R |\sin \varphi_1| / B$  — a linear filter's rotation of the fundamental, never memory; for a linear device, whose peak is its fundamental's, it reduces to  $(\pi/4) |\sin \varphi_1|$ .

5. **The nonlinear residual.** The ellipse's output is subtracted and the lobes of what remains are summed:

$$\text{nonlinear} = \frac{\text{lobes}(\bar{x}, \bar{y}_b - \bar{y} - a_y \cos \theta_b - b_y \sin \theta_b)}{B}, \quad (16)$$

the **nonlinear loop area**: the harmonic content's own lobes, the headline memory figure. A memoryless clipper's harmonics are in phase with the drive and retrace; harmonics rotated by a filter, or by genuine memory, enclose area here.

6. **The static curve.** The cycle is split at the bins of the input's minimum and maximum into a rising branch and a falling one, each reduced to a strictly ascending polyline in  $\bar{x}$ , interpolated linearly onto a common grid of inputs from  $\min \bar{x}$  to  $\max \bar{x}$ , and the two interpolants are averaged — per-branch resampling, never binning by input value, whose sample density under a sinusoidal drive is  $\propto 1/\sqrt{A^2 - x^2}$  and starves the centre.
7. **The memory verdict.** Taken on the nonlinear area (of the compensated figure when the removal is accepted, §4.4): below the first threshold the device reads memoryless-ish, above the second it reads memory, and between them the reading is hedged deliberately — tone filtering *ahead* of the clipper rotates every harmonic by  $k$  times the drive's rotation, and a single fundamental cannot separate that from mild memory. Both thresholds are literals in the shipped verdict; the manifest reads them by bisecting it.

Every derived quantity is recomputed from the stored cycle whenever a record is read, so a stored record carries the cycle and nothing else that a fix would have to migrate.

#### 4.4 Read-time predicates on the result

Three judgements are made against the record's paired Harmonic Distortion analysis — the same device, the same setting, the swept measurement of §3 — when one exists; none is persisted.

**Orientation.** A plugin-path capture is aligned by correlation against the tone and can land half a cycle off, which reverses the figure and, for an asymmetric device, pairs  $x(t)$  with  $y(t + P/2)$ . The **orientation resolution** reads the cycle's fundamental phase  $\psi_1 = \angle Y[1] - \angle X[1]$  (direct DFTs of the two cycles); when it sits within the lock tolerance of  $\{0, \pi\}$  the capture is detectably locked, and when the paired  $H_1$ 's **polarity** confidently disagrees with  $\cos \psi_1 < 0$ , the output cycle is advanced by  $n/2$  bins and the figure recomputed. Every other case displays the figure as captured and implies nothing. Polarity itself is the energy-weighted mean unit phasor of  $H_1$  over the low band,

$$m = \frac{\sum_b |H_1[b]| \operatorname{Re} H_1[b]}{\sum_b |H_1[b]|^2}, \quad \text{inverting} \Leftrightarrow m < 0, \quad \text{confidence} = |m|, \quad (17)$$

the real part and never the resultant length: a quadrature response is coherent yet polarity-mute.

**Fundamental presence.** The memory verdict, the polarity statement and the pairing verdict all key off a fundamental; the **fundamental presence** reading is taken from the paired record at  $f_0$  (which separates orders far better than one averaged cycle) and from the cycle's own DFT only when the paired reading is undetermined. Where the fundamental is absent the page draws the raw figure and states none of the three.



**The phase removal.** Harmonics generated *behind* pre-clipper dispersion inherit  $k \varphi_{\text{pre}}(f_0)$ , while an  $H_1$ -based rotation would remove  $\varphi_{\text{pre}}(kf_0)$ ; no single shift reconciles the two, so the removal is fitted against the paired record's own branch phases  $\angle H_k(kf_0)$ , which carry each harmonic's true burden  $k \varphi_{\text{pre}}(f_0) + \varphi_{\text{post}}(kf_0)$  by construction. The steps:

1. **Fittable terms.** The cycle's harmonic phases are read from its DFT, input-relative:  $\text{cyc}_k = \angle Y[k] - k \angle X[1]$ . Order  $k$  votes when the paired record's **validity bound** accepts  $(k, f_0)$  and, for  $k \geq 2$ , both  $|Y[k]|/|Y[1]|$  and  $|H_k(kf_0)|/|H_1(f_0)|$  clear the magnitude floor; its weight is  $w_k = \min(1, |Y[k]|/|Y[1]|)$  and its phase difference

$$d_k = \text{wrap}(\text{cyc}_k - \angle H_k(kf_0) - \gamma_k), \quad \gamma_k = \text{wrap}((k-1)\pi/2), \quad (18)$$

$\gamma_k$  being the Novák factor between the cycle's convention and the deconvolver's. Fewer than the minimum number of terms means insufficient evidence, never a verdict.

2. **The frame fit.** The two captures each measured their own latency, so a shift between their frames is legitimate and must never be applied as rotation. One shift  $\Delta l$  (samples) is fitted by maximizing the weighted phase coherence

$$C(\Delta l) = \sum_k w_k \cos(d_k - k \omega_0 \Delta l), \quad \omega_0 = 2\pi f_0 / f_s, \quad (19)$$

by a coarse scan over  $\pm P/2$  and a fine scan around its maximum (both steps are literals, stated in the parameter table): the **frame fit**. The residuals  $r_k = \text{wrap}(d_k - k \omega_0 \Delta l)$  give the **pairing residual**, the weighted RMS over  $k \geq 2$  (the fundamental can always be absorbed into  $\Delta l$ , so consistency lives in the harmonics); above the tolerance the outcome is a pairing mismatch.

3. **The anchored sequential fold.** The frame-mapped branch phase  $m_k = \text{cyc}_k - r_k$  is what the paired record predicts for harmonic  $k$  in the cycle's own frame. Its burden over the memoryless **memoryless lattice**  $\{0, \pi\}$  is known only modulo  $\pi$  (the multiple removed is the device's own harmonic sign, which the figure keeps), and the folds must come out mutually consistent. The fundamental is anchored: its lattice point is  $\pi$  when the paired polarity reads inverting and 0 when not (the nearest point only when polarity is not confident, and then the sign is quotable only inside the lattice-sign confidence), giving  $e_1 = \text{wrap}(m_1 - \sigma_1)$ . Each higher voting order, in ascending order, folds against a trend:  $k e_1$  for the first, and for every later one  $ak + F(k)$ , where  $F(k) = \phi_1^u(k) - \phi_1^u(1) + (e_1 - \omega_0 \Delta l)$  is the paired  $H_1$ 's unwrapped phase *shape* anchored at the fundamental's burden (the post-clipper term) and the slope  $a$  is refit after each fold over the orders already folded (the pre-clipper term; the fundamental never votes in it):

$$\beta_k = m_k - \pi \text{round}\left(\frac{m_k - \text{trend}_k}{\pi}\right), \quad a = \frac{\sum_{j \leq k} w_j j (\beta_j - F(j))}{\sum_{j \leq k} w_j j^2}. \quad (20)$$

The per-order assumption is that the curvature beyond the ramp is under  $\pi/2$  — true of the resistor-capacitor networks pedals are built from, and weakest at the lowest order, where it holds across every device class measured.

4. **The all-pass.** The correction is a smooth function of frequency, applied to every bin of the cycle's spectrum at unity magnitude: the deviations  $\beta_k - ak - F(k)$  at the voting orders are interpolated by a shape-preserving cubic (Fritsch-Carlson, constant beyond the end nodes), and for  $1 \leq k < n/2$

$$Y[k] \leftarrow Y[k] e^{-j\theta_k}, \quad \theta_k = ak + F(k) + \text{dev}(k), \quad (21)$$

with DC and the real Nyquist bin untouched. The compensated cycle is the inverse real DFT and the figure is recomputed from it. Content above the voting orders survives phase-corrected rather than being resynthesized away, which is what removed the truncation ripple an earlier per-order resynthesis drew on hard-clipped plateaus. The correction is *measured* up to the support  $k_{\max}f_0$ , the highest voting order times the drive note; above it the ramp is extended,  $F$  follows  $H_1$ 's phase as far as its grid holds a value, and the deviation holds its last knot — an extrapolation the page's caption states.

5. **The outcome.** One of five: compensated; a pairing mismatch (no shift reconciles the harmonic phases); insufficient evidence (too few fittable terms); a negligible raw loop (the total under the dead zone, decided after the fit ran so polarity still rides it, and outranking a mismatch — a noise-dominated tiny loop must not wear a warning); and a tripped metric guard, when a removal that passed the residual still grew a *decomposed* area beyond its tolerance (the unsigned total is never compared: the ellipse and the harmonic lobes partially cancel in it). A pairing mismatch has two possible causes the residual alone does not separate: a session mismatch — another setting or another time, resolved by re-measuring both kinds together — and a structural one, a device whose harmonics have no fundamental-anchored phase relationship (a frequency doubler), which no re-measurement resolves.
6. **Polarity, confirmed.** The figure states polarity only when a fit ran, its residual confirms the pairing, and the fundamental's lattice sign resolved; the robust  $H_1$  statistic supplies the sign, the fit the trust. Absence is honest; a guessed sign is not.

An older correction — rotating harmonic  $k$  by  $\angle H_1(kf_0)$  alone — remains in the kernel with no consumer, and is not the method this document describes.

## 4.5 Parameters

Every value below is read from the manifest, which read it from the shipped code; the two verdict thresholds were located by bisecting the shipped verdict, and the scan steps, the bin count, the grid, the polarity band and the analyzer's unused default are literals the manifest carries as rule strings.

| Parameter                                                 | Value                                       |
|-----------------------------------------------------------|---------------------------------------------|
| <b>Stimulus</b>                                           |                                             |
| standard note / frequency $f_0$                           | E2 / 82.4 Hz                                |
| standard amplitude $A$ (re full scale)                    | 0.05 (-26.02 dBFS)                          |
| tone duration $T$ / pre-roll / tail                       | 1.5 s / 0.2 s / 0.2 s                       |
| fade-in and fade-out $t_{\text{fade}}$                    | 0.02 s                                      |
| skipped cycles (the app's call)                           | 8 (the analyzer's own default is 5, unused) |
| sample rate $f_s$ (the manifest's)                        | 96 000 Hz                                   |
| samples per cycle $P$ at $f_0$                            | 1165.0485                                   |
| probe notes offered                                       | E2, A2, E3, A3, E4, A4, E5, A5, A6, C7      |
| note-range capture                                        | E2, A3, A4                                  |
| drive level bounds (hardware / plugin ceiling)            | -60 to -6 dBFS / 0 dBFS                     |
| <b>Binning and the figure</b>                             |                                             |
| phase bins $n$                                            | 256                                         |
| static-curve grid points                                  | 101                                         |
| lobe floor (fraction of the box)                          | 0.0002                                      |
| <b>Memory verdict (bisected from the shipped verdict)</b> |                                             |
| memoryless-ish below                                      | 0.05                                        |
| memory effects from                                       | 0.15                                        |

| Parameter                                   | Value                           |
|---------------------------------------------|---------------------------------|
| <b>Phase removal (§ 6.5)</b>                |                                 |
| dead zone $\varepsilon$ (raw total area)    | 0.01                            |
| pairing residual tolerance                  | 0.25 rad                        |
| fit magnitude floor (both sides)            | 0.005 (-46 dB)                  |
| minimum fittable terms                      | 3                               |
| lattice sign confidence                     | 0.6 rad                         |
| metric-guard tolerance (decomposed areas)   | 0.02                            |
| frame-fit scan: coarse / fine step          | 0.05 / 0.001 samples (literals) |
| <b>Orientation and polarity</b>             |                                 |
| half-cycle lock tolerance                   | 0.05 rad                        |
| polarity confidence threshold               | 0.35                            |
| polarity band                               | 20 Hz – 2 kHz (literals)        |
| <b>Fundamental presence</b>                 |                                 |
| analyzer separation bound                   | 76 dB                           |
| <b>Oracle (analysisdump transfer-synth)</b> |                                 |
| biquad default $Q$                          | 0.70710678                      |
| truth series orders $K$                     | 200                             |
| truth quadrature points                     | 262 144                         |

## 4.6 The result

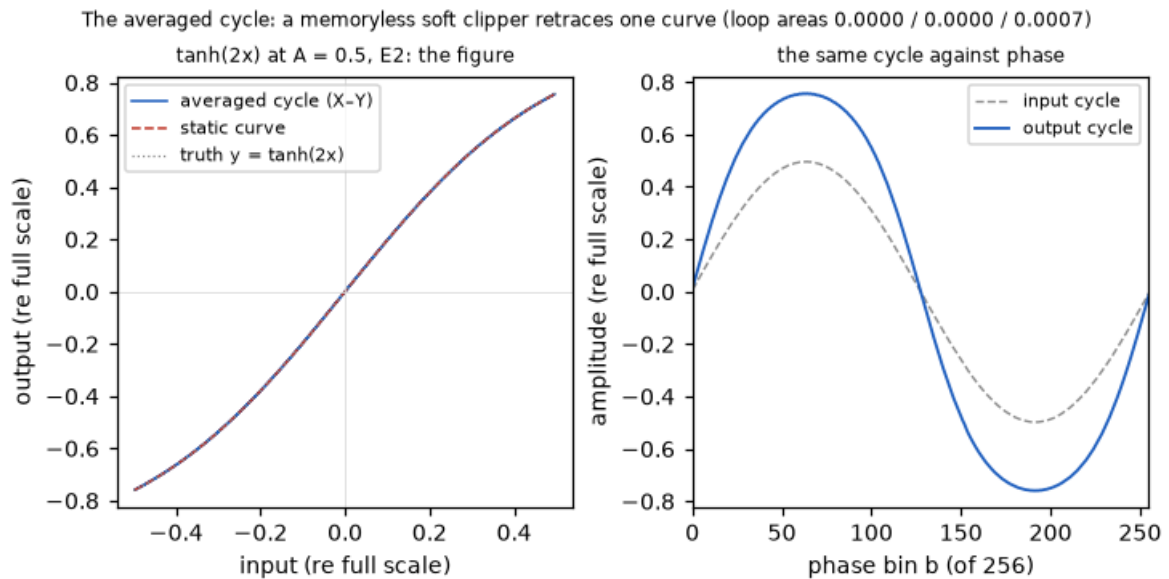


Figure 7: The worked example,  $y = \tanh(2x)$  at  $A = 0.5$  and E2, no filter: the averaged cycle as the X-Y figure with the static curve and the exact curve overlaid (left; the three coincide), and the same cycle against phase (right). Every loop area is zero to the bin floor.

The app’s own rendering of the same measurement kind, from the user guide’s figure pipeline (a simulated device, rendered headlessly from the chart view with the removal on):

## 4.7 Worked example and parity

The [parity test](#) runs 21 synthetic cases through both implementations, each a known device — one of the reference nonlinearities between an optional pre-filter and an optional post-filter, each

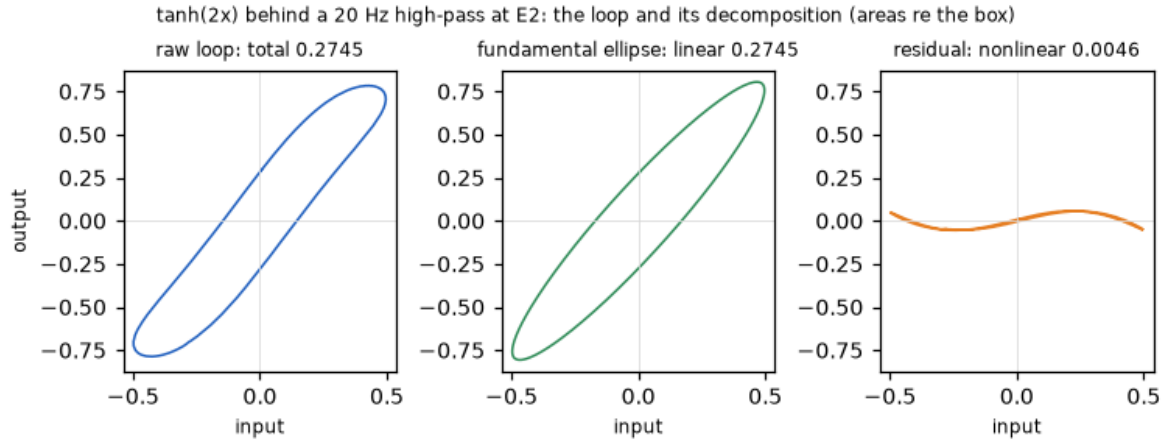


Figure 8: The same clipper behind a 20 Hz second-order high-pass at E2: the raw loop, the fitted fundamental ellipse and the residual after the ellipse's output is subtracted, each with its normalized area. The filter draws the ellipse; the residual holds only the harmonics' own rotation.

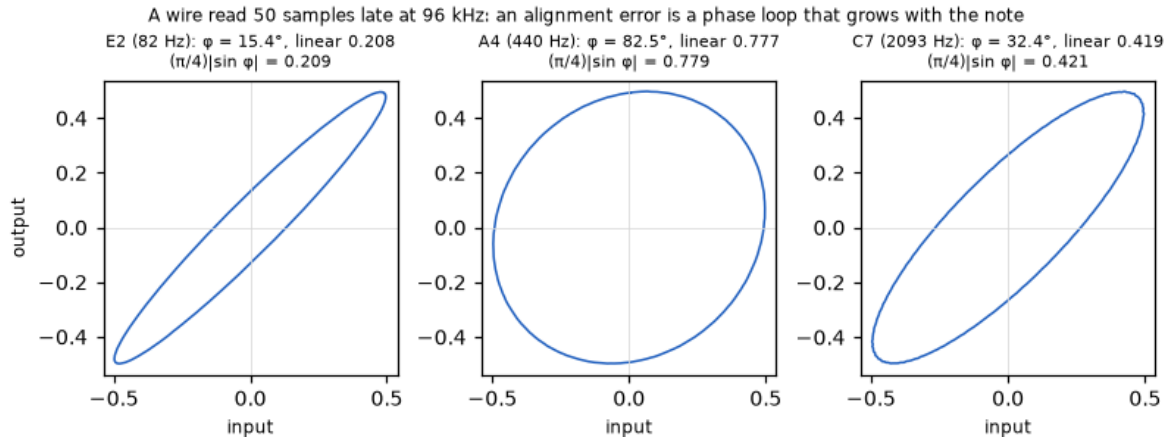


Figure 9: A wire, analyzed 50 samples late at 96 kHz, at three notes: an alignment error is a phase loop that grows with the note. Each title gives the phase the error implies at that note, the measured linear area and its closed form  $(\pi/4)|\sin \varphi|$ .

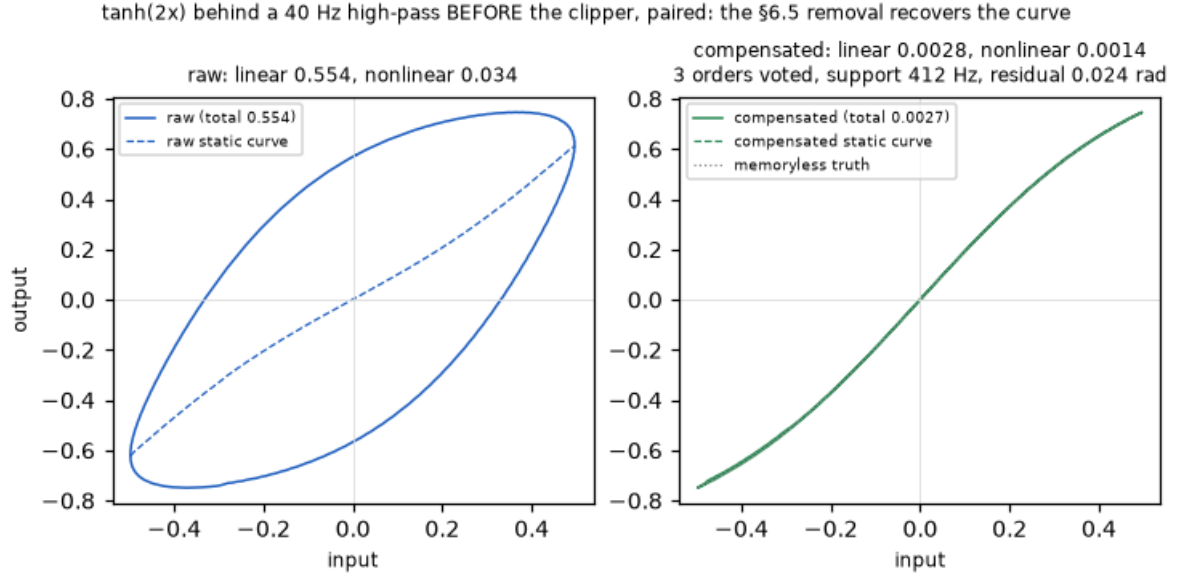


Figure 10: The soft clipper behind a 40 Hz high-pass placed *before* the clipper — the pre-dispersive case an  $H_1$ -only rotation cannot handle — paired with its own swept analysis. Left: the raw figure and its static curve. Right: the compensated figure over the memoryless truth, with the orders that voted, the support and the residual.

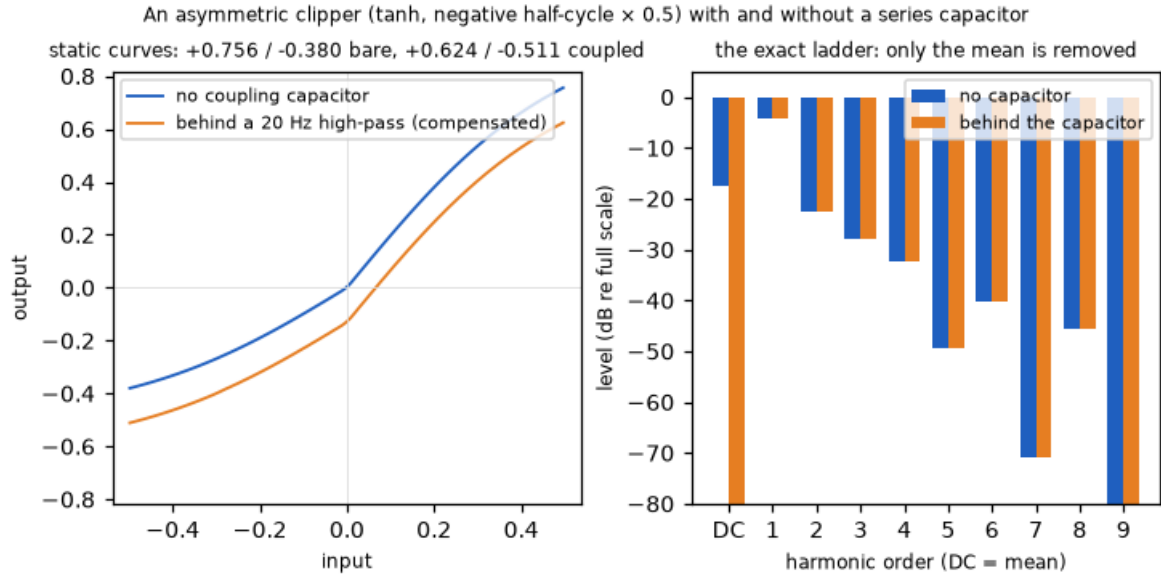


Figure 11: An asymmetric clipper (the negative half-cycle scaled by one half) with and without a series capacitor: the static curve's unequal clip levels are centred by the capacitor while the exact harmonic ladder loses only its mean — the even harmonics survive.

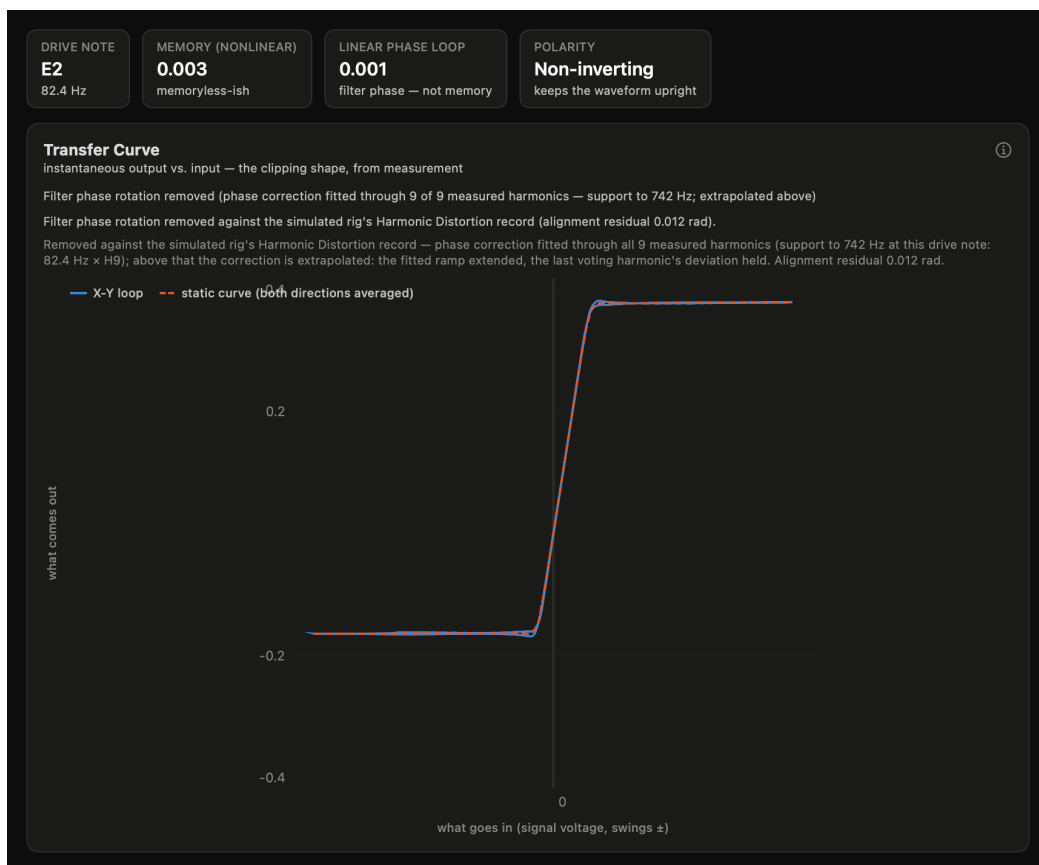


Figure 12: The Transfer Curve view as the app draws it, from the guide's simulated fixture, with the phase removal applied and its caption stating the support and the residual.

a second-order (Butterworth) low-pass or high-pass — driven by the plan’s own tone and, where paired, measured by the swept method of §3 for its reference. Beside every measured quantity the oracle prints a truth computed by quadrature: for a memoryless device with no filter the curve  $y(x)$  itself; otherwise the Fourier series of the clipper’s output at the pre-filter’s delivered drive, each order scaled and rotated by the post-filter’s analytic response and by any planted alignment error, and — for the compensated figure — the same series with every phase returned to the lattice and every magnitude kept. The test asserts:

- Swift against Python: every cycle value, static-curve value, compensated value, area,  $\psi_1$ , residual and fit term within 1e-10 (measured maximum 7.4e-13 on this run), the frame shift within 1e-09 samples (measured 0), and every discrete reading — class, flip, polarity, tile, verdict, presence — identical;
- the fundamental ellipse against its closed form  $(\pi/4)|\sin \varphi_1|$  on a linear device: a ratio within 0.01 of one at the plan’s tone (measured deficit at worst 0.73 %) and within 0.001 at ten times its length (measured 0.05 %);
- the raw static curve against the truth, as a fraction of the output peak: 0.01 for smooth memoryless devices (measured 0.003 on the soft clipper, 4.4e-05 at the standard drive), 0.05 for the hard clip (measured 0.027: a 256-bin cycle rounds a corner by about an eighth of a bin’s input span), 0.05 for an open ellipse (the branch-mean curve of an open loop is least trustworthy where the branches meet); and a memoryless device’s nonlinear area under 0.005 (measured 0.002);
- every paired filtered device compensates, finds the planted frame shift within 0.5 samples (measured 0.39), keeps its residual under 0.05 rad (measured 0.0237), recovers the memoryless truth to 0.01 of peak on the static curve (measured 0.0055) and 0.02 on the cycle, and reads both compensated areas under 0.005 (measured 0.0028).

The removal’s parity is established: every paired filtered case compensates on both sides and the two implementations agree on the compensated figure to the Swift-vs-Python tolerance.

The memoryless cases, at the loudest hardware drive and (one case) the standard drive, where the soft clipper is within its linear region:

| case        | $A$  | total   | linear   | nonlinear | static vs truth (of peak) |
|-------------|------|---------|----------|-----------|---------------------------|
| a-tanh      | 0.5  | 0       | 5.95e-06 | 0.000719  | 0.00298                   |
| a-tanh-plan | 0.05 | 0       | 4.52e-08 | 0.000612  | 4.41e-05                  |
| a-hardclip  | 0.5  | 0.00132 | 9.68e-05 | 0.00196   | 0.0266                    |
| a-asym      | 0.5  | 0       | 0.000119 | 0.000458  | 0.00337                   |
| a-param     | 0.5  | 0       | 1.41e-05 | 0.000939  | 0.00747                   |

The linear devices — the identity behind the low-pass at three notes and under a 50-sample alignment error at three notes — at the plan’s tone and at ten times its length:

| case        | $f_0$ (Hz) | $\varphi_1$ (°) | measured | $(\pi/4) \sin \varphi_1 $ | ratio   |
|-------------|------------|-----------------|----------|---------------------------|---------|
| b-lp-E2     | 82.4       | -4.3            | 0.0586   | 0.058997                  | 0.99327 |
| b-lp-A6     | 1 760      | -100.2          | 0.77136  | 0.7729                    | 0.998   |
| b-lp-C7     | 2093       | -113.4          | 0.71634  | 0.72094                   | 0.99362 |
| c-delay-E2  | 82.4       | 15.5            | 0.20771  | 0.20923                   | 0.99275 |
| c-delay-A4  | 440        | 82.5            | 0.77706  | 0.77868                   | 0.99792 |
| c-delay-C7  | 2093       | 32.4            | 0.41926  | 0.42128                   | 0.9952  |
| b-lp-E2-15s | 82.4       | -4.3            | 0.058976 | 0.058997                  | 0.99965 |
| b-lp-A6-15s | 1 760      | -100.2          | 0.77283  | 0.7729                    | 0.9999  |
| b-lp-C7-15s | 2093       | -113.4          | 0.72058  | 0.72094                   | 0.9995  |

The paired cases: two post-filters and the pre-filter on the soft clipper, the asymmetric clipper behind a DC-blocking high-pass, an inverted device, the half-cycle lock and the unfiltered clipper (whose raw loop is under the dead zone, so the removal is a no-op and the fit still confirms polarity):

| case          | class           | voted | support<br>(Hz) | residual<br>(rad) | $\Delta l$<br>(samples) | comp. lin. | comp. nonlin. | comp. vs truth<br>(of peak) |
|---------------|-----------------|-------|-----------------|-------------------|-------------------------|------------|---------------|-----------------------------|
| d-tanh-lp     | compensated     | 3     | 412             | 0.00322           | -0.172                  | 0.00049    | 0.00054       | 0.0044                      |
| d-tanh-hp     | compensated     | 3     | 412             | 0.00945           | -0.388                  | 0.00024    | 0.00066       | 0.0022                      |
| e-tanh-prehp  | compensated     | 3     | 412             | 0.0237            | -0.272                  | 0.0028     | 0.0014        | 0.004                       |
| f-asym-hp     | compensated     | 7     | 659.2           | 0.0108            | -0.102                  | 0.0015     | 0.0008        | 0.0055                      |
| g-tanh-inv-lp | compensated     | 3     | 412             | 0.00322           | -0.172                  | 0.00049    | 0.00054       | 0.0044                      |
| g-tanh-shift  | negligible loop | 3     | 412             | 0.00833           | 0.401                   | —          | —             | —                           |
| g-tanh-paired | negligible loop | 3     | 412             | 0.00816           | -0.0763                 | —          | —             | —                           |

Two readings from those tables are worth stating in words. The pre-dispersive case’s raw nonlinear area, 0.0343, sits in the hedged band of the verdict with no memory in the device at all, and the removal takes it to 0.0014 — the mid-band hedge in the pipeline is exactly this case. And the half-cycle lock is undone: the captured  $\psi_1$  read 3.144 rad, 0.0026 from the lattice, the paired polarity read non-inverting, and the resolved figure ascends.

#### 4.8 Limits and failure modes

- **The normalization is the box’s, and the constant is  $\pi/4$  only for a linear device.** The ellipse area is  $\pi AR|\sin \varphi_1|/(4A y_{\max})$ ; on a clipped output the peak is not the fundamental’s and the figure is  $(\pi/4)(R/y_{\max})|\sin \varphi_1|$ , which can exceed  $(\pi/4)|\sin \varphi_1|$  (a square wave’s fundamental is  $4/\pi$  of its peak). A reader converting a linear-device figure to an angle uses  $\varphi_1 = \arcsin(4\text{linear}/\pi)$ . On the synthetic single-pole check the shipped code reads the closed form to 0.73 % at the plan’s tone and 0.05 % at ten times its length, so a deficit of several percent on a hardware capture is in the measurement chain, not in the metric’s constant.
- **The fade-out is binned.** The skipped opening cycles protect the start; every sample to the end of the tone is binned, the last  $t_{\text{fade}}$  under its ramp, a phase-dependent weighting whose share of the average falls with the tone’s length. It is the whole of the deficit above (the E2 ratio moves from 0.9933 to 0.99965 between the two lengths), and it lowers the captured input peak by a few tenths of a percent at the plan’s length.
- **A residual alignment error is a loop.** An [residual alignment error](#) of  $e$  samples rotates the fundamental by  $\varphi = 2\pi f_0 e / f_s$  and draws a linear area of  $(\pi/4)|\sin \varphi|$  on a wire — growing with the note (Figure 9). The integer latency the app removes leaves at least the fractional part, and a rig can leave more; on a device, that loop is indistinguishable from the device’s own linear-phase loop. Two notes measured on one rig let a reader test the hypothesis: a single  $e$  must reproduce both deficits. The nonlinear area is untouched by it on a linear device.
- **The coupling capacitor removes level asymmetry, not harmonic asymmetry.** A clipper clamped at  $+a$  and  $-b$  at roughly equal duty carries a mean near  $(a - b)/2$ ; a series capacitor at the output strips it, leaving levels of  $\pm(a + b)/2$  — equal magnitudes from an unequal circuit (Figure 11). The shape asymmetry remains, and the even harmonics in the paired swept analysis are where the asymmetry is read. Because every pedal has one, a transfer curve essentially never shows the raw asymmetry of an asymmetric clipper.
- **The support ends at the highest voting order.** A harmonic too quiet to clear the magnitude floor on either side does not vote (the soft clipper at this drive seats 3 orders,



the asymmetric clipper 7); above the support the correction is the extended ramp, the  $H_1$  shape and a held deviation, and a hard-clipped device has content far above it.

- **The branch phase is read at the nearest bin.** The paired record's phase at  $kf_0$  is the nearest bin's, so a read can be off by up to half a bin width times the filter's phase slope, which the frame fit absorbs as a fictitious shift of a fraction of a sample (measured up to 0.39 samples on the parity cases) and leaves in the higher orders' residuals.
- **The bins round a corner.** With  $n$  bins per cycle a hard clip's corner is averaged over a bin's input span, and the static curve departs from the exact curve there by a few percent of the clip level; the smooth devices read under one percent of peak.
- **The static curve of an open loop.** The branch-mean estimate is least trustworthy where the two branches meet, at the input's extremes; on a wide ellipse it departs from the true line by a few percent of peak there. The app draws the static curve dashed on every figure; on an open loop its ends are where the estimate is least trustworthy.
- **A declined removal names its class, not its cause.** The residual separates a genuine pairing from a mismatch; it does not separate a session mismatch from a structural one, and the two need different remedies (§4.4).

## 5 Chord IMD

### 5.1 Purpose, and what the measurement cannot tell you

A [two-tone measurement](#) plays two sines at a musical interval through the device at once. A nonlinearity does not treat them separately: it multiplies them, and the output carries every [intermodulation product](#)  $mf_1 + nf_2$  for integer  $m, n$ , whose [product order](#)  $|m| + |n|$  is the lowest polynomial power that can produce it. The result is the level of every product up to the enumeration ceiling  $K$  (the parameter table gives it, and the count of recipes it admits), each read at one bin of one long spectrum and quoted in [dBc](#) against the louder played tone; the two tone amplitudes as the device delivered them; three capture-time percentage sums (two borrowed from laboratory two-tone standards and one total, kept for the record as the frozen tile definitions of earlier builds); and the raw spectrum beside each played tone, stored for the [coherence check](#). At read time every product gets one [product reading](#), the capture states its own [capture floor](#), the products are sorted by [harmonic role](#) on the grid of the requested interval, and the drive level is stated from the stored stimulus.

What it cannot tell you, stated once: (a) anything about other drive levels — one capture is one operating point, and the products of a clipper scale with drive faster than anything else about it; (b) whether a resolved product is the device’s or the rig’s — an interface’s converters intermodulate too, and only a control capture with nothing in the loop separates the two, which the floor cannot do and does not claim to; (c) which stage moved the notes when the coherence check fails — a verdict without an attribution, by design (§5.4); (d) anything below the capture’s own floor, and, for the three percentage sums, anything at all about a floor: they are capture-time sums over the products the gate passed and carry no bound (§5.8); (e) anything about products above order  $K$  — the enumeration is a small-signal census, and a hard clipper at playing levels has energy at orders it does not walk; and (f) anything about what the distortion sounds like.

### 5.2 The stimulus

In this chapter  $f_1 < f_2$  are the two played tones (not the sweep band edges of §3),  $N$  is the [analysis window](#) length in samples,  $\Delta f = f_s/N$  the bin width,  $k_1, k_2$  the bins the two tones sit on,  $K$  the enumeration ceiling, and  $(m, n)$  a product’s recipe. The stimulus is two equal sines at a linear combined peak  $A$ :

$$x[n] = \frac{A}{2} \sin(2\pi f_1 n / f_s) + \frac{A}{2} \sin(2\pi f_2 n / f_s), \quad n = 0 \dots \lfloor T f_s + \frac{1}{2} \rfloor - 1, \quad (22)$$

scaled over its first and last  $\lfloor t_{\text{fade}} f_s \rfloor$  samples by the raised cosine  $\frac{1}{2}(1 - \cos(\pi i / N_{\text{fade}}))$ . The standard peak is the fingerprint sweep’s drive, so the products and the other measurement kinds describe one operating point; the worked example below uses the loudest drive the hardware sheet offers, where the reference clippers clip. The notes are named in twelve-tone equal temperament with A4 = 440 Hz (the parameter table gives the default pair), and the measure sheet offers one probe: the interval. Two laboratory probes — an SMPTE-style pair at a fixed ratio and a CCIF twin-tone — existed in the code and were retired before 1.0, because a standard is never moved onto the lattice described next and neither could be given one; the two percentage keys that carry their names are sums over recipe sets and are unrelated to the retired probes.

**Coherent sampling, the premise — stated once.** Both tones are moved onto exact bin centres of the  $N$ -point window before anything plays, so each completes a whole number of cycles inside the segment the analyzer reads, and the segment is windowed *rectangularly*: a tone that fits the window exactly leaks into no other bin, so there is nothing to taper and no taper is used, and the floor beneath a record is noise, not a window’s skirt. [coherent sampling](#) is inherited by every product: with the tones on bins  $k_1, k_2$  the product  $(m, n)$  lands on bin  $mk_1 + nk_2$  exactly, because

bin numbers add as frequencies do. That set of bins is the [analysis lattice](#). Every step below is a consequence of this one premise, and the price of it is the coherence check: a rectangular window’s sidelobes fall as  $1/(\pi d)$  at  $d$  bins — excellent, with no margin — so a tone that drifts during the capture spills its own skirt across the lattice.

**Two numbers describe the lattice.** Write  $k_1 = ga$ ,  $k_2 = gb$  with  $\gcd(a, b) = 1$ : the [reduced tone ratio](#)  $a : b$  and the [lattice spacing](#)  $g = \gcd(k_1, k_2)$ . They answer two different questions. Two recipes share a bin when  $(m - m')a = -(n - n')b$ , which with  $a, b$  coprime forces  $m - m' = -jb$ ,  $n - n' = ja$  for some integer  $j \neq 0$ , so  $|m - m'| + |n - n'| = |j|(a + b) \leq 2K$ ; hence

$$a + b \geq 2K + 1 \quad (23)$$

admits no collision up to order  $K$  (the bound is tight: at  $K = 7$  the ratio  $1 : 13$  collides and every ratio at 15 is clean). And the integer combinations of  $k_1$  and  $k_2$  are exactly the multiples of  $g$  (Bézout), so two distinct lattice points differ by at least  $g$  bins, and some pair by exactly  $g$ : no product can sit nearer than  $g$  bins to another product or to either tone, at any order. A lattice can pass (23) and still be crowded — the probe every record before the current length was captured on shared no bins at a spacing of two — which is why the snap buys spacing and not only collision-freedom.

**The lattice snap.** The [lattice snap](#) is decided in two steps, so that the interval a record asks is one interval at every rate and length. First the ratio is chosen *once*, against every standard grid at the same time (the four supported rates, each at the length it is meant to analyze at): over  $a = 1 \dots \lfloor f_1 \rho / (\beta \Delta f_{\max}) \rfloor$  with  $\rho = 2^{c/1200}$  the cents budget  $c$  as a ratio and  $\beta$  the resolvable-spacing bar, and for each  $a$  over  $b = \max(a + 1, \lceil ar/\rho^2 \rceil) \dots \lfloor ar\rho^2 \rfloor$  with  $r = f_2/f_1$ , a coprime pair satisfying (23) is a candidate; it survives when on *every* standard grid some multiple  $g \geq \beta$  places both  $ga \Delta f$  and  $gb \Delta f$  inside  $\pm c$  of the requested tones (the placement on one grid is the in-budget  $g$  with the smallest worst-case pitch error, ties to the smaller  $g$ ). The survivors are ranked by the interval nearest the requested one, then the wider minimum spacing, then the smaller  $a + b$ , then  $(a, b)$ . The default fifth lands on the ratio the parameter table quotes, a few cents flat of equal temperament and the same at every rate; the snap table in §5.5 is that search run by the shipped code on every grid. Second, on the grid at hand, the standard ratio is scaled: the in-budget multiple with the smallest pitch error at *any* spacing — keeping the interval outranks clearing the bar, so a grid too coarse to resolve it (Simulation Mode’s interactive preview) still asks the same musical question, in a degraded regime the read-time predicates carry. Only when no multiple of the ratio fits the budget does the older search run: every bin pair  $(k_1, k_2)$  inside the budget (the half-widths searched are derived from the budget and the bin width, never fixed), ranked by spacing, then worst-case cents, then index; the best pair clearing both (23) and  $g \geq \beta$  is emitted as *resolvable*, else the best clearing (23) alone as *degraded* — which at the old length is the corpus probe, field for field — else the plan *refuses* before anything plays, quoting the widest lattice the budget reached with its reduced ratio, spacing and sum against the bound. The plain snap is never returned as a fallback: an exactly degenerate probe drops colliding products through the enumeration’s tie-break and reports a suspiciously clean record.

**The window is a duration, and the stimulus is sized to hold it.** The standard analysis length is stated at the standard rate as a sample count, but what it fixes is a *duration*: each rate takes the power of two nearest  $N_{\text{std}} f_s / f_{s,\text{std}}$ , capped at the plan’s ceiling, so the four rates form two grid classes with the same bin width in pairs. The stimulus duration is then derived from the length it must hold,

$$T = t_{\text{fade}} + t_{\text{settle}} + N/f_s + t_{\text{fade}} + t_{\text{settle}}, \quad (24)$$

never set by hand: a hand-set duration is how an earlier fixed window came to overrun the fade-out at two of the four rates (Figure 14). The analysis length a stored stimulus is read at is derived by the same rule in reverse — from the ceiling, halve while the window would overrun the steady state — so the stimulus is snapped to exactly the length it is analyzed at, and a shorter snap grid is a subset of a longer window’s bins, never the reverse.

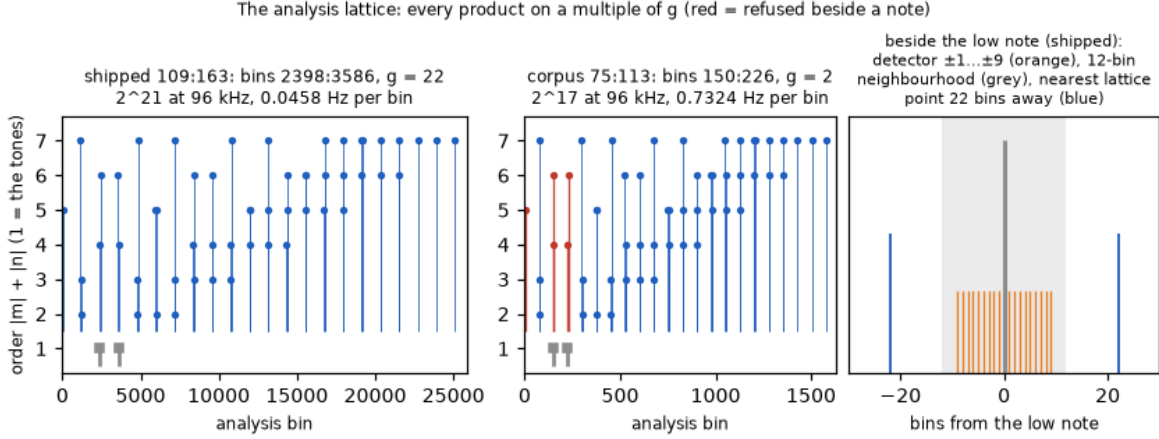


Figure 13: The analysis lattice: every enumerated recipe as a stem at its bin and order on the shipped fifth (left) and on the corpus probe every earlier record was captured on (middle), the two tones as squares at order one. On the corpus probe the four even-order recipes two bins from the played notes and the one two bins from DC draw red: refused on position (§5.4). Right: the bins beside the low note on the shipped lattice — the detector’s offsets, the played-tone neighbourhood, and the nearest lattice point a spacing away.

### 5.3 The pipeline

Given the captured output  $y[n]$  and the integer loop latency  $\ell$  (the app measures and removes it per capture; the oracle renders at zero latency and, when asked, analyzes with a deliberate offset):

1. **The window.** The segment starts at  $s = \ell + \lfloor t_{\text{fade}} f_s \rfloor + \lfloor t_{\text{settle}} f_s \rfloor$  and runs for exactly  $N$  samples, rectangularly windowed and zero-padded if the capture ends first; the steady state ends at  $e = \ell + \lfloor T f_s + \frac{1}{2} \rfloor - \lfloor t_{\text{fade}} f_s \rfloor$ , and the *overrun*  $s + N - e$  is positive when the segment would swallow the fade-out. Latency shifts  $s$  and  $e$  alike, so a window that fits at zero latency fits at any, and because the tones are bin-centred a shift is a phase: no magnitude moves (§5.7 measures it). The analyzer never shortens a window it is handed — the tones are centred on the length the stimulus was snapped to, and a shorter rectangular window would leak far worse than the fade it avoided.

2. **The spectrum.** With  $c$  the number of captured samples inside the segment,

$$M[b] = \frac{2}{\min(N, c)} \left| \text{DFT}_N \{ y[s+i] \} [b] \right|, \quad b = 0 \dots N/2, \quad (25)$$

so a bin-centred sine of amplitude  $a$  reads  $a$ .

3. **The enumeration and the lattice.** With  $\text{bin}(f) = \text{round}(f/\Delta f)$ , the set  $S = \{\text{bin}(f_1), \text{bin}(f_2), 0\}$  is seeded, and the walk runs  $\text{order} = 2 \dots K$ ,  $m = -\text{order} \dots \text{order}$ ,  $n = +(\text{order} - |m|)$  then  $-(\text{order} - |m|)$ : a recipe with  $\Delta f < m f_1 + n f_2 < 0.95 f_s/2$  (both edges literals) whose bin is not yet in  $S$  is admitted and its bin joins  $S$ . The lowest order

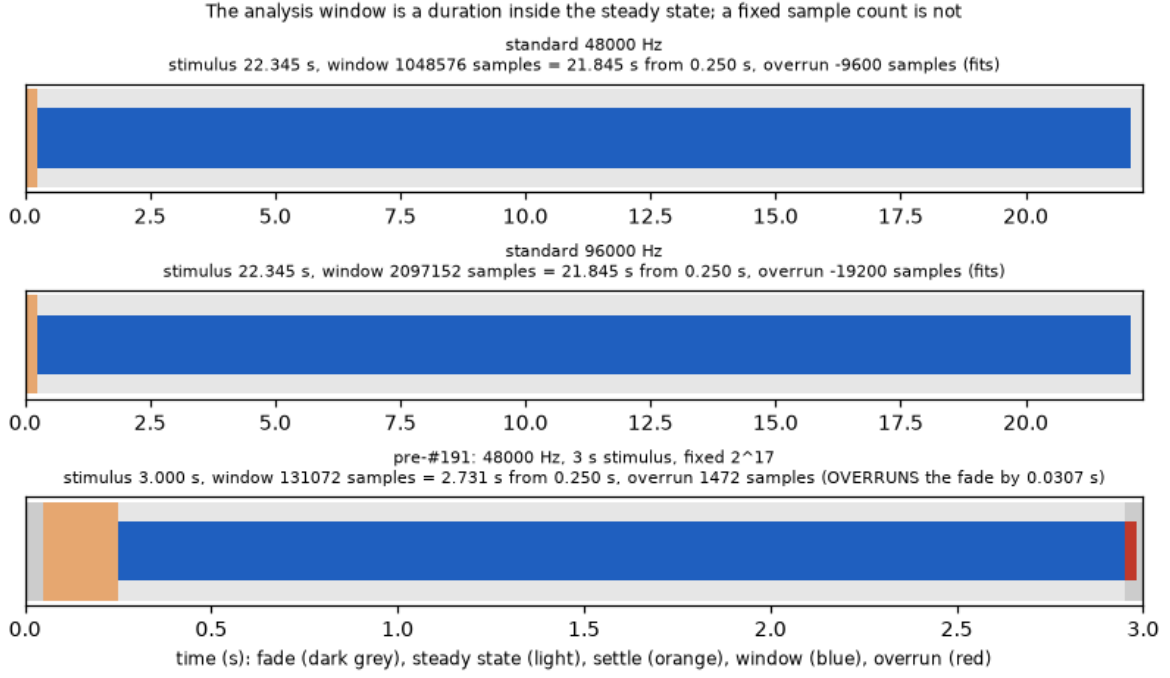


Figure 14: The analysis window inside the stimulus at the two 96 kHz-class rates, and the historical fixed-length window on a hand-set stimulus at 48 kHz, which overran the fade-out: periodicity broke and the window’s own skirt flooded the low bins. The fade-contaminated separation bound in the parameter table exists for captures made that way.

therefore keeps a shared bin — on a degenerate lattice a surviving label names the enumeration’s tie-break, not a fact — and a clean lattice admits  $\sum_{k=2}^K 2k$  recipes. The lattice is collected *before* any floor is read, so a floor’s neighbourhood can leave out every bin the enumeration claims.

4. **The reads.** Each product’s amplitude is  $M[\text{bin}(mf_1 + nf_2)]$ , the one bin, exactly; the tone amplitudes are  $M[k_1]$  and  $M[k_2]$ . An older read took the largest of the bin and its two neighbours “to tolerate drift” there was none of: it left a strong product untouched and lifted every read at the noise by a measured mean of about four decibels, and the separation bound in the parameter table is deliberately left where it was measured under that read, because a stored record may carry it.
5. **The local floor and the gate.** Each product is judged against the median magnitude of its own neighbourhood: the bins from the inner offset to the outer offset on both sides, inside the spectrum, with every bin the lattice claims removed by name — every product bin, both tone bins, DC, and the coherence detector’s bins beside each tone. The inner offset is a bin quantity (the product’s own skirt under lost coherence); the outer offset is a *frequency* span converted to bins at the record’s bin width and never fewer than a minimum count (a noise density is local in hertz, and a median wants samples). The median is the upper middle of the sorted values, and zero when the neighbourhood is empty. The [presence gate](#) is

$$\text{present at capture} \Leftrightarrow a_{m,n} > 4\text{floor}_{m,n}, \quad (26)$$

a literal factor of four (the table states it in decibels), decided once and *frozen* in the payload as the product’s flag. That is the [local floor](#); the capture floor of §5.4 is a read-time summary that can never change it.

6. **The percentage sums.** Over the products inside the audible band  $B$  (the band is stored

with the record),

$$\begin{aligned}
\text{pct}(P, R) &= 100\sqrt{P/R} \quad (R > 0), \\
\text{SMPTE} &= \text{pct}\left(\sum_{|n|=1, 0 < |m| \leq 4} a_{m,n}^2, a_{f_2}^2\right), \\
\text{CCIF} &= \text{pct}\left(\sum_{(m,n) \in \{(-1,1), (2,-1), (-1,2)\}} a_{m,n}^2, a_{f_1}^2 + a_{f_2}^2\right),
\end{aligned} \tag{27}$$

and the total is the same over every in-band product the gate passed. These are the capture-time scalars; the read-time derivation of §5.4 re-runs them with the lattice’s refusals left out.

7. **The coherence detector.** For each tone bin the raw magnitude at every detector offset on both sides, inside the spectrum, and the local floor beside the tone (the same neighbourhood and exclusion about the tone bin) are stored with the window length: the [coherence detector](#). Nothing else keeps those bins, and the read-time check needs no derived length to place them.
8. **Storage.** The products are stored sorted by descending amplitude with their recipe, frequency, amplitude, order and gate flag; beside them the stimulus as delivered, the two tone amplitudes, the three sums, the band and the detector. Nothing read-time is persisted.

## 5.4 Read-time predicates on the result

Everything below is computed from the frozen payload when a record is read, so a corrected predicate corrects every stored record. A consumer that knows the analysis length passes it; one that does not passes nothing, and nothing is placed on the lattice for it — an absent argument is a smaller claim, stated rather than defaulted.

**Levels and the capture floor.** A product’s level is  $L = 20 \log_{10} (a / \max(a_{f_1}, a_{f_2}))$  dBc, undefined when either amplitude is zero. The *noise reads* are the levels of every product the gate did *not* pass and that the lattice does not refuse — not inside a reference bin’s neighbourhood, not under the band’s low edge, not beside a louder product (each defined below) — sorted ascending; bins under the loudest product’s skirt deliberately stay in, because on a record like a rectifier’s every empty bin is under it and a vanished floor would read as “every product resolved”. With at least the minimum count of reads the [capture floor](#) is the upper median, with the count and the spread of the loudest read above the median; fewer reads is honestly no floor. The two floors run one way: the gate is capture-time against the raw spectrum, the capture floor is read-time over bins already flagged, so excluding a refused bin changes which flagged bins are averaged and never which are flagged.

**Positions on the lattice.** *Proximity:* a product’s bin against the three reference bins —  $\text{bin}(f_1)$ ,  $\text{bin}(f_2)$  and 0 — the nearest wins (ties in that order), and the product is inside the neighbourhood when the absolute offset is at most the played-tone neighbourhood width (a bin quantity: sidelobes fall as  $1/(\pi d)$  in bins). *Band:* the stored frequency against the stored band’s *low* edge only; a product above the band’s top is inaudible but perfectly attributable and reads like any other. *Louder neighbour:* among the products that passed the gate and are strictly louder, those within the same width of this product’s bin (offset nonzero) — the nearest, ties to the louder; a neighbour that is itself noise casts no skirt and refuses nothing. *Spacing:*  $g = \text{gcd}(\text{bin}(f_1), \text{bin}(f_2))$  on the analysis length. *Loudest skirt:* a record-level condition, present only when the loudest *product* (the maximum of  $L$  over every stored product, never a quantity that

folds the tone in) stands above 0 dBc and the coherence reading is absent or does not interpret for a device; it carries  $g$  and the skirt depth  $20\log_{10}(\pi g)$ , computed never typed. The *analyzer window* bound is the fade-contaminated separation of the parameter table, asserted only when the window overruns the fade (§5.3, step 1) and otherwise nil — a record captured at the old fixed length at one of the two rates where it overran gets that bound; every capture the app makes now gets none.

**The reading.** One *product reading* per product, decided in this order and stopping at the first that applies:

1. inside a reference bin’s neighbourhood (a played tone, or DC) — *cannot resolve, beside a played tone*, naming the tone and the signed offset: a bin inside a note’s own skirt holds the note, and a bin beside DC holds the capture’s own offset and drift; decided on position before any level is consulted, because the gate samples the floor from offsets that lie mostly outside the skirt the bin is sitting in, and would pass such a bin honestly at tens of decibels of margin;
2. under the band’s low edge — *out of band*: a frequency fact, not a limit, so its sentence never merges with a lattice’s or the instrument’s;
3. no level (a zero amplitude or reference) — *unmeasurable*;
4. a louder neighbour inside the neighbourhood — *cannot resolve, beside a louder product*, naming it: at that separation the quieter bin holds the louder product’s skirt; the louder of the pair is readable, the quieter is not;
5. depth below the loudest component at or beyond the analyzer’s bound, when one is asserted — *cannot resolve, analyzer window*;
6. deeper below the loudest product than its skirt reaches, on a record where the loudest-skirt condition holds — *cannot resolve, under the loudest skirt*: unread, never present and never absent;
7. the gate passed — *present*, with the margin over the capture floor when one exists;
8. no capture floor — *cannot resolve, floor not observed*, carrying the number of noise reads;
9. otherwise — *absent*, with the bin’s own level as the bound: nothing here above it, a bound and never a value.

A refusal is never silently absent, and “absent” is never a value. The verdict counts present, absent, unresolved (any *cannot resolve*) and out-of-band readings with the noise-read count; a nil floor is stated by its reason — *every product resolved* only when nothing was refused, else *too few empty bins* with the counts — in the one floor sentence every surface prints.

**Shared bins, and the lattice a record was delivered on.** Every capture path snaps both tones, so distinct lattice points differ by at least one bin width while collisions agree to the ulp; a stored product’s bin is *contested* when another enumerated recipe (the default ceiling raised to the record’s own highest stored order, since the ceiling is not stored) lands at the same frequency to a relative tolerance, and the contestants are listed ascending in order. The delivered lattice — the reduced ratio of the two stored tones and the grid  $f_1/a$  — is recovered from the stored signal by the smallest denominator that reproduces the pair to the same tolerance, which no snapped record can fail.



**The derived percentages and the partner check.** The three sums of (27) are re-run over the stored products with the recipes inside a reference bin’s neighbourhood left out, the band as stored (every product on a legacy record with no band), and the excluded list names only what the band would have counted — the recipe beside DC sits under the band and was never in any sum. The partner check is the second-order cross-check: a quadratic term produces  $f_1 + f_2$  with  $f_2 - f_1$  at equal level at any tone balance, and  $2f_1$  with  $2f_2$  at twice the measured *output* tone gap; only *excess* over the partner beyond the tolerance flags, a contested operand refuses, and a flag says the bin holds something beyond a clean second-order product — the energy is real, the label and rank are not to be trusted.

**The coherence check.** The detector’s bins are read as levels in dBc, each pair (the sidebands about one tone) against the local floor beside that tone in the same unit. On a record with no detector — every capture before the detector existed — the same reading is built from the stored products inside a played tone’s neighbourhood (never DC’s) against the capture floor at the derived length: the legacy path, which the detector outranks whenever present. The lattice facts:  $g$  on the detector’s own length, the widest stored offset, and the *clearance*  $g - d_{\max}$ ; the lattice is *clear* when the clearance is at least the analyzer’s inner offset. The detector offsets are  $\pm 1 \dots \pm d_{\max}$  with  $d_{\max}$  *derived* as the resolvable-spacing bar minus one minus the inner offset — the widest offset that stays the inner-edge clearance from the nearest lattice point on the narrowest lattice the snap calls resolvable — and every offset is read, odd and even: a wander at an even number of cycles per window puts its sidebands on even offsets only, and an odd-only detector read the floor on the very fixture whose stored two-bin products read tens of decibels above it.

Clearance is necessary and not sufficient, because distance is a proxy for level. Both a tone and a product spill a  $1/(\pi d)$  skirt under lost coherence, so a bin at offset  $d$  beside a tone reads the tone only while the loudest product, assumed at the nearest lattice point  $g - d$  bins the other way, sits under the tone’s skirt there by the stated margin  $\mu$ :

$$L_p \leq 20 \log_{10} \frac{g - d}{d} - \mu = \text{ceiling}(d, g), \quad (28)$$

$-\infty$  when a lattice point sits at or inside the offset. The *readable* offsets are those whose ceiling the loudest product clears (every offset on the legacy path or an uncleared lattice); the ceiling falls with  $d$ , so the set is always  $\pm 1 \dots \pm k$  and the nearest bins survive last. The figure is the loudest sideband over the readable offsets of each pair, falling back to every sideband only when the level rule withheld every offset (and then the reading does not interpret); the excess is that figure over its pair’s floor, the worst over the two pairs, and the bar is a floor-relative excess (the table’s value) — an absolute bar was right at one length only, since spilled carrier energy does not scale with bin width while noise drops with it. Beside the excess the line prints what noise alone would put the worst of the  $n$  read bins over the floor: a noise bin’s power is exponential, the floor a median, and the expected worst of  $n$  over the median is

$$10 \log_{10}(H_n / \ln 2), \quad H_n = \sum_{i=1}^n 1/i, \quad (29)$$

the exact form, chosen over the  $\sqrt{2 \ln n}$  asymptote which under-reads by nearly two decibels at the legacy path’s four bins — a reference, never a bar. The [sideband pair statistic](#) is the mean of the two innermost sidebands of each pair and the difference between the two tones’ pairs, printed beside  $20 \log_{10}(f_2/f_1)$ : a shared fractional frequency wander separates the pairs by that figure, while amplitude modulation or a frequency-independent phase modulation leaves them equal — a quantity the sentence states, and no verdict is drawn from it (equality is never read as amplitude modulation, a biconditional the shape has not earned).

The reading *interprets* when a floor exists and either the lattice is clear with at least one readable offset — a statement about the played tones whatever was in the loop — or the lattice is not



clear and the subject is a loop-only capture, which held nothing that could put a product beside a note. The subject (loop only, or a device in the loop) is stated by the caller, which knows the record's identity; it is context on a cleared lattice and the deciding fact only on an uncleared one. The verdict is *not trustworthy* when the reading interprets and exceeds the bar: the played notes did not stay stationary across the window, from some source in the chain the reading cannot name — a pedal with a dying battery moves a note over twenty seconds as surely as a rig — so the whole spectrum is smeared and this capture's floor and product levels are not trustworthy. Nothing is refused, discarded or re-run; the capture is saved as measured and the sentence rides it. Where the level rule withheld every offset the sentence names the signal that would have been read instead and draws no verdict; where the lattice is not clear and a device was in the loop the number is shown and labelled not attributable.

**The harmonic roles.** The tiles partition the products by what each does to the chord that was *requested* — classified at the requested interval, measured at the lattice. The request is not on the record, so the nominal notes are the nearest equal-tempered pitches to the delivered tones, unambiguous inside the snap's cents budget; this is the one place the delivered frequencies are consulted, and only to name the notes. The interval's [just ratio](#)  $a : b$  is the simplest five-limit ratio (every prime factor 2, 3 or 5) within the pitch tolerance of the equal-tempered interval, found by search — smallest  $a + b$ , ties to the nearer cents — five-limit because a seven-limit ratio can sit nearer by cents and put the grid's fundamental somewhere the music never goes. The grid's fundamental is  $f_1/a$  (55 Hz for the default fifth at 2 : 3; 27.5 Hz for the major third at 4 : 5), and a recipe lands on grid index

$$k = ma + nb, \tag{30}$$

exact in integers. Its pitch class relative to the lower note is read from the odd parts (factors of two stripped — octave reduction, exact in integers): cents =  $1200 \log_2(\text{odd}(k)/\text{odd}(a)) \bmod 1200$ , the nearest semitone, the deviation from it, and the class modulo twelve; a point is a chord tone when  $\text{odd}(k)$  equals  $\text{odd}(a)$  or  $\text{odd}(b)$ , and between the frets when its deviation exceeds the tolerance. The roles: a chord tone at  $k \geq a$  is *Thickening*, below  $a$  *Growl (sub-octave)*; otherwise a point between the frets is *Sourness (clash)*, a consonant class (the parameter table lists them; the ninth is the one deliberate departure from the classical list) is *Added colour*, and any other class is *Sourness (clash)*;  $k \leq 0$  has no role. The tolerance is derived from the grid: the widest gap between a five-limit just interval and its equal-tempered class is under it and the nearest higher-limit point a two-tone lattice produces is beyond it, an empty band with the bar inside. The role table in §5.5 is the shipped classifier run on the default fifth's grid. A role's figure is the power sum, in dBc re the louder played note, of its present, in-band products counted *once per analysis bin* (loudest first, so a duplicated bin keeps its louder read); an absent product contributes nothing and is bounded by the page's floor; a product the instrument could not read was never looked at, so a role that holds one is a *lower bound* on its energy, stated beside the figure; a product under the band, or at grid index at most zero, is placed nowhere and marks nothing. *All products* is the same sum over every role. The record also states what it enumerated — the recipe count and the orders spanned — so a tone absent from the list can be told from one that was measured and silent.

**Drive level and summaries.** The drive line is read from the frozen stimulus: the combined peak  $a_1 + a_2$  and each tone's own amplitude in dBFS ( $20 \log_{10} \max(a, 10^{-12})$ ), volts only when the record's own embedded calibration carries the factor (the per-tone volts rounded to the label's digits, the combined figure derived from the rounded per-tone figures so the printed arithmetic holds), and a note that the two tones are equal — every order relation above assumes it. The summaries consume the present products at the derived length and one ratio of the cross products to the harmonic ones; a record that resolved nothing says so through the same verdict.

## 5.5 Parameters

Every value below is read from the manifest, which read it from the shipped code; the gate's factor, the enumeration's frequency edges, the pitch reference, the just-ratio search's numerator span, the contest tolerance and the spectrum's scale are literals the manifest carries as rule strings.

| Parameter                                                 | Value                                                                                        |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Plan</b>                                               |                                                                                              |
| default interval                                          | A2 + E3                                                                                      |
| default combined peak $A$ (re full scale)                 | 0.05 (-26.02 dBFS; each tone $A/2$ )                                                         |
| standard rate / supported rates                           | 96 000 Hz / 44 100, 48 000, 88 200, 96 000 Hz                                                |
| standard analysis length (the ceiling) $N$                | 2 097 152 ( $2^{21}$ )                                                                       |
| minimum analysis length                                   | 4 096                                                                                        |
| fade-in and fade-out $t_{\text{fade}}$                    | 0.05 s                                                                                       |
| settle $t_{\text{settle}}$                                | 0.2 s                                                                                        |
| <b>The lattice search</b>                                 |                                                                                              |
| tone cents budget                                         | $\pm 15$ cents                                                                               |
| resolvable spacing bar ( $g \geq$ )                       | 13 bins                                                                                      |
| enumeration ceiling (max order $K$ )                      | 7                                                                                            |
| minimum reduced sum $a + b$ at $K$                        | 15                                                                                           |
| delivered-lattice denominator limit                       | 8 192                                                                                        |
| standard ratio $a : b$ (the fifth)                        | 109 : 163 = 696.653 cents (requested 700; 26 ratios survived; runner-up 105 : 157 at 696.45) |
| <b>The analyzer</b>                                       |                                                                                              |
| enumerated recipes (orders 2– $K$ )                       | 54                                                                                           |
| local floor: inner offset / span / minimum offsets        | 3 bins / $\pm 8.789$ Hz / 12 per side                                                        |
| local floor: outer offset at $2^{21}$ (96 kHz) / $2^{17}$ | 192 / 12 bins                                                                                |
| presence gate                                             | amplitude $> 4\times$ the local floor (12.04 dB, a literal)                                  |
| enumeration ceiling in frequency                          | $0.95 f_s/2$ (a literal)                                                                     |
| <b>The read-time predicate</b>                            |                                                                                              |
| local floor separation                                    | 12.04 dB                                                                                     |
| minimum floor samples                                     | 3                                                                                            |
| fade-contaminated separation bound                        | 32.5 dB                                                                                      |
| played-tone / DC neighbourhood                            | 12 bins                                                                                      |
| skirt depth $20 \log_{10}(\pi g)$ at $g = 22$ / 24        | 36.79 / 37.55 dB                                                                             |
| partner-check tolerance                                   | 6 dB                                                                                         |
| <b>The coherence check</b>                                |                                                                                              |
| minimum clearance (the inner edge)                        | 3 bins                                                                                       |
| detector offsets                                          | $\pm 1 \dots \pm 9$ bins (derived: $13 - 1 - 3$ )                                            |
| bar over the local floor                                  | 30 dB                                                                                        |
| product skirt margin                                      | 10 dB                                                                                        |
| per-offset ceilings at $g = 22$ (dBc)                     | 16.4, 10, 6.03, 3.06, 0.63, -1.48, -3.38, -5.14, -6.81                                       |
| expected noise excess, 4 / 36 bins                        | 4.78 / 7.8 dB                                                                                |
| <b>The harmonic roles</b>                                 |                                                                                              |
| pitch tolerance                                           | 20 cents                                                                                     |
| added-colour semitone classes                             | 2, 3, 4, 5, 7, 8, 9                                                                          |
| just-ratio search limit                                   | 32 (denominator; numerator to $16\times$ )                                                   |
| <b>Drive level and pitch</b>                              |                                                                                              |
| volts label digits                                        | 3                                                                                            |

| Parameter                              | Value                           |
|----------------------------------------|---------------------------------|
| pitch reference                        | A4 = 440 Hz, 12-TET (a literal) |
| audible band                           | 20 Hz – 20 000 Hz               |
| <b>Oracle (analysisdump imd-synth)</b> |                                 |
| torus quadrature points per axis       | 4 096                           |

The lattice the shipped search chooses on every grid — the standard ratio placed on each, and the fallback on the two grids too coarse to place it — and the major third at the standard grid; candidates are the resolvable and the non-degenerate counts the search saw:

| probe                                   | $f_s$ (Hz) | $N$             | tier                | $k_1 : k_2$   | $g$ | $a : b$   | cents         | candidates  |
|-----------------------------------------|------------|-----------------|---------------------|---------------|-----|-----------|---------------|-------------|
| fifth 44100 Hz                          | 44 100     | 2 <sup>20</sup> | standard resolvable | 2 616 : 3 912 | 24  | 109 : 163 | 0.334 / -3.01 | 121 / 3 061 |
| fifth 48000 Hz                          | 48 000     | 2 <sup>20</sup> | standard resolvable | 2 398 : 3 586 | 22  | 109 : 163 | -3.6 / -6.94  | 109 / 2 523 |
| fifth 88200 Hz                          | 88 200     | 2 <sup>21</sup> | standard resolvable | 2 616 : 3 912 | 24  | 109 : 163 | 0.334 / -3.01 | 121 / 3 061 |
| fifth 96000 Hz                          | 96 000     | 2 <sup>21</sup> | standard resolvable | 2 398 : 3 586 | 22  | 109 : 163 | -3.6 / -6.94  | 109 / 2 523 |
| fifth corpus 96000 Hz / 2 <sup>17</sup> | 96 000     | 2 <sup>17</sup> | degraded            | 150 : 226     | 2   | 75 : 113  | -2.15 / 7.48  | 0 / 8       |
| fifth Simulation Mode interactive       | 48 000     | 2 <sup>15</sup> | degraded            | 75 : 113      | 1   | 75 : 113  | -2.15 / 7.48  | 0 / 2       |
| major third A2 + C#3 96000 Hz           | 96 000     | 2 <sup>21</sup> | standard resolvable | 2 400 : 3 024 | 48  | 50 : 63   | -2.15 / -2.04 | 99 / 2 127  |

The window on every grid, from the shipped geometry on the snapped default probe (sample indices; a negative overrun is the margin before the fade-out):

| grid                                                   | $f_s$ (Hz) | $N$             | bin (Hz) | stimulus (s) | start  | steady end | overrun |
|--------------------------------------------------------|------------|-----------------|----------|--------------|--------|------------|---------|
| standard 44100 Hz                                      | 44 100     | 2 <sup>20</sup> | 0.042057 | 24.2772      | 11 025 | 1 068 421  | -8 820  |
| standard 48000 Hz                                      | 48 000     | 2 <sup>20</sup> | 0.045776 | 22.3453      | 12 000 | 1 070 176  | -9 600  |
| standard 88200 Hz                                      | 88 200     | 2 <sup>21</sup> | 0.042057 | 24.2772      | 22 050 | 2 136 842  | -17 640 |
| standard 96000 Hz                                      | 96 000     | 2 <sup>21</sup> | 0.045776 | 22.3453      | 24 000 | 2 140 352  | -19 200 |
| corpus (pre-#254 B2a) 96000 Hz / 2 <sup>17</sup>       | 96 000     | 2 <sup>17</sup> | 0.73242  | 1.86533      | 24 000 | 174 272    | -19 200 |
| Simulation Mode interactive 48000 Hz / 2 <sup>15</sup> | 48 000     | 2 <sup>15</sup> | 1.4648   | 1.18267      | 12 000 | 54 368     | -9 600  |

The role table for the default fifth: every grid point’s musical frequency, nearest pitch and deviation, semitone class relative to the root, and the role the classifier assigns it:

| $k$ | Hz  | pitch | cents | class | role               |
|-----|-----|-------|-------|-------|--------------------|
| 1   | 55  | A1    | 0     | 0     | Growl (sub-octave) |
| 2   | 110 | A2    | 0     | 0     | Thickening         |
| 3   | 165 | E3    | 1.96  | 7     | Thickening         |
| 4   | 220 | A3    | 0     | 0     | Thickening         |
| 5   | 275 | C#4   | -13.7 | 4     | Added colour       |
| 6   | 330 | E4    | 1.96  | 7     | Thickening         |
| 7   | 385 | G4    | -31.2 | 10    | Sourness (clash)   |
| 8   | 440 | A4    | 0     | 0     | Thickening         |
| 9   | 495 | B4    | 3.91  | 2     | Added colour       |
| 10  | 550 | C#5   | -13.7 | 4     | Added colour       |
| 11  | 605 | D#5   | -48.7 | 6     | Sourness (clash)   |
| 12  | 660 | E5    | 1.96  | 7     | Thickening         |
| 13  | 715 | F5    | 40.5  | 8     | Sourness (clash)   |

## 5.6 The result

The app’s own rendering of the same measurement kind, from the user guide’s figure pipeline (the bundled simulated rig fed the default fifth at the standard drive, rendered headlessly from the chart view):

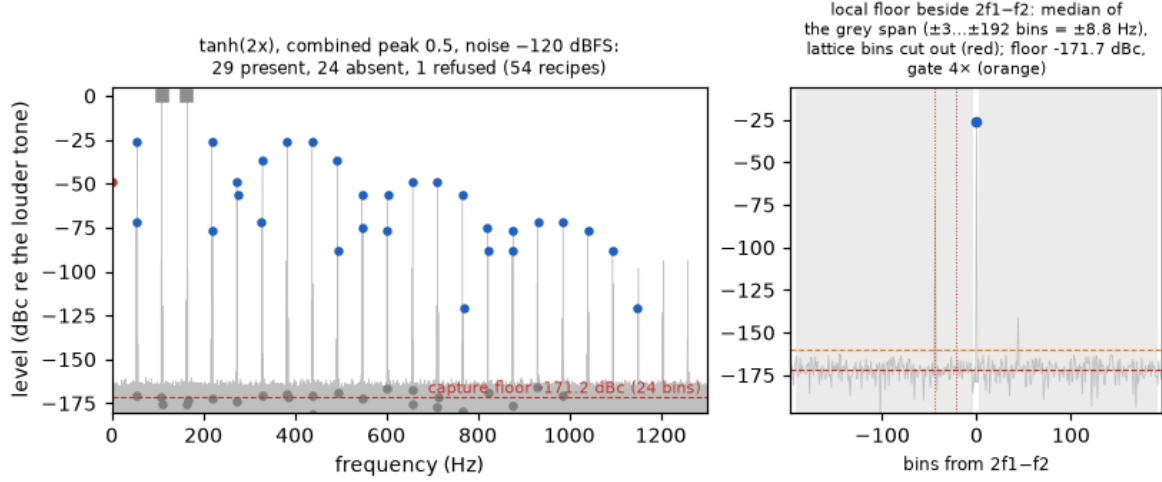


Figure 15: The worked example,  $y = \tanh(2x)$  at a combined peak of 0.5 on the shipped fifth through seeded noise: the analysed spectrum with every enumerated product marked present (blue), absent (grey) or refused (red), the two tones as squares, and the capture floor as the dashed rule (left). Right: the local-floor neighbourhood of one product, the median of the grey span with every lattice bin cut out, and the gate four times above it.

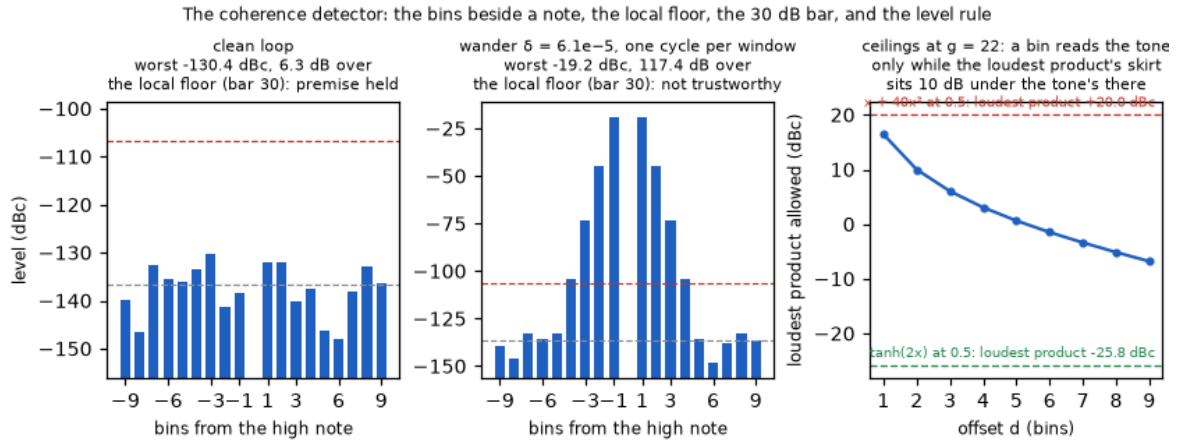


Figure 16: The coherence detector beside the high note on a clean identity (left) and on the validated wander fixture (middle), each against the local floor beside the note and the bar above it; right, the per-offset ceilings of (28) on the shipped spacing, with the worked example's loudest product far below every ceiling and a squarer driven above its own tones clearing none.

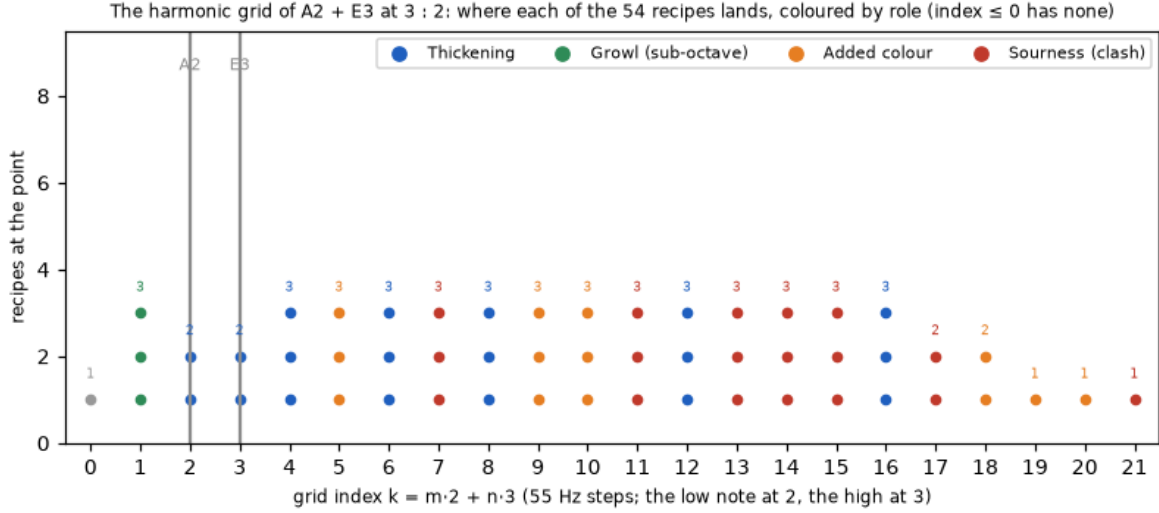


Figure 17: The harmonic grid of the requested fifth at its just ratio: every enumerated recipe at its grid index (30), stacked where several share a point and coloured by the role the classifier assigns the point; the two played notes marked. Index zero and below has no role.

## 5.7 Worked example and parity

The [parity test](#) runs 28 cases through both implementations, and each device case again as its [phantom capture](#) twin (9 twins): the lattice choice alone at every standard grid and both fallback grids, for the fifth, an octave, a major third and a refused pair; the polynomial devices and the two clippers at the loudest hardware drive on the shipped fifth; a phantom holding all  $\sum_{k=2}^K 2k$  recipes at a level ladder; the identity through seeded noise at two levels and two lengths; a squarer behind a post-filter and the soft clipper behind a pre-filter; the corpus lattice with a squarer and a cubic; the wander fixture through the legacy path and through the detector; a squarer driven until its products stand above the played tones; and the worked example analyzed 100 samples late. The stimulus is the plan’s own, snapped by the shipped search, so the lattice choice is exercised, not assumed; the analysis is the app’s own call. Beside every product the oracle prints a truth: for a memoryless device on an exact bin lattice the output is periodic on the  $(\theta_1, \theta_2)$  torus, so every product’s amplitude is an exact two-dimensional Fourier coefficient,

$$a_{m,n} = 2 |c_{m,n}| |H_{\text{post}}(mf_1 + nf_2)|, \quad c_{m,n} = \frac{1}{4\pi^2} \iint f(A_1 \sin \theta_1 + A_2 \sin \theta_2) e^{-i(m\theta_1 + n\theta_2)} d\theta_1 d\theta_2, \quad (31)$$

with  $A_i = a_i |H_{\text{pre}}(f_i)|$  the drive after any pre-filter, computed by the trapezoid rule on the grid the parameter table sizes (exact for the periodic integrand’s orders below the grid minus  $K$ ), never by hand; the polynomial device’s closed form (the binomial expansion in two sines) is printed beside the quadrature as a check on it; the tone bins carry the linear response plus the odd self-terms (a cubic’s  $\frac{3}{4}a_3A_1^3 + \frac{3}{2}a_3A_1A_2^2$  on tone 1); the identity’s every product is zero and the phantom’s are its listed levels. The test asserts:

- Swift against Python: every header and row quantity in decibels within 1e-07 (measured maximum 1.8e-08 on this run) on a capture with no filter in it, and within 0.001 (measured 0.00016) on one rendered through a biquad, whose direct-form recursion accumulates rounding differently in the two languages over two million samples; every discrete reading — bin, gate, reading and its evidence, grid index, role, contest, partner — identical, and every sentence identical after last-bit rounding; the two truths within 1e-08 relative (measured 2.6e-10), the closed form against the quadrature within 1e-10 (measured 4.4e-14);
- every analytically empty recipe on a noiseless capture at or below -200 dBc on both sides

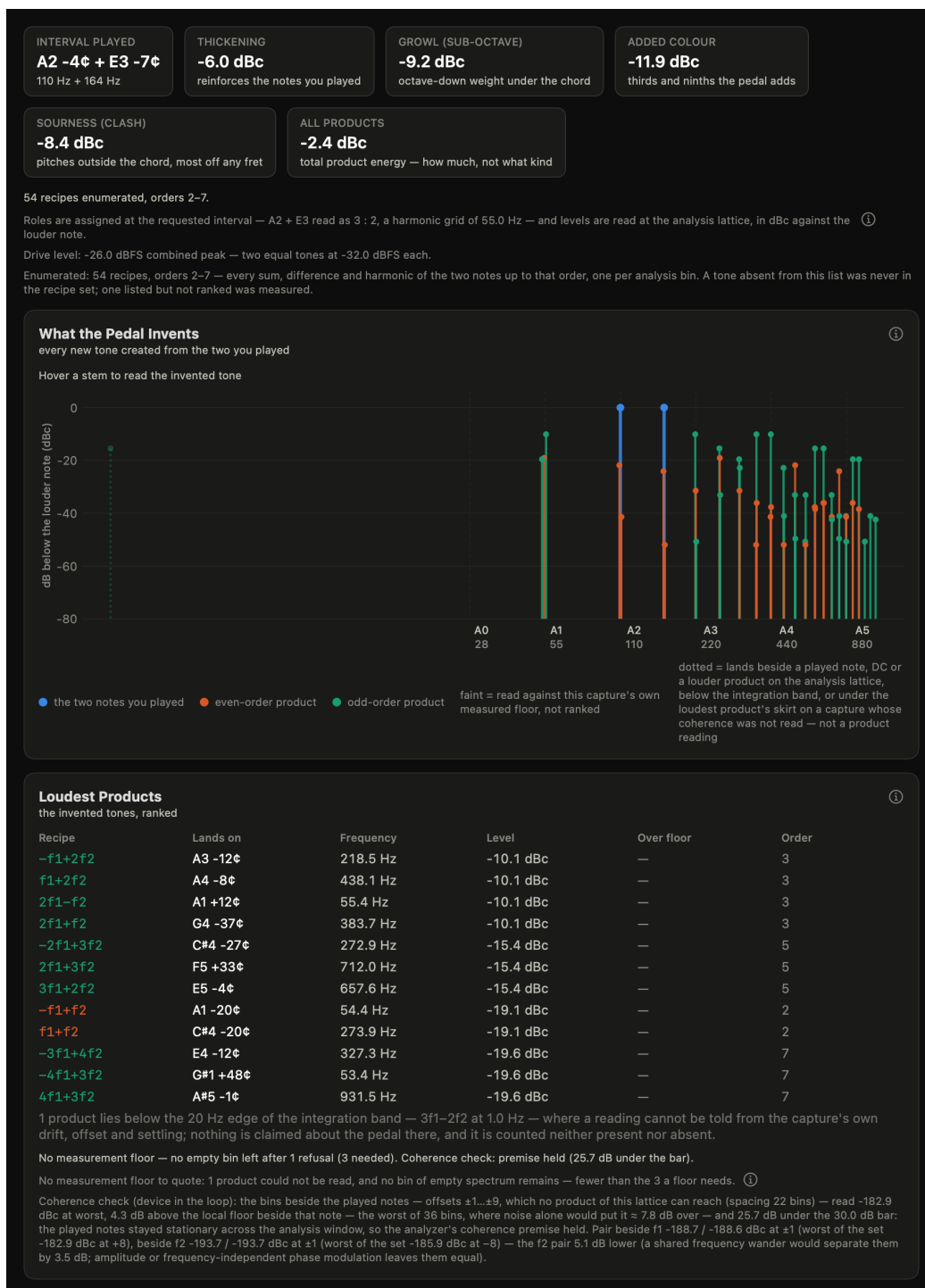


Figure 18: The Chord Roughness view as the app draws it, from the guide's simulated fixture: the pitch-axis spectrum, the role tiles in dBc, the drive line and, under the table, the floor sentence and the coherence line.

(measured worst -273.8): the depth at which a bin-centred rectangular window separates nothing from the loudest component is double precision's, not a window's;

- the shipped read against the truth on every content recipe, after the noise allowance (a bin holds its noise too, so a read may sit up to  $20 \log_{10}(1 + 10^{(\text{floor} + \text{spread} - L)/20})$  from the truth by the noise alone): within 0.001 for the bandlimited devices (measured residue at worst -1.2e-07), within 0.05 for the two kinked ones, and within 1e-08 for every phantom (measured 2.5e-10 on the full ladder);
- the lattice each case played is the row the manifest recorded by running the shipped search on that grid, field for field, and the refused pair is refused with its numbers on both sides;
- the identity through noise reads every product absent, its capture floor within 3 dB of the noise's analytic per-bin median (measured deficits 0.21 and 0.14 dB at the two lengths — a median of a few dozen Rayleigh draws), and its coherence excess within 6 dB of (29) (measured 6.3 against 7.8);
- the late analysis moves the window by exactly its offset, keeps the overrun, and moves no content product's level by more than 1e-05 dB (measured 2.2e-06: the noise stream is not shifted with the window, so the late segment exchanges a hundred noise samples for a hundred new ones).

Part B's parity is established: on every predicate case — the corpus lattice's refusals by position and beside a louder product, the wander fixture through the legacy path and through the detector, and the device whose products stand above its tones — the two implementations agree on every reading, every count and both registers' sentences.

The lattice cases — the shipped fifth at the four rates, the two fallback grids, the octave at two lengths, the major third, and the refusal:

| case                | $f_s$  | $N$      | tier                | $k_1 : k_2$ | $g$ | $a : b$   | interval (c) | tones (c)     |
|---------------------|--------|----------|---------------------|-------------|-----|-----------|--------------|---------------|
| a-fifth-96k         | 96 000 | $2^{21}$ | standard resolvable | 2398 : 3586 | 22  | 109 : 163 | 696.65       | -3.6 / -6.94  |
| a-fifth-48k         | 48 000 | $2^{20}$ | standard resolvable | 2398 : 3586 | 22  | 109 : 163 | 696.65       | -3.6 / -6.94  |
| a-fifth-88k         | 88 200 | $2^{21}$ | standard resolvable | 2616 : 3912 | 24  | 109 : 163 | 696.65       | 0.334 / -3.01 |
| a-fifth-44k         | 44 100 | $2^{20}$ | standard resolvable | 2616 : 3912 | 24  | 109 : 163 | 696.65       | 0.334 / -3.01 |
| a-fifth-corpus      | 96 000 | $2^{17}$ | degraded            | 150 : 226   | 2   | 75 : 113  | 709.63       | -2.15 / 7.48  |
| a-fifth-interactive | 48 000 | $2^{15}$ | degraded            | 75 : 113    | 1   | 75 : 113  | 709.63       | -2.15 / 7.48  |
| a-octave-96k        | 96 000 | $2^{21}$ | standard resolvable | 2418 : 4823 | 13  | 186 : 371 | 1195.3       | 10.8 / 6.12   |
| a-octave-corpus     | 96 000 | $2^{17}$ | degraded            | 150 : 302   | 2   | 75 : 151  | 1211.5       | -2.15 / 9.35  |
| a-third-96k         | 96 000 | $2^{21}$ | standard resolvable | 2400 : 3024 | 48  | 50 : 63   | 400.11       | -2.15 / -2.04 |
| a-refusal           | 96 000 | $2^{16}$ | refused             | 25 : 50     | 25  | 1 : 2     | 1 200        | -4.11 / -4.11 |

The device cases, each beside its alias-free twin. A twin is noiseless, so its empty bins are numerical dust and its “present” count is dust against dust — the verdict of a noiseless case is not compared, and the column shows why; a device case carries seeded noise far below its content. The residue is what remains of the read's error against the truth once the noise allowance is subtracted, clamped at zero:

| case              | content | present | S-P (dB) | floor (dBc) | vs truth (dB) | residue (dB) | at      |
|-------------------|---------|---------|----------|-------------|---------------|--------------|---------|
| b-poly            | 10      | 10      | 1.7e-09  | -166.84     | 7.53e-06      | 0            | 3f2     |
| b-poly-twin       | 10      | 16      | 7.1e-15  | -334.09     | 4.82e-14      | 4.8e-14      | 3f2     |
| b-squarer         | 4       | 4       | 1.9e-09  | -166.53     | 2.86e-07      | 0            | 2f2     |
| b-squarer-twin    | 4       | 15      | 3.6e-15  | -333.84     | 1.06e-14      | 1.1e-14      | 2f1     |
| b-asym            | 54      | 53      | 2.3e-09  | —           | 0.0298        | 0.03         | 7f1     |
| b-asym-twin       | 54      | 53      | 3.4e-12  | —           | 1.48e-10      | 1.5e-10      | 7f1     |
| c-tanh            | 30      | 29      | 1.6e-09  | -171.17     | 0.0224        | 0            | 7f1     |
| c-tanh-twin       | 30      | 33      | 2.1e-12  | -318.75     | 2.91e-10      | 2.9e-10      | 7f1     |
| c-hardclip        | 30      | 29      | 5.8e-11  | -156.53     | 0.0128        | 0.013        | f1+4f2  |
| c-hardclip-twin   | 30      | 46      | 1.4e-14  | -305.72     | 1.5e-12       | 1.5e-12      | 5f1–2f2 |
| d-phantom         | 54      | 53      | 1.3e-13  | —           | 2.48e-10      | 2.5e-10      | 5f1–2f2 |
| f-squarer-hp      | 4       | 4       | 0.00016  | -166.53     | 3.04e-07      | 0            | 2f2     |
| f-squarer-hp-twin | 4       | 15      | 7.1e-15  | -333.82     | 1.16e-14      | 1.2e-14      | 2f1     |
| f-squarer-lp      | 4       | 4       | 2.1e-07  | -166.53     | 2.09e-07      | 0            | 2f2     |
| f-squarer-lp-twin | 4       | 15      | 3.6e-15  | -333.81     | 1.06e-14      | 1.1e-14      | 2f1     |
| f-tanh-prehp      | 30      | 29      | 4.1e-05  | -171.17     | 0.0294        | 0            | 7f1     |
| f-tanh-prehp-twin | 30      | 33      | 2.3e-12  | -318.91     | 2.96e-10      | 3e-10        | 7f1     |
| j-latency         | 30      | 29      | 1.6e-09  | -171.16     | 0.0223        | 0            | 7f1     |
| j-latency-twin    | 30      | 33      | 2.1e-12  | -318.75     | 2.91e-10      | 2.9e-10      | 7f1     |

Three readings from that table are worth stating in words. The two *kinked* devices — the hard clip, a near-square wave whose coefficients fall as  $1/k$ , and the asymmetric clipper, whose slope breaks at zero — applied per sample at  $f_s$  fold what they make above Nyquist back onto the lattice, where a folded high-order product lands *exactly* on a low-order bin (every lattice point is a multiple of  $g$ ): 0.0128 dB at f1+4f2 and 0.0298 dB at 7f1, against 1.5e-12 and 1.5e-10 on their phantom twins. That is chapter one’s finding on a two-tone lattice, and the same corollary holds: a per-sample synthetic device is never the analytic oracle for a product near the top of the band; the phantom is. The bandlimited devices read the truth inside the allowance on every case. And the phantom of every recipe reads back 53 present and 1 out of band — the recipe the shipped fifth puts under the band’s low edge, whose level still reads back to 2.5e-10 dB and whose reading is a refusal on frequency, by the band rule that outranks the gate.

The worked example,  $y = \tanh(2x)$  at a combined peak of 0.5 through seeded noise: 29 present, 24 absent, a floor of -171.2 dBc from 24 bins, a coherence excess of 6.3 dB over the local floor, and a loudest product of -25.83 dBc. Its third-order difference products  $2f_1 - f_2$  and  $2f_2 - f_1$  read -25.832 and -25.832 dBc against truths of -25.832 and -25.832 (both present);  $3f_1$  reads -37.078 against -37.078; and the second-order  $f_1 + f_2$ , analytically empty for an odd device, reads -173.79 dBc — the noise — and is absent, with the bin’s own level as its bound.

The identity through noise: the capture floor against the analytic median  $2\sigma\sqrt{\ln 2/N}$  of the same noise through the analyzer’s scale, and the distance of that per-bin median below the noise’s broadband rms,  $10\log_{10}(N/(4\ln 2))$  — 58.79 dB at the standard length and 46.75 at the corpus length, the relation that decides whether a broadband floor can bound a bin (it cannot without an assumption about how the noise is spread, and the assumption is worth tens of decibels in the refusing direction):

| case                 | $N$      | noise (dBFS) | median (dBc) | floor (dBc) | bins | per-bin – broadband (dB) | excess (dB) |
|----------------------|----------|--------------|--------------|-------------|------|--------------------------|-------------|
| e-identity-80        | $2^{21}$ | -80          | -106.75      | -106.53     | 53   | 58.79                    | 6.3         |
| e-identity-110       | $2^{21}$ | -110         | -136.75      | -136.53     | 53   | 58.79                    | 6.3         |
| e-identity-corpus-80 | $2^{17}$ | -80          | -94.705      | -94.843     | 49   | 46.75                    | 8.65        |

The predicate cases. On the corpus lattice a squarer through noise has 5 recipes refused beside the notes and DC and 8 refused beside a louder product, and its floor rests on 37 bins with those left out. The wander fixture — a shared fractional frequency wander of both tones, the validated generator of the kernel’s own test, which reproduces the one real fault the reference bench has captured — reads through the legacy path at -49.25 dBc, 75.6 dB over the capture floor, with the pair difference 3.56 dB against the 3.56 a frequency wander predicts; through the detector on the shipped lattice at one cycle per window it reads -19.23 dBc, 117 dB over the local floor, over



the bar in both registers. The squarer driven above its tones has a loudest product of 20 dBc, clears no ceiling, does not interpret, and marks 49 products unread under a skirt of 36.8 dB at the shipped spacing:

| case             | present | absent | refused | floor bins | coherence source | excess / bar (dB) | readable |
|------------------|---------|--------|---------|------------|------------------|-------------------|----------|
| g-corpus-squarer | 4       | 37     | 13      | 37         | detector         | 8.65 / 30         | 1..9     |
| g-corpus-cubic   | 7       | 28     | 19      | 28         | detector         | 8.65 / 30         | 1..9     |
| h-wander-legacy  | 0       | 49     | 5       | 49         | stored products  | 75.6 / 30         | 2        |
| h-wander-shipped | 0       | 53     | 1       | 53         | detector         | 117 / 30          | 1..9     |
| i-loud-squarer   | 4       | 0      | 50      | 49         | detector         | 6.3 / 30          | none     |

## 5.8 Limits and failure modes

- **The floor cannot separate the device from the rig.** A resolved product stands above this capture’s noise; the measurement does not say what made it. An interface’s converters intermodulate, and a bare loop resolves a handful of second-order products of its own. The capture floor is the right decision floor for presence (a per-bin read at the product’s own frequency), and the wrong instrument for attribution; a control capture with nothing in the loop is the only thing that does that job, and no rig-side broadband floor composes into a per-bin decision — the per-bin median of white noise sits 58.79 dB below the same noise’s broadband rms at the standard length, so a broadband bound applied per bin would refuse genuine products by tens of decibels.
- **The percentage sums carry no floor bound.** The three capture-time sums are broadband aggregates over the products the gate passed, and a broadband quantity is exactly what an operative rig floor legitimately bounds — so a bound is available and unbuilt: a percentage summed from products that each passed their own per-bin gate can still be quoted below what the rig supports as a whole. The read-time derivation subtracts refused products and bounds nothing.
- **A coherence verdict names no culprit, and reads nothing beside a loud product.** A detector bin responds to modulation of the played tone at periods comparable to the window itself; a pedal’s thermal drift, a sagging supply or a dying battery moves a note as surely as a clock, so “not trustworthy” is a statement about the capture, never about a stage. And the check is a level rule as much as a distance rule: a device whose loudest product stands above the ceiling of (28) for even the nearest offset — a full-wave rectifier’s products stand above its own fundamentals — is not read at all, and the line says why. A device whose sidebands are roughly constant in absolute level and fall in dBc as its output rises is consistent with something after the level control modulating the notes; the check reports the excess, the pair shape and the level, and the design question of whether it should say more when that dependence is clean is open.
- **What a wander looks like, and what it does not distinguish.** The pair statistic separates a shared fractional frequency wander (the pairs differ by  $20 \log_{10}(f_2/f_1)$ , measured on the fixture to 3.56 against 3.56) from amplitude or frequency-independent phase modulation (the pairs equal); it does not separate the latter two from each other, and no verdict is drawn from the shape. A  $1/d$  fall of the sidebands with offset is the rectangular skirt of a slow, non-periodic drift; a lift that does not fall with offset is broadband. A drop in a device’s even-order products between a short stimulus and the current long one, alongside a skirt that follows the device position, is consistent with an operating point that shifts over the window (a rectified mean through a coupling capacitor, say); the oracle reproduces the skirt from a stated wander and does not reproduce the product drop from any mechanism it models, so that half stays a bench question.

- **Aliasing in a simulated device.** A device applied per sample folds its content above Nyquist onto the lattice, exactly on low-order bins; the kinked devices' residues above quantify it. Hardware behind a converter's anti-alias filter does not do this.
- **The enumeration is a small-signal census.** Orders two through  $K$ ; at the standard drive the reference clippers' content sits in orders two and three with trace in four and five, but a hard clipper at playing levels has real energy at orders the walk does not reach. An empty row above order five means measured and nothing above the floor at this drive, never that the device makes nothing there.
- **The corpus lattice is crowded, and every record captured on it carries the marks.** At the old length the fifth's spacing was two bins: four even-order recipes sit two bins from the played notes and one two bins from DC, all refused on position, and products cluster two bins apart so the quieter of a pair is refused beside the louder. The old three-bin read applies to those records too; the separation bound was measured under it and is deliberately not raised. On an exact small ratio (the original plain snap of the fifth, an exact  $2 : 3$ ) recipes share bins and the surviving labels are the enumeration's tie-break, marked at read time as contested; those records can never be re-measured onto a clean lattice.
- **The sub-band recipe.** On the shipped fifth one recipe,  $3f_1 - 2f_2$ , lands about one hertz up: under the band's low edge, where nothing a pedal does can be told from the capture's own drift, offset and settling. It reads out of band on every record — named in its own line, never in a sum, and never in a tile — and the phantom parity shows the level under it still reads back exactly.
- **The stored gate is frozen.** The presence flag and the three sums are written at capture; a change to the gate's factor or the neighbourhood moves future captures only, and a read-time predicate can change which flagged bins are averaged into a floor but never which bins are flagged. The window length is likewise not stored: a consumer derives it from the stored stimulus by the plan's own rule, and a record captured at the old fixed length at a rate where it overran is asked the post-fix question and answers with no bound — never a certificate that its window was clean.
- **The interval is a few cents off, on purpose.** Keeping the products separable costs pitch inside the stated budget, disclosed on every record; a consonant interval is a simple ratio, and a simple ratio is a crowded lattice, so a request that cannot escape an exact small ratio inside the budget is refused before anything plays rather than measured on a lattice whose labels are not facts.

## 6 Gain Map

### 6.1 Purpose, and what the measurement cannot tell you

A [gain map](#) is the Harmonic Distortion measurement of §3 repeated at each [drive level](#) of a [level lattice](#) — one sweep per level, each level analyzed on its own — and summarized on one grid: for every level  $i$  and every column  $j$  of an [export grid](#) of fundamentals, the [total harmonic distortion](#) over the audible band and the level of every stored [harmonic order](#), each a [cell](#). It answers what one sweep cannot: how the harmonic content moves with how hard the device is driven — whether it cleans up, where, and whether its recipe holds across the range. At read time the record states a [peak tile](#) (the loudest cell that is a measurement, or a bound), a [cleanup reading](#) per column and one tile at the Compression probe's note, and a [fundamental presence](#) verdict over the map. What it cannot tell you, stated once: (a) anything about the device's output level or gain — the cells are ratios to each level's own fundamental, and the drive axis is the signal driving the

circuit, never a knob on it (to map a knob is to measure one map per position); (b) anything between two levels — the rows are separate captures seconds apart, and a circuit that drifts or sags between them shows the drift as row-to-row wobble; (c) whether a cell near its floor is the device or the rig — below the [composed margin](#) a cell is quoted as a bound; (d) anything at the two [edge mask](#) columns, which are the sweep’s own fade ramps; (e) anything a [truncated column](#) lost above the top of the band; and (f) anything about what the distortion sounds like.

## 6.2 The stimulus

The lattice is  $N$  levels evenly spaced in dB from a floor to a ceiling inclusive,

$$\ell_i = \ell_{\min} + (\ell_{\max} - \ell_{\min}) \frac{i}{N-1}, \quad A_i = 10^{\ell_i/20}, \quad i = 0 \dots N-1, \quad (32)$$

a count of one being the ceiling alone. Two [drive windows](#) exist. The hardware window is anchored to gain staging: with the interface’s input staged so that pedal peaks land in the staging target at the loudest planned drive, the ceiling is the loudest drive that cannot clip the converter, and the floor sits a fixed travel below it (a step of 5.143 dB on the default lattice). The plugin window’s ceiling is digital full scale — the float render path has no converter to protect — and its floor is lower, where high-gain preamp knees were measured to sit (a step of 8.571 dB). The parameter table gives both.

Each level’s stimulus is the synchronized sweep of §3.2 at that level’s amplitude — the same band, the same requested length, the same fades — between a pre-roll of silence (the per-level noise region) and a tail of silence:

$$s_i[n] = [\underbrace{0, \dots, 0}_{\lfloor t_{\text{pre}} f_s \rfloor}, x_i[0], \dots, x_i[S-1], \underbrace{0, \dots, 0}_{\lfloor t_{\text{tail}} f_s \rfloor}], \quad (33)$$

with  $S$  the sweep’s sample count. The sweep length is the measure sheet’s pick from a short list and is *not stored* in the record; the consequence is a floor derivation that must take the worst case over the list (§6.4). The default length is shorter than the fingerprint sweep’s, so every extraction window is shorter and the analyzer separates orders less well at the band’s low edge (§6.7 measures it).

## 6.3 The pipeline

Given the  $N$  captured outputs  $y_i[n]$  and the loop’s integer latency  $\ell$  (the calibration’s own measured figure on the hardware path; zero on the plugin path, which measures and removes its latency per capture):

1. **The aligned cut.** From each capture the sweep and its tail are cut at the pre-roll plus the latency,  $y_i[\lfloor t_{\text{pre}} f_s \rfloor + \ell \dots \lfloor t_{\text{pre}} f_s \rfloor + \ell + S + \lfloor t_{\text{tail}} f_s \rfloor]$ , clipped to the capture’s end — one arithmetic for every caller.
2. **The per-level analysis.** Each aligned capture goes through the analysis of §3.3 with the level’s own sweep and the journey’s configuration: the same harmonic count, a response DFT half the fingerprint analyzer’s length (twice its bin width), the same window cap. The levels are analyzed in parallel — the analyses are pure — and the entries sort by ascending amplitude. Nothing is compensated: the loop’s gain and response stay in every order.
3. **The summary grid.** On the export grid  $f_j = f_1(f_2/f_1)^{j/(n-1)}$ ,  $j = 0 \dots n-1$  —  $n$  fundamentals log-spaced from the sweep’s start to its end *inclusive*, so the first and last

columns sit at the sweep's edges — each level's THD over the audible band  $B$  and each stored order's level:

$$\text{thd}[i][j] = \frac{\sqrt{\sum_{k \geq 2, kf_j \in B, \text{valid}_i(k, f_j)} |H_k^{(i)}(kf_j)|^2}}{|H_1^{(i)}(f_j)|}, \quad h[k][i][j] = 20 \log_{10} |H_k^{(i)}(kf_j)|, \quad (34)$$

nil where the fundamental reads zero or the harmonic sits at or above Nyquist;  $\text{valid}_i$  is the level's own **validity bound** — the sweep's alias-image bound and nothing else, since the journey is uncompensated and has no calibrated band. Because the cells are uncompensated, a cell's **delivered fundamental**,  $\ell_i + h[1][i][j]$ , is the level the note actually reached at the converter in absolute dBFS; the floor's drive term reads it.

4. **The models.** Beside the grid the record carries one **audition model** per level — a Hammerstein fit of that level's analysis, bounded in level by its own amplitude. The Hear-it panels consume them and nothing measured does; a render between two levels blends two fits and the quieter one clamps, inventing a bracket-shaped clipping the device does not have, measured at  $-25$  dB on this lattice — a property of what the models *render*, stated here and measured in a later chapter.
5. **Storage.** The grid, the levels, the orders, the models, the sample rate and the integration band. The last two are additive fields: a legacy record without them derives no validity, no fade, no masks and no analyzer residue, and its cells summed everything below Nyquist.

## 6.4 Read-time predicates on the result

Everything below is computed from the frozen grid when a record is read.

**The validity fade and the truncated columns.** The summary's validity is the grid's own span with the alias-image bound: a harmonic's cells past  $f_j \leq f_s/(2k+1)$  are faded on display. A **truncated column** is one where some order  $k \geq 2$  has  $kf_j$  inside the stored band and fails the bound: its THD cells under-read their declared band, and the column is flagged and excluded from the tiles, never corrected.

**The edge masks.** The two columns within a quarter tone of the grid's edges —  $f_j \leq f_0\rho$  and  $f_j \geq f_{n-1}/\rho$ ,  $\rho$  the guard ratio — are the sweep's fade-in and fade-out ramps: excited under a ramp, they describe the measurement and not the device (a jumpered wire reproduces them). They are derived from the grid alone, faded, excluded from the tiles and from the presence vote, and never touch the colour scale.

**Fundamental presence.** Per cell, the reference is the loudest order *stored* for that cell, the fundamental included; the cell is absent when the fundamental sits at or beyond the analyzer's separation bound of §3.4 below it, present otherwise, undetermined without a fundamental read; no rig floor enters. The map's verdict is the majority over every cell outside the two edge columns, and it outranks both peak states. The vote is a majority for a reason the worked example shows: a cell whose fundamental bin holds the loop's noise, or — beside the low edge — the second harmonic's own leakage into the fundamental's window, reads present on a device that has no fundamental at all.

**The floor, composed from three sources.** A cell's THD is quoted as a measurement only where it stands clear of *every* bound the rig and the analyzer put under it; the applicable floor is the largest, and the largest floor is the smallest margin.

The first source is the [calibration stamp](#): the embedded calibration’s THD-versus-frequency curve of the loopback, one read per point at log-spaced fundamentals, measured once at the calibration’s own drive. It is read through the [median read](#): each point replaced by the running median of its neighbours’ dB over an odd window before the interpolation on the  $(\log f, \text{dB})$  plane. The window is a fixed count of points on the stamp’s own grid ( $\pm 0.25$  octave), narrower than the map’s third-of-an-octave columns — nothing a column resolves is smoothed away — and wide enough that a single-read trough cannot set the floor. The stamp is a ratio re the calibration’s delivered fundamental, while the cells are uncompensated, so the rig’s absolute noise is the calibration’s level plus its measured chain gain, and a cell’s floor ratio is that against the cell’s delivered fundamental:

$$F_{\text{rig}}[i][j] = \text{stamp}_{\text{med}}(f_j) + \max(0, (\ell_{\text{cal}} + G) - (\ell_i + h[1][i][j])), \quad (35)$$

the [drive term](#). Quieter than the calibration the ratio rises by the difference, because the noise is absolute; louder, it stays at the stamp’s own ratio — never extrapolated past what the calibration measured, because the rig’s own distortion rises with drive and the stamp measured their sum at one drive (the bare loop at the loudest level read a few thousandths of a percent at 6–16 dB “clear” of a floor extrapolated 14 dB below anything the calibration saw). The calibration source contributes only with both a usable stamp and a chain gain; omitting the gain left every margin one input-knob setting optimistic.

The second source is a plugin record’s [operative floor](#) — the larger of its silence render and its A/A residual, an rms re full scale — entering as  $20 \log_{10}(\sqrt{2} \text{rms}) - (\ell_i + h[1][i][j])$ , an absolute bound with no chain-gain term, because the grid is already absolute. It governs where the render-path calibration is numerically silent. The larger of the two ratios is the rig-noise floor; nil when neither source is usable, since a floor known to be missing a term is worse than none.

The third source is the [analyzer residue](#), and it is what a noise floor cannot see. The plan’s own sweep, handed back as its own capture with the tail’s silence — nothing in the loop — goes through the shipped analyzer at the journey’s configuration, and its THD is read per column of the record’s grid: the THD of nothing. One column above the sweep’s start the fade-in ramp leaks *coherently* into the second-harmonic window: on the shipped grid the residue at 26.2 Hz reads -70.11 dB on the default sweep, -46.32 on the short one and -86.67 on the long one, level-invariant — a ratio — and a column up it is -103.6 dB, under every real stamp. The sweep length is not stored, so the record takes the worst (loudest) residue per column over the offered lengths, -46.32 dB at that column; the residue is computed at read time from the record’s own geometry and cached per grid:

| $j$ | $f_j$ (Hz) | 2 s    | 5 s    | 10 s   | worst  |
|-----|------------|--------|--------|--------|--------|
| 0   | 20         | -21.63 | -27.74 | -34.13 | -21.63 |
| 1   | 26.2       | -46.32 | -70.11 | -86.67 | -46.32 |
| 2   | 34.33      | -74.85 | -103.6 | -117   | -74.85 |
| 3   | 44.99      | -92.87 | -117.2 | -129.3 | -92.87 |
| 4   | 58.94      | -104   | -127.2 | -140.3 | -104   |
| 5   | 77.23      | -114.2 | -136.3 | -150.4 | -114.2 |
| 6   | 101.2      | -123   | -144.4 | -160   | -123   |
| 7   | 132.6      | -130.2 | -152.1 | -169   | -130.2 |

The composition is per cell,

$$m[i][j] = \min\left(20 \log_{10} \text{thd}[i][j] - F_{\text{rig}}[i][j], 20 \log_{10} \text{thd}[i][j] - R[j]\right), \quad (36)$$

the [composed margin](#); a cell with no rig-noise margin keeps nil, a column with no residue keeps the rig-noise margin.

**The peak tile.** The truncated and edge columns masked, the cells whose presence reads absent dropped, the map-level verdict first (absent  $\Rightarrow$  no percentage at all, only the median depth). Pass one: a cell is *clear* when it is unmasked, measured, and its composed margin reaches the minimum margin — every unmasked measured cell when there are no margins. Pass two: a clear cell is a *corroborated peak* when the cell one column either side, the same level a third of an octave away, is also clear. The shoulder is asked for in frequency and never in level: a nonlinearity acting at one drive acts on the neighbouring notes too and THD moves slowly across a third of an octave, while across one level step it can move by tens of dB, so a genuine onset peak at the loudest level may have no clearing level neighbour — and a noise pair in adjacent levels of a single-track column is as likely as any other pair (measured: two of five bare-loop nulls paired that way). The tile is the loudest corroborated cell as a peak; else the loudest unmasked cell as a bound, quoted with  $\leq$ ; else nothing. Without floor information — Simulation Mode, a record with no calibration — every unmasked measured cell counts as clear and corroborated, and the raw maximum is stated as a peak: reported, not changed.

**The cleanup reading.** A column's  $(\ell_i, \text{thd}[i][j])$  pairs, ascending, cells without a THD dropped, read as one of three states: the first point already at or above the threshold is *never clean* (within this map's range); the first  $i$  with  $\text{thd}_i \geq \tau$  gives *cleans up* at

$$\ell_{i-1} + \frac{\tau - \text{thd}_{i-1}}{\max(\text{thd}_i - \text{thd}_{i-1}, \epsilon)} (\ell_i - \ell_{i-1}), \quad (37)$$

else *always clean* — the endpoint of the measured range is never dressed up as a crossing. The threshold is a default-argument literal in the kernel, read for the manifest by bisecting the shipped initializer;  $\epsilon$  is a literal guard. The tile reads the column nearest, in log frequency, the Compression probe's own note, so the two records' cleanup figures are about one pitch, and it is gated on the presence verdict. The summaries' journey fallback, used only when the Compression kind supplied no cleanup, reads the column nearest the same note in *linear* frequency with at least three levels, the same interpolation and a fixed confidence (all literals). The page adds one rule outside the kernel: a never-clean tile is scoped to the map's own drive range in its subtitle, and when the pedal's variant-matched Compression record finds a genuine crossing below the map's floor the subtitle points there.

## 6.5 Parameters

Every value below is read from the manifest, which read it from the shipped code; the residue capture's drive, the summaries' reference note and confidence and level count, the crossing guard, the calibrator's grid start, point count and cut, and the chain gain's grid are literals the manifest carries as rule strings.

| Parameter                                          | Value                         |
|----------------------------------------------------|-------------------------------|
| <b>Drive windows</b>                               |                               |
| hardware window (floor / ceiling / levels)         | -48 / -12 dBFS / 8            |
| plugin window (floor / ceiling / levels)           | -60 / 0 dBFS / 8              |
| hardware lattice step                              | 5.143 dB                      |
| <b>Plan</b>                                        |                               |
| sweep band $f_1 - f_2$                             | 20 Hz – 10 000 Hz             |
| sweep length (default) / offered                   | 5 s / 2, 5, 10 s (not stored) |
| actual duration / rate constant $L$ at the default | 4.97169 s / 0.8 s             |
| pre-roll / tail silence                            | 0.3 s / 0.5 s                 |
| sample rate $f_s$ (the manifest's)                 | 96 000 Hz                     |



| Parameter                                                        | Value                                                        |
|------------------------------------------------------------------|--------------------------------------------------------------|
| sweep samples / pre-roll / tail at $f_s$                         | 477 282 / 28 800 / 48 000                                    |
| analyzer: harmonic count $K$ / response DFT $M$ / half-width cap | 9 / 32 768 / 0.2 s                                           |
| export grid columns                                              | 24 (20–10 000 Hz inclusive)                                  |
| <b>Summary</b>                                                   |                                                              |
| integration band                                                 | 20 Hz – 20 000 Hz                                            |
| edge mask guard ratio (a quarter tone)                           | 1.0293022                                                    |
| masked columns on the default grid (bottom / top / truncated)    | 0 / 23 / none                                                |
| <b>Floor</b>                                                     |                                                              |
| stamp median window (the Gain Map) / per-harmonic path           | 7 points / 1 (verbatim)                                      |
| calibration analyzed sweeps / level                              | 1 / -26 dBFS                                                 |
| calibration sweep (the app's)                                    | 20 Hz to round( $0.45 f_s$ ), 10 s, 0.5 s before / 1 s after |
| residue capture drive                                            | -26 dBFS (a literal; the read is a ratio)                    |
| <b>Peak tile</b>                                                 |                                                              |
| minimum margin                                                   | 6 dB                                                         |
| <b>Cleanup</b>                                                   |                                                              |
| threshold (bisected from the shipped initializer)                | 0.01 (1 %)                                                   |
| tile column (nearest in log frequency)                           | 220 Hz                                                       |
| summaries' fallback: reference / minimum levels / confidence     | 220 Hz (linear nearest) / 3 / 0.7 (literals)                 |
| crossing interpolation guard                                     | 1e-12 (a literal)                                            |
| <b>Oracle (<code>analysisdump journey-synth</code>)</b>          |                                                              |
| ladder quadrature points                                         | 262 144                                                      |

## 6.6 The result

The app's own rendering of the same measurement kind, from the user guide's figure pipeline (Figure 24).

## 6.7 Worked example and parity

The [parity test](#) runs 15 cases through both implementations and two device cases again as their [phantom capture](#) twins (2 twins). The Python imports chapter one's reimplementations for the sweep, the device, the phantom, the noise and the analyzer, and adds only what the map adds. Every case is a known device through a known LOOP: a delay, one seeded noise stream referred to the interface's input, and a chain gain applied after the device and the noise, as an input-gain knob is; the calibration is the app's own sweep on the identity loop through the same gain and noise, analyzed by the shipped calibrator, whose *measured* latency cuts every level — the app's flow. Beside every cell the oracle prints its truth: the memoryless device's closed-form ladder at that level's drive (chapter one's quadrature, one pass for all orders), scaled by a pre-filter's magnitude at  $f_j$ , a post-filter's at  $kf_j$  and the chain gain, and the truth THD as (34) applied to that ladder; and the read-time half is the shipped predicates run on a summary built from the truth cells — the definition applied to the ladder, never a second derivation.

The instrument every truth comparison rests on is the [leakage map](#) (Figure 21): the pure phantom [1], whose truth is zero at every order above the first, read on the journey's geometry. Its higher-order cells are the analyzer's own leakage into each order's window per column — -70.13 dB into

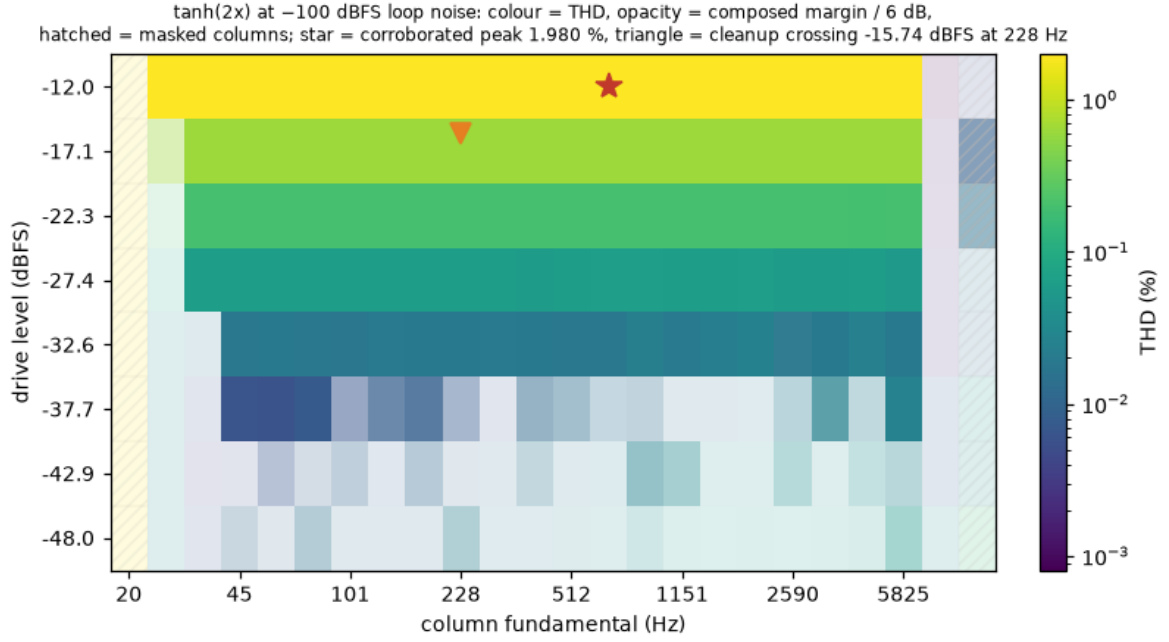


Figure 19: The worked example's map:  $y = \tanh(2x)$  on the hardware lattice through a loop with seeded noise, every cell coloured by its THD and drawn at an opacity that follows its composed margin, the two edge columns hatched, the corroborated peak starred and the cleanup crossing marked at the tile's column.

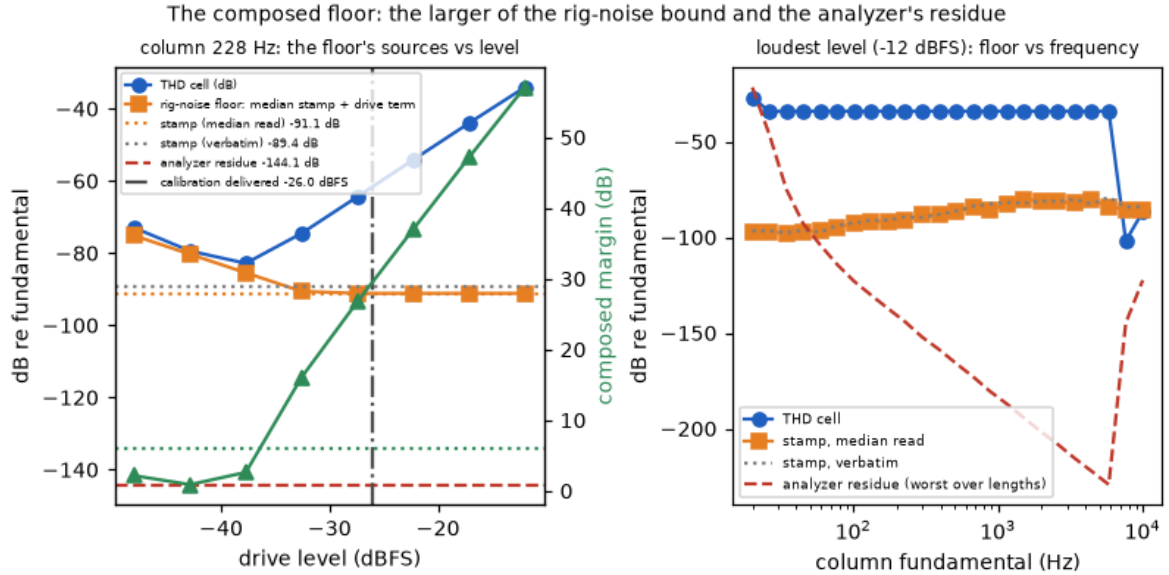


Figure 20: The composed floor at one column against level (left): the stamp read verbatim and through the median, the drive term rising below the calibration's delivered level and clamped above it, the analyzer's residue, and the composed margin against the minimum; and at the loudest level against frequency (right).



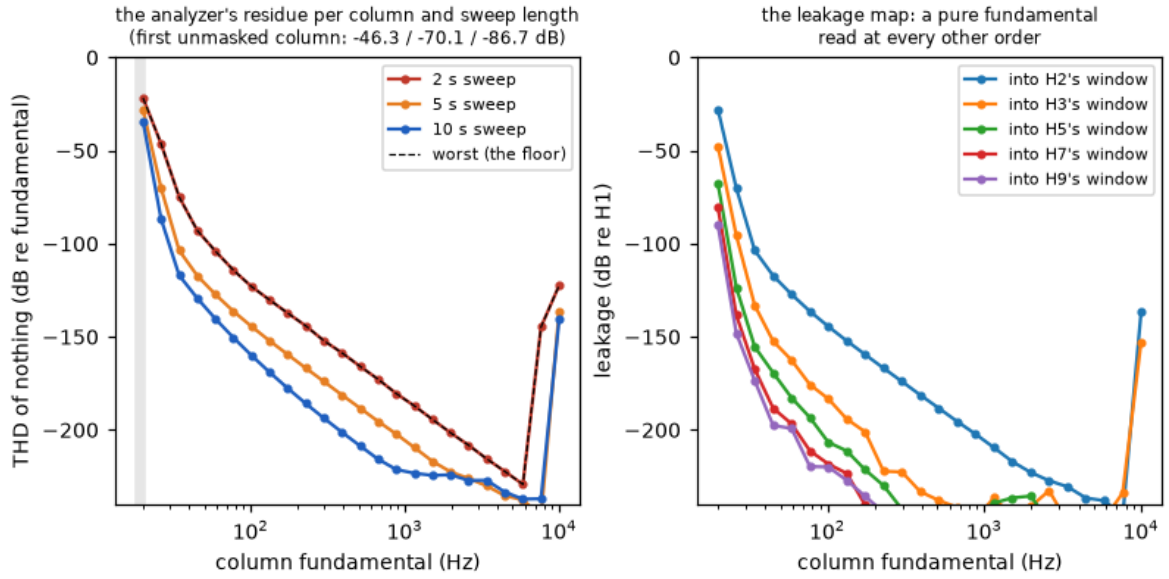


Figure 21: The analyzer's residue per column at the three offered sweep lengths (left) — the fade-in leak at the first unmasked column and the worst envelope the record takes — and the leakage map (right): a pure fundamental read at every other order, the parity's own allowance.

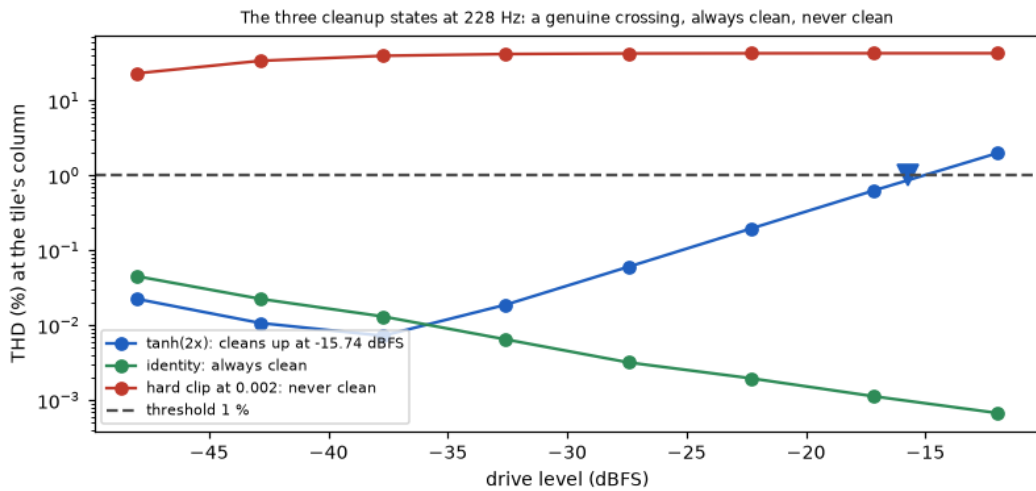


Figure 22: THD against level at the tile's column for the three cleanup states: a genuine crossing interpolated between two levels, a column under the threshold throughout, and a clipper whose threshold sits under the lattice's floor.

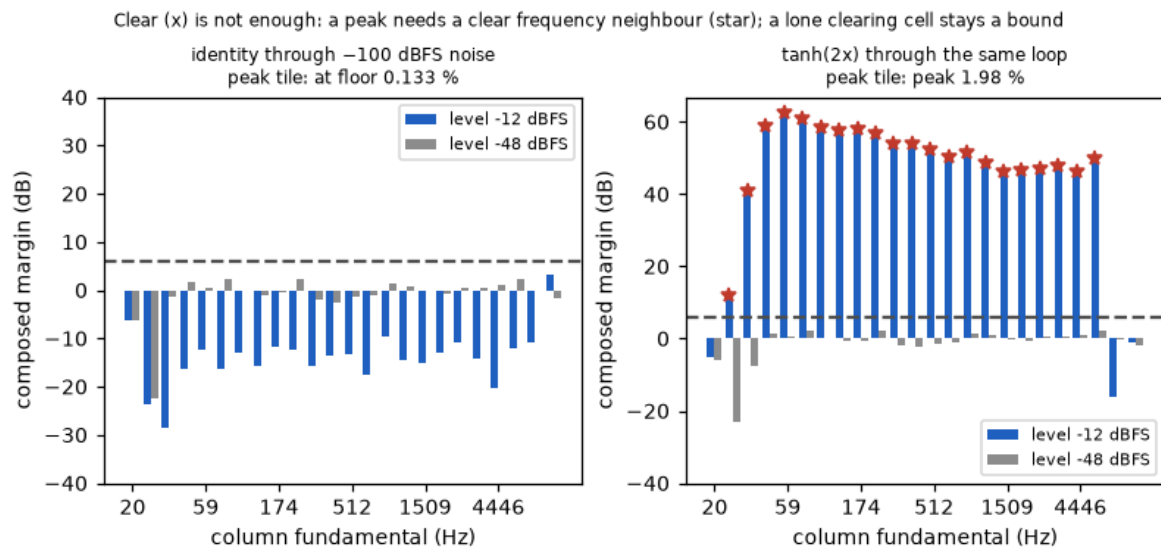


Figure 23: The corroboration rule: composed margins per column at the loudest and the quietest level for the identity through noise (left, nothing clears and the tile is a bound) and for the soft clipper through the same loop (right, every clear cell has a clear neighbour and the tile is a peak).

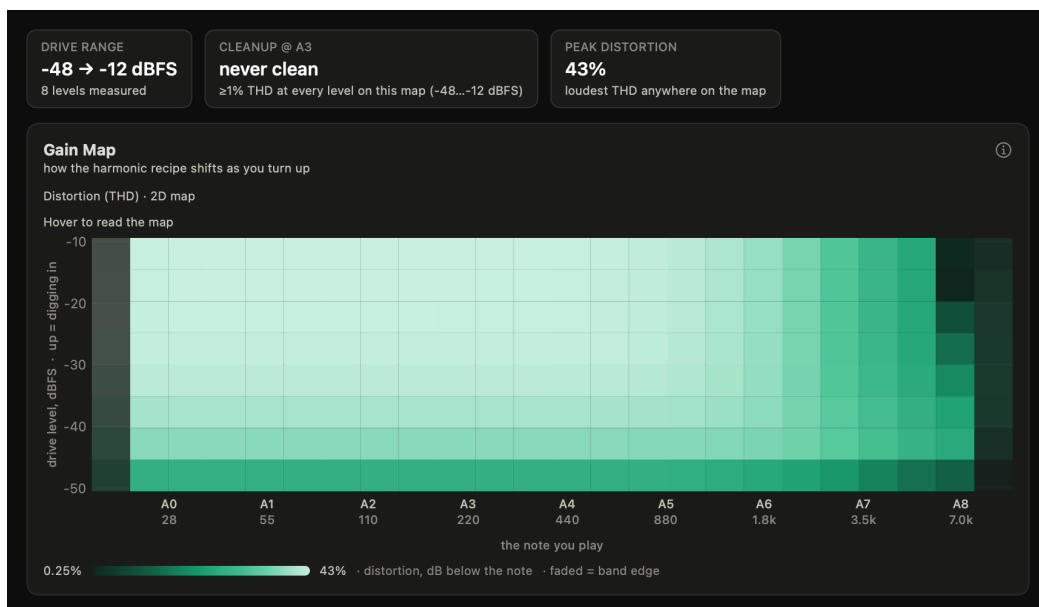


Figure 24: The Gain Map view as the app draws it, from the guide's simulated fixture: the THD map over note and level, the outermost columns drawn faded, quiet cells dimmed by their margin.

$H_2$ 's at the first unmasked column, -95.55 into  $H_3$ 's, -123.5 into  $H_5$ 's; a column up, -133.1 into  $H_3$ 's:

| $k$ | 26.2 Hz | 34.33 Hz | 44.99 Hz | 58.94 Hz | 173.7 Hz |
|-----|---------|----------|----------|----------|----------|
| 2   | -70.13  | -103.6   | -117.2   | -127.2   | -159.5   |
| 3   | -95.55  | -133.1   | -152.3   | -162.5   | -201     |
| 4   | -112.1  | -145.6   | -163.2   | -176.4   | -217.4   |
| 5   | -123.5  | -155.3   | -169.3   | -182.7   | -221.4   |
| 6   | -131.9  | -162.7   | -181.1   | -194.2   | -230.5   |
| 7   | -138.3  | -167.3   | -188.3   | -196.7   | -240.2   |
| 8   | -143.6  | -171.3   | -191.5   | -196.7   | -233.4   |
| 9   | -148    | -173.5   | -197.6   | -199.2   | -235.6   |

A cell whose truth stands  $c$  dB above the leakage landing in its window can read up to  $20 \log_{10}(1 + 10^{-c/20})$  from the truth by the analyzer alone — 0.27 dB at  $c = 30$ , 0.0864 at 40 — and that allowance is subtracted before the bar is applied; what remains is the method's own error. The test asserts:

- Swift against Python: every header quantity and every content cell in decibels within 1e-06 (measured maximum 5.1e-08 on this run) on a capture with no filter in it, and within 0.0001 (measured 1.8e-06) through a biquad; every discrete reading — mask, presence, clear, corroborated, every tile's state, every column's cleanup state — identical; the stamp point for point; the residue per column per length wherever it is a reading (a column two hundred decibels down is dust on both sides, stated and not compared, as is a harmonic cell whose truth sits more than 140 dB under the fundamental);
- the shipped read against the ladder on every noiseless case, after the allowance: within 0.05 for the smooth devices (measured residue 0.012), within 0.05 for the twins (0.04), within 0.3 through a pre- or post-filter (0.15 — the truth treats each column's fundamental as a steady tone through the filter, and a sweep is not steady); the raw error tracks the bound: at 30 dB of clearance the worst cell reads 0.179 dB against the bound's 0.27, at 40 dB 0.0681 against 0.0864;
- the THD cells the same way, the allowance built from the residue at that column for the case's own sweep length: within 0.001 (measured 6.4e-05) and 0.2 (0.09);
- the cleanup crossing per unmasked column against the truth curve's crossing within 0.3 dB (measured 0.197, at the first unmasked column in every case, where the residue's  $H_2$  leak sits inside the summed THD);
- the identity through noise at the three offered lengths: the residue at the first unmasked column reads the law's three figures from the Python's own analysis (-46.32 / -70.11 / -86.67 against -46.3 / -70.1 / -86.7, within 0.05), every cell's fundamental present, the peak tile a bound (0.133 % at the default length), the cleanup tile always clean, and no cell in the loudest row quoted clear;
- the #100 term: the same device and noise through a loop 12 dB louder moves  $H_1$  by 12 dB, the stamp's chain gain to 12, and no composed margin by more than 5e-11 dB (bar 1e-06);
- a capture 100 samples late: the calibrator finds the latency and every harmonic cell moves by 0 dB (bar 1e-09);
- Part B on the corpus grids: the Python's tiles against the kernel's, the peaks within 0.1 dB and the bounds to a relative 1e-09.

Part B’s parity is established: on every read-time case — the identity through noise at the three sweep lengths, the device with no fundamental, the never-clean clipper, the loop 12 dB louder, the operative floor on a silent stamp, and the plugin window — the two implementations agree on every margin, every flag, every tile and every verdict, and on the corpus grids the reimplementations reproduces the kernel’s own tiles to the digit.

The device cases, each with its content-cell count, its Swift-vs-Python figure, its worst raw error and where, its residue after the allowance, its worst raw error at the two clearance bars, and its THD residue:

| case          | cells | S–P (dB) | raw (dB) | at       | residue (dB) | 30 dB bar | 40 dB bar | THD residue |
|---------------|-------|----------|----------|----------|--------------|-----------|-----------|-------------|
| a-tanh        | 426   | 5.1e-08  | 0.348    | H5 (4,2) | 0.012        | 0.0978    | 0.0681    | 4e-08       |
| a-tanh-twin   | 479   | 1.6e-08  | 0.341    | H5 (4,2) | 0.0051       | 0.0981    | 0.068     | 4e-08       |
| a-poly        | 456   | 3.5e-08  | 1.33     | H3 (0,2) | 0.00037      | 0.17      | 0.0534    | 6.4e-05     |
| a-poly-twin   | 504   | 1.6e-08  | 1.52     | H3 (0,2) | 0.04         | 0.179     | 0.0545    | 1.7e-05     |
| j-latency     | 426   | 5.1e-08  | 0.348    | H5 (4,2) | 0.012        | 0.0978    | 0.0681    | 4e-08       |
| k-tanh-prehp  | 421   | 1.2e-06  | 0.215    | H5 (5,2) | 0.15         | 0.215     | 0.155     | 0.09        |
| k-poly-postlp | 456   | 1.8e-06  | 1.34     | H3 (0,2) | 0.00038      | 0.169     | 0.0544    | 7.1e-05     |

The read-time cases on the Swift side — the tiles the page would print — with the two implementations’ agreement:

| case              | peak tile           | value (%) | cleanup tile | level (dBFS) | absent | S–P (dB) |
|-------------------|---------------------|-----------|--------------|--------------|--------|----------|
| b-identity-2s     | at floor ( $\leq$ ) | 0.4834    | always clean | –            | 0      | 1.6e-08  |
| b-identity-5s     | at floor ( $\leq$ ) | 0.1332    | always clean | –            | 0      | 1.6e-08  |
| b-identity-10s    | at floor ( $\leq$ ) | 0.1245    | always clean | –            | 0      | 1.6e-08  |
| c-absent          | no fundamental      | –         | none         | –            | 1      | 1.6e-08  |
| d-never-clean     | peak                | 43.69     | never clean  | –            | 0      | 1.6e-08  |
| e-tanh-gain0      | peak                | 1.98      | cleans up    | -15.74       | 0      | 5.1e-08  |
| e-tanh-gain12     | peak                | 1.98      | cleans up    | -15.74       | 0      | 5.1e-08  |
| f-operative       | peak                | 1.98      | cleans up    | -15.74       | 0      | 5.1e-08  |
| g-plugin-hardclip | peak                | 41.57     | cleans up    | -25.05       | 0      | 1.6e-08  |

The worked example,  $y = \tanh(2x)$  noiseless on the hardware lattice: 426 content cells compared, the worst raw error 0.348 dB at H5 (4,2) and 0.012 dB after the allowance, 0.0978 dB at the 30 dB bar and 0.0681 at 40, a THD residue of 4e-08 dB; its peak tile reads 1.979% against the truth grid’s 1.979%, and its cleanup tile -15.74 dBFS at 227.6 Hz against the truth curve’s -15.74 (a difference of 0.044 dB). The device with no fundamental — a phantom whose fundamental sits 80 dB under its second harmonic — reads a majority verdict of absent (0.79 of the cells) at a median depth of 80.03 dB and a tile with no percentage; the cells beside the low edge read present, because the fundamental’s window there holds the second harmonic’s own leakage. The operative-floor case: the render-path stamp reads -101 dB at its loudest point, the stated floor enters at -76.99 dBFS, and the peak reads 1.98%. The plugin window with the hard clipper: a peak of 41.57% and a crossing at -25.05 dBFS. The never-clean clipper: 43.69% at every level of its tile column.

Part B on the corpus — the kernel test’s own transcribed grids and stamps, the read-time half’s one stored-input exception (a stored record holds results, not a capture, but for the read-time half the stored grid *is* the input, and these grids are the store’s own figures pinned to the digit by the kernel’s test):

| record | tile                | Python (%) | kernel (%) |
|--------|---------------------|------------|------------|
| 59     | at floor ( $\leq$ ) | 0.964566   | 0.964566   |
| 428    | at floor ( $\leq$ ) | 0.778677   | 0.778677   |
| 65     | peak                | 11.2116    | 11.2116    |
| 446    | peak                | 6.21905    | 6.21905    |
| 404    | at floor ( $\leq$ ) | 0.130628   | 0.130628   |
| 410    | at floor ( $\leq$ ) | 0.106096   | 0.106096   |
| 416    | at floor ( $\leq$ ) | 0.119273   | 0.119273   |
| 422    | at floor ( $\leq$ ) | 0.130136   | 0.130136   |
| 464    | at floor ( $\leq$ ) | 0.105552   | 0.105552   |

## 6.8 Limits and failure modes

- **The sweep length is not stored, and the floor pays for it.** The residue is taken as the worst over the offered lengths, so a record captured at the default length carries the short sweep's residue at the first unmasked column, -46.32 dB against its own -70.11: that column's margin is about 24 dB pessimistic — a bound in the honest direction, and a stored length would sharpen it (a stored-format decision, not a read-time one).
- **The fundamental grid is a level, not a normalizer.** The cells are uncompensated, so  $h[1][i][j]$  is the fundamental as the device delivered it through the loop, gain and all. On a device with a series resistance ahead of its clipper it is also a load gauge: the same reference position measured on two interfaces with different input impedances delivers a different fundamental, and every ratio to it is taken against that delivered level. The map does not separate the device's gain from the loop's, and it does not claim to.
- **The analyzer's separation is geometry-dependent, and the journey's geometry is the coarser one.** The default sweep is half the fingerprint sweep's length and the response DFT half its length, so the leakage of the loudest order into each window at the low edge is worse than chapter one's measured bound; the leakage map states it per column and order, and a cell whose content sits inside it is the analyzer's, not the device's — which is what the residue and the margins say in a real capture, and why a device with no fundamental reads present beside the low edge.
- **Aliasing in a simulated device.** A device applied per sample folds its content above Nyquist onto the reads; the smooth devices' residues above are inside the allowance, the phantom twins are the alias-free control, and a kinked device (a hard clipper at playing levels) would not be the oracle for a cell near the top of a band.
- **A raw maximum is a peak where no floor exists.** Without a calibration or an operative floor every unmasked cell counts as clear and corroborated, and the tile states the map's loudest cell as a peak — Simulation Mode and every record with no calibration are in that state; reported, not changed.
- **The per-harmonic path reads the stamp verbatim.** The §3 distance and the fundamental-presence margin read the calibration stamp point by point; the single-read exposure the median read removes from this map exists there and is reported, not changed — moving every distance figure in the library is its own measured change.
- **The audition models invent.** The per-level fits are consumed by the listening panels only, and a render between two levels blends a clamped quieter fit into the louder one; nothing measured reads a fit, and the invention's size is a later chapter's measurement.

- **The rows are separate captures.** Each level is its own sweep seconds apart; a circuit whose operating point moves between them shows it as row-to-row wobble, which the map draws and does not name.

## 7 Glossary

Generated from `Docs/Methods/terms.yaml`; the plain-language companion prints the same entries in plain language.

**alias image** Content above half the sample rate folds back below it in a sampled capture; order  $k + 1$ 's image collides with the read of order  $k$  once  $f_0 > f_s/(2k + 1)$ .

**analysis lattice** The set of bins the two tones and every product occupy: with tone bins  $k_1 = ga$  and  $k_2 = gb$  ( $a : b$  coprime), the product  $(m, n)$  lands on bin  $mk_1 + nk_2$  exactly, so every lattice point is a multiple of  $g$  and the closest two products can come is  $g$  bins. Chosen once by the ratio search and scaled to each grid.

**analysis window ( $N$ )** Exactly  $N$  captured samples, rectangularly windowed, starting after the fade-in and the settle and ending before the fade-out;  $N$  is the longest power of two under the ceiling whose segment fits the steady state, and the stimulus is sized to hold it —  $2^{21}$  samples (21.8 s) at 96 kHz.

**analyzer residue** The THD the shipped analyzer reads with nothing in the loop — the plan's own sweep handed back as its capture — per column of the record's grid, the worst over the sweep lengths the sheet offers, computed at read time from the record's geometry. One column above the sweep's start the fade-in ramp leaks coherently into the second-harmonic window, and a noise floor cannot see a leak that scales with the signal.

**audition model** The per-level Hammerstein fit the record carries beside the grid, consumed by the Hear-it panels only: no measured quantity reads it, and a render between two of its levels invents a bracket-shaped clipping the device does not have.

**averaged cycle** The stimulus and the capture folded onto one period of the drive note in  $n$  equal phase bins, each bin holding the mean of every sample whose phase fell in it, from the skipped opening cycles to the end of the tone; the figure's two coordinates are the two bins' means.

**box normalization** Every loop area is divided by  $4 x_{\max} y_{\max}$ , the rectangle that bounds the figure, so the areas are dimensionless and a pure ellipse from a phase shift  $\varphi$  reads  $(\pi/4) |\sin \varphi|$ .

**calibration stamp** The embedded calibration's THD-versus-frequency curve: one read of the loopback's own distortion-plus-noise at each of a log-spaced set of fundamentals, measured once at the calibration's own drive, stored in every record.

**capture floor** The read-time summary of a capture's empty bins: the upper median of the levels of every product judged not present and not refused on lattice position or band, with the spread above it. It bounds what the capture could see and can change no product's verdict — one direction, no feedback.

**cell** One (level, column) of the summary grid: a THD figure over the declared band and one figure per stored harmonic order, each read from that level's own analysis at the column's fundamental.

**cleanup reading** A column's THD-versus-level curve read as one of three states: the interpolated level at which it first reaches the threshold (cleans up); at or above the threshold already at the quietest level (never clean, within this map's range); under it at every level (always clean). The tile reads the column nearest the Compression probe's note.

**clipping** A memoryless nonlinearity that limits the output amplitude: the output follows the input up to a threshold and is compressed or held beyond it (hard: a corner into a flat ceiling; soft: a gradual bend). An odd-symmetric clipper produces odd harmonics only; any asymmetry between the two half-cycles adds even harmonics.

**coherence** ( $\Gamma$ )  $|\sum_i z_i| / \sum_i |z_i|$  over lagged products  $z_i = H(b_i + \text{lag}) \overline{H(b_i)}$ : shaped content is a phasor whose phase advances smoothly with frequency, noise scatters isotropically, and the statistic is  $1/\sqrt{n}$  for pure noise.

**coherence check** Whether the played tones stayed stationary across the window: the worst of the detector bins beside each tone, read against the local floor beside that tone, against a measured 30 dB bar — interpreted only over the offsets where the tone’s own skirt outweighs the loudest product’s, and only as a verdict about the chain, never a culprit.

**coherence detector** The raw spectrum at offsets  $\pm 1 \dots \pm 9$  bins beside each played tone, stored at capture with the local floor beside the tone; the widest offset is derived from the narrowest resolvable spacing ( $13 - 1 - 3$ ), and every offset is read because an even-cycle wander lights the even ones only.

**coherent sampling** Both tones complete a whole number of cycles in the analysis window, so a rectangular window leaks nothing: every product sits on one bin and the floor beneath a record is noise, not a window’s skirt. The premise has no margin — a tone that drifts during the capture spills a  $1/(\pi m)$  skirt across the lattice.

**composed margin** A cell’s THD in dB over the applicable floor, the floor being the larger of the rig-noise bound (the median-read stamp with its drive term, or the operative floor) and the analyzer residue — the larger floor is the smaller margin. Nil where neither rig source is usable.

**corroborated peak** A cell that stands clear of its floor by the minimum margin AND whose neighbour one column either side — the same level, a third of an octave away — also stands clear; a real peak has a shoulder in frequency, a floor-riding read has none. Corroboration is never asked for in level.

**coupling capacitor** A series capacitor at a device’s output: a first-order high-pass whose corner sits far below the probe note, so it removes the output’s mean and leaves every harmonic’s magnitude essentially untouched — an asymmetric clipper’s unequal clip levels are centred by it while its even harmonics survive.

**dBc** Decibels relative to a carrier — here the fundamental, or the louder of two played tones — so that a product’s level is stated as a ratio to the signal that produced it.

**dBFS** Decibels relative to digital full scale:  $20 \log_{10}$  of a linear amplitude in which 1.0 is the largest value the converter or file can represent.

**decibel (dB)** A logarithmic ratio:  $20 \log_{10}$  of an amplitude ratio (equivalently  $10 \log_{10}$  of a power ratio). +6 dB doubles an amplitude, −20 dB is one tenth, −40 dB one hundredth; the reference is named by a suffix or by context.

**deconvolution** Dividing the captured response’s spectrum by the sweep’s spectrum — here, multiplying by the analytic inverse filter — to recover the device’s impulse response, in which the harmonic pulses appear at their time advances.

**delivered fundamental** A cell’s drive level plus its stored  $H_1$ , in absolute dBFS. The journey’s harmonic cells are uncompensated — the loop’s gain is in them — so this is the level the fundamental actually reached at the converter, the quantity the floor’s drive term compares against.



**derotation** Multiplying every harmonic response by  $e^{+j2\pi f\delta/f_s}$  on its own output-frequency axis to remove a capture delay  $\delta$  — a unity-magnitude rotation, so magnitudes are untouched at every bin.

**drive level** The digital level of one level’s sweep,  $20\log_{10} A$  in dBFS, where  $A$  is the sweep amplitude the device is driven with. On a hardware record it is the interface’s output level; on a plugin record it is digital full scale itself. The lattice is a list of these.

**drive term** The floor’s adjustment for a cell delivered quieter than the calibration: the stamp’s ratio rises by the difference, because the rig’s noise is absolute; for a cell delivered louder it stays at the stamp’s own ratio — never extrapolated past what the calibration measured, since the rig’s own distortion rises with drive.

**drive window** The floor and ceiling of the lattice. The hardware window’s ceiling is the loudest drive that cannot clip the converter once the input is staged for the pedal, its floor a fixed travel below; the plugin window’s ceiling is digital full scale (the float render path has no converter) and its floor is lower, where high-gain preamp knees sit.

**edge mask** The two columns that sit within a quarter tone of the grid’s edges — the sweep’s fade-in and fade-out ramps — masked from the tiles and the presence verdict; derived from the grid alone, because the artifact reproduces on a jumpered wire.

**excited band** The range of fundamentals the sweep drove at full amplitude: from the instantaneous frequency where the fade-in ends to the one where the fade-out begins. Outside it the device was driven under a ramp and a read describes the measurement, not the device.

**export grid** The summary’s frequency columns: a fixed count of fundamentals log-spaced from the sweep’s start to its end inclusive, so the first and last columns sit at the sweep’s edges, inside its fade ramps.

**extraction window** ( $T_k$ ) The raised-cosine window that cuts order  $k$ ’s pulse out of the impulse response; its half-widths are half the gap to each neighbouring pulse, capped, and its duration sets the order’s spectral resolution.

**frame fit** The single time shift  $\Delta l$  between the transfer capture’s alignment and the paired record’s, found by maximizing the magnitude-weighted phase coherence of every fittable harmonic; its residual over the harmonics is the pairing statistic.

**fundamental** ( $f_0$ ) The excitation frequency whose harmonics are being read; in a swept measurement every  $f_0$  in the band is excited in turn, and the  $k$ -th harmonic of  $f_0$  is read at the output frequency  $kf_0$ .

**fundamental ellipse** The least-squares ellipse through the trajectory: the projection of both cycles onto the fundamental. Its area is the linear-phase loop — what a filter’s rotation of the fundamental contributes, never memory.

**fundamental presence** Whether  $H_1(f_0)$  is a measurement at all: absent when it sits at or beyond the analyzer’s measured separation bound below the loudest valid in-band order, or at or below its rig-floor margin. Where absent, no ratio to it is quoted.

**gain map** The level-resolved harmonic measurement (§6.2’s gain journey): the Harmonic Distortion sweep repeated at each drive level of a lattice, each level analyzed on its own, and the results summarized on one export grid as  $\text{thd}[i][j]$  and  $H_k[i][j]$  — a map of the device’s distortion against level and note.

**harmonic order** ( $k$ ) The integer multiple of the fundamental at which a harmonic sits; order 1 is the fundamental itself. Odd orders are produced by symmetric nonlinearities, even orders require asymmetry.

**harmonic pulse** In the deconvolved response, the impulse response belonging to one harmonic order, arriving  $L \ln k$  seconds before the linear one; each is cut out with its own window.

**harmonic response** ( $H_k(f)$ ) The complex, input-normalized amplitude with which the device emits energy at output frequency  $f$  as its  $k$ -th harmonic; the harmonic of a fundamental  $f_0$  is read at  $f = kf_0$ .

**harmonic role** The tiles' partition of the resolved products by what each does to the requested chord: the recipe  $(m, n)$  is placed at index  $ma + nb$  on the harmonic grid of the interval's simplest 5-limit just ratio  $a : b$ , its pitch class read off the odd part of the index; on a played note's class it thickens (at or above the low note) or adds a sub-octave; on a consonant other class it adds colour; elsewhere, or between frets, it clashes. Classified at the requested interval, measured at the lattice.

**harmonic shaping** The read-time predicate deciding whether a harmonic read is content the circuit deterministically shaped or noise that landed in that order's window, from the coherence of the complex response along frequency.

**impulse response** ( $h[n]$ ) The deconvolved response, circular over the analysis length  $N$ : the linear pulse at index 0 and order  $k$ 's pulse at  $N - \text{round}(L \ln k \cdot f_s)$ .

**input-normalized** Divided by the sweep amplitude  $A$ , so that a unit-gain linear device reads  $H_1 = 1$  (0 dB) and every  $H_k$  is a ratio to the input, not an absolute level.

**instantaneous frequency** ( $f(t)$ ) The frequency the sweep is passing through at time  $t$ :  $f(t) = f_1 e^{t/L}$ , the derivative of the phase divided by  $2\pi$ .

**intermodulation product** ( $mf_1 + nf_2$ ) A component the nonlinearity places at  $mf_1 + nf_2$  for integers  $m, n$ ; its recipe is the pair  $(m, n)$  and its order  $|m| + |n|$  is the lowest polynomial power that can produce it. Read at one bin, exactly, and quoted in dBc re the louder played tone.

**inverse filter** ( $\tilde{X}^{-1}(f)$ ) The closed-form stationary-phase inverse of the synchronized sweep,  $2\sqrt{f/L} \exp(j(\pi/4 - 2\pi fL(1 - \ln(f/f_1))))$ , tapered to zero below  $f_1$ ; using the closed form extrapolates the phase law above  $f_2$ , so harmonics above the sweep's top deconvolve cleanly.

**just ratio** The simplest ratio  $a : b$  with every factor 2, 3 or 5 within 20 cents of the requested 12-TET interval, found by search (3 : 2 for the fifth, 5 : 4 for the major third): the harmonic grid the roles are placed on has fundamental  $f_1/a$ , 55 Hz for A2 + E3.

**lattice snap** Placing the two tones on bin centres so that a whole number of cycles of each fits the window: the reduced ratio is chosen once against every standard grid (no collisions, spacing at or above the bar, both tones inside a  $\pm 15$  cent budget, then the interval nearest the request) and scaled to the grid at hand; a grid too coarse for it falls back to the widest collision-free lattice in budget, and a request that cannot escape exact degeneracy is refused before anything plays.

**lattice spacing** ( $g$ )  $\text{gcd}(k_1, k_2)$ : the integer combinations of the two tone bins are exactly the multiples of  $g$  (Bézout), so no product can sit nearer than  $g$  bins to another or to either tone. 22 bins on the shipped fifth at 96 and 48 kHz, 24 at 88.2 and 44.1 kHz.

**leakage map** The parity’s own instrument: the pure phantom [1] — a fundamental and nothing else — through the journey’s geometry, whose every higher-order read is the analyzer’s leakage into that order’s window at that column; the allowance every truth comparison subtracts.

**level lattice** The drive levels a journey measures, evenly spaced in dB from a floor to a ceiling inclusive, one sweep per level; the hardware and plugin windows set the two ends and the level count.

**local floor** The capture-time noise estimate beside one bin: the median magnitude over the bins from 3 out to about  $\pm 8.8$  Hz out on either side, with every bin the lattice claims — products, tones, DC, the detector’s bins — removed by name. A product is called present when it stands more than four times (12.04 dB) above it; the verdict is stored and never revisited.

**measurement floor** The level below which a read is the rig rather than the device, derived at read time from the embedded calibration’s pooled distortion stamp with per-track, drive and averaging adjustments; a read at or below it is reported as a bound.

**median read** How the Gain Map reads the calibration stamp: each point replaced by the running median of its neighbours over an odd window before the interpolation, so a single-read trough cannot set the floor; the per-harmonic path reads the stamp verbatim (a window of one).

**memory verdict** The three-way reading taken on the nonlinear loop area (of the compensated figure when the phase removal is accepted): memoryless-ish below the first threshold, memory effects at and above the second, and a deliberately hedged band between them, where tone filtering ahead of the clipper rotates every harmonic by  $k$  times the drive’s rotation and a single fundamental cannot separate that from mild memory. Both thresholds are literals in the shipped verdict, which the manifest reads by bisection.

**memoryless lattice** The set of harmonic phases  $\{0, \pi\}$  (in the cycle’s convention, after the Novák factor  $(k - 1)\pi/2$ ) at which a memoryless device’s harmonics sit; each measured harmonic’s excess over its nearest lattice point, folded consistently from low order to high, is the filter burden the removal takes out.

**methods manifest** A machine-readable JSON file, emitted by the shipped measurement kernel through `analysisdump manifest`, listing every parameter that determines a measurement’s result — each value obtained by calling the function that owns it. The documents’ tables are generated from it and CI fails when the committed copy drifts from a fresh one.

**nonlinear loop area** The unsigned lobe area of the residual trajectory after the fundamental ellipse’s output is subtracted — the harmonic content’s own lobes, the headline memory figure the verdict is taken on.

**operative floor** A plugin record’s own noise bound: the larger of its silence render’s rms and its A/A residual, stamped at run initiation; it enters the floor as an absolute dBFS bound with no chain-gain term, and governs where the render-path calibration is numerically silent.

**orientation resolution** A plugin-path capture is aligned by correlation against a tone and may land half a cycle off; when the cycle’s fundamental phase sits on the  $\{0, \pi\}$  lattice and the paired polarity confidently disagrees, the output cycle is advanced half a period, restoring the true input–output pairing.

**pairing residual** The magnitude-weighted RMS phase residual, in radians, over the harmonics of order two and above after the frame fit; a genuine pairing sits well under the accept bar

and a record of another session or setting, or a device whose harmonics have no fundamental-anchored phase relationship, does not.

**parity test** A continuous-integration test that runs the shipped code and the independent Python reimplement on identical synthetic devices and asserts, at every valid point, agreement between the two to a stated tolerance and agreement of both with analytic ground truth to a stated tolerance.

**peak tile** The map’s headline figure, in three states: the loudest corroborated clear cell as a peak; else the loudest unmasked cell as a bound (“at or below”); else, when the map’s fundamental is absent, no percentage at all.

**phantom capture** A synthetic capture  $\sum_k m_k A \sin(k\varphi(t))$  built from the sweep’s own phase law, so that its  $k$ -th harmonic response is exactly  $m_k$  by construction, with no content above the listed orders and no aliasing; the oracle’s exact case.

**phase removal** The §6.5 correction: the capture’s harmonic phases are fitted against the paired Harmonic Distortion record’s branch phases with one frame shift, each harmonic’s burden over the memoryless lattice is folded out sequentially, and the correction is applied as a full-band unity-magnitude all-pass, so filter phase before and after the clipper is removed and only unpredicted phase — memory evidence — remains.

**polarity** Whether the device inverts: read from the paired record’s  $H_1$  as the energy-weighted mean unit phasor over the low band, whose real part is the confidence (a quadrature response is coherent yet polarity-mute); stated on the figure only when the frame fit confirms the pairing.

**presence gate** The analyzer’s per-bin criterion, amplitude  $> 4\times$  the local floor ( $20\log_{10} 4 = 12.04$  dB), decided at capture and frozen in the payload as the product’s `aboveFloor` flag.

**product order** ( $|m| + |n|$ ) The order of the recipe  $(m, n)$ : the polynomial power that first produces it. The analyzer enumerates every recipe of order two through seven — 54 on a lattice with no collisions — and a shared bin is kept by its lowest order.

**product reading** One of four outcomes per product, decided in a fixed order: refused on lattice position (beside a played tone or DC), out of band (under 20 Hz), refused beside a louder product, refused by the analyzer’s window or under the loudest product’s skirt, then present with its margin over the capture floor or absent with the bin’s own read as a bound. A refusal is never silently absent.

**rate constant** ( $L$ ) The time in seconds over which the sweep’s instantaneous frequency grows by a factor of  $e$ :  $f(t) = f_1 e^{t/L}$ . Fixed by the synchronization condition from the requested duration.

**read scatter** The one-sigma uncertainty, in dB, of a magnitude read at its estimated signal-to-noise ratio — the Rician mean-of-dB statistic, pinned by a Monte-Carlo table and saturating at the pure-noise value.

**reduced tone ratio** ( $a : b$ ) The tone bins in lowest terms. Two recipes of order at most  $K$  can share a bin only when  $a + b \leq 2K$ , so the probe insists on  $a + b \geq 2K + 1$ ; the ratio alone fixes the delivered interval, whatever the rate or length. The shipped fifth is 109 : 163.

**residual alignment error** Any difference between the integer loop latency removed before binning and the true delay, in samples; it rotates the output’s fundamental by  $2\pi f_0 e / f_s$  and harmonic  $k$  by  $k$  times that, opening the figure into a phase loop that grows with the note.

**sideband pair statistic** The mean of the two innermost detector bins about each tone, and the difference between the two tones' pairs against  $20 \log_{10}(f_2/f_1)$ : a shared fractional frequency wander separates the pairs by that figure, while amplitude or frequency-independent phase modulation leaves them equal — a shape the line states without a verdict.

**static curve** The output-versus-input estimate that ignores the loop: the cycle is split at the input's extremes into a rising and a falling branch, each is interpolated onto a common input grid, and the two are averaged — never binned by input value.

**sweep averaging** Combining  $N$  back-to-back sweeps from one capture by complex, bin-wise averaging of their harmonic responses after each repeat is derotated by its fitted capture delay; the noise power falls as  $1/N$  where the residual is noise, and magnitudes are never averaged.

**synchronization condition** The requirement that  $f_1 L$  be an integer, met by rounding the rate constant; it makes the sweep's phase at every harmonic's arrival a whole number of cycles, so harmonic responses separate with meaningful phase.

**synchronized swept sine** ( $x(t)$ ) A sine whose frequency rises exponentially from  $f_1$  to  $f_2$  over the duration  $T$ , with its rate constant chosen so that  $f_1 L$  is an integer; under that condition each harmonic a memoryless device generates is an exact time-advanced copy of the sweep, advanced by  $L \ln k$ .

**tanh** The hyperbolic tangent, as the saturating function  $y = \tanh(gx)$ : linear for small inputs and bending smoothly toward  $\pm 1$ , which it never reaches — the textbook soft clipper. It is odd-symmetric, so it produces odd harmonics only, and its ladder falls off steeply with order. The worked example throughout these documents is  $\tanh(2x)$  at a sweep amplitude of 0.5, whose exact harmonic amplitudes are computed by quadrature.

**total harmonic distortion** (THD)  $\sqrt{\sum_{k \geq 2} |H_k(kf_0)|^2} / |H_1(f_0)|$  over the orders that are in the audible band and pass the validity bound — THD, not THD+N; undefined where the fundamental is not present.

**transfer curve** The X–Y figure of a device under a steady low-frequency sine: instantaneous output against instantaneous input over one phase-averaged cycle. A memoryless device retraces a single curve; linear filter phase and genuine memory open it into a loop.

**truncated column** A column where the validity bound removed an order that the declared band would have counted, so its THD cells under-read their band; the column is flagged and excluded from the tiles, never corrected.

**two-tone measurement** Chord IMD: two simultaneous sines at a musical interval, snapped to exact bins of the analysis window, driven through the device; the result is the level of every intermodulation product up to the seventh order, the three capture-time percentage sums, and the coherence detector's reads beside each tone.

**unsigned lobe area** The total loop area: the closed trajectory is split at every self-crossing and the absolute area of every lobe is summed, so opposite-winding lobes add instead of cancelling; lobes under a floor fraction of the box are dropped as bin noise.

**validity bound** Where a harmonic read is trustworthy:  $f_1 \leq f_0 \leq f_2$  and  $f_0 \leq f_s/(2k+1)$  (the alias-image bound), plus  $f_0 \leq f_{\text{cal}}/k$  on a compensated record. Reads outside it are excluded from every fit, feature, summary and comparison.

## A Scripts, tests and how to reproduce

Paths are relative to Docs/Methods/ in the repository, except the tool, which lives under Tools/. The repository: <https://github.com/PhaseDog/PedalScope>

| Path                       | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tools/analysisdump         | the shipped-code oracle: <b>synth</b> (a known device through the shipped analyzer, with exact truth beside it), <b>transfer-synth</b> (a known device through the shipped transfer path, paired, with its truth beside it), <b>imd-synth</b> (a known device or a phantom through the shipped two-tone path, with the torus-quadrature truth beside every product), <b>journey-synth</b> (a known device through a known loop and the shipped Gain Map path, the ladder beside every cell), <b>windows</b> , <b>manifest</b> |
| generated/manifest.json    | the manifest, committed, drift-checked                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| py/harmonic_distortion.py  | the reimplementations; <b>-help</b> lists the same options as <b>synth</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| py/parity_hd.py            | the comparison and the tolerances, with their provenance                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| py/test_parity_hd.py       | the CI parity test (pytest)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| py/transfer_curve.py       | the Transfer Curve reimplementations (the options of <b>transfer-synth</b> )                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| py/parity_transfer.py      | its comparison and tolerances, with provenance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| py/test_parity_transfer.py | its CI parity test                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| py/chord_imd.py            | the Chord IMD reimplementations (the options of <b>imd-synth</b> )                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| py/parity_imd.py           | its comparison and tolerances, with provenance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| py/test_parity_imd.py      | its CI parity test                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| py/gain_map.py             | the Gain Map reimplementations (the options of <b>journey-synth</b> ); imports <b>harmonic_distortion.py</b>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| py/parity_journey.py       | its comparison and tolerances, with provenance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| py/test_parity_journey.py  | its CI parity test                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| py/make_figures.py         | the figures, seeded, byte-stable, with TSV sidecars of the plotted values                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| py/gen_docs.py             | the generated fragments this document inputs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| terms.yaml                 | the terms source; <b>py/test_terms.py</b> is its guard                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| make methods-test          | parity and terms guards                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| make methods-docs          | manifest → figures → fragments → both PDFs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Toolchain: Python 3 with the versions pinned in Docs/Methods/requirements.txt; L<sup>A</sup>T<sub>E</sub>X through latexmk with pdf<sub>l</sub>atex, pinned in Docs/Methods/tex/latexmkrc. The Swift tool builds with the repository's own toolchain.

## References

- [1] Angelo Farina. Simultaneous measurement of impulse response and distortion with a swept-sine technique. In *Audio Engineering Society Convention 108*, Paris, February 2000. Preprint 5093.
- [2] Antonin Novák, Laurent Simon, František Kadlec, and Pierrick Lotton. Nonlinear system identification using exponential swept-sine signal. *IEEE Transactions on Instrumentation and Measurement*, 59(8):2220–2229, 2010. doi:10.1109/TIM.2009.2031836.

- [3] Antonin Novák, Pierrick Lotton, and Laurent Simon. Synchronized swept-sine: Theory, application, and implementation. *Journal of the Audio Engineering Society*, 63(10):786–798, 2015. doi:[10.17743/jaes.2015.0071](https://doi.org/10.17743/jaes.2015.0071).
- [4] S. O. Rice. Mathematical analysis of random noise. *Bell System Technical Journal*, 23(3): 282–332, 1944.